GLOBAL SYSTEMS FOR MOBILE

GSM

GSM (**Global System for Mobile Communication**) is **open**, **digital** cellular technology used for transmitting mobile voice and data services

Originally (***Groupe Special Mobile***) is a standard set developed by the European Telecommunication Standards Institute (**ETSI**) to describe protocols for 2G digital cellular networks used by mobile phones. It became de facto global standard for mobile communication with over **80%** market share.

The GSM standards was developed as a replacement for 1G analog cellular networks and originally described a digital circuit-switched network optimized for full duplex voice telephony. This was expanded over a time to include data communication, first by Circuit Switched Transport, then Packet Data Transport via **GPRS** (General Packet Radio Services) & **EDGE** (Enhanced Data Rates for GSM Evolution) or **EGPRS**

GSM uses a Variation of **TDMA**

In the **US :- Sprint, Verizon** and **US Cellular** use **CDMA** and

AT&T and **T-Mobile** use GSM

Most of The Rest of The World uses **GSM**

The Global Spread of GSM came about Because in **1987**

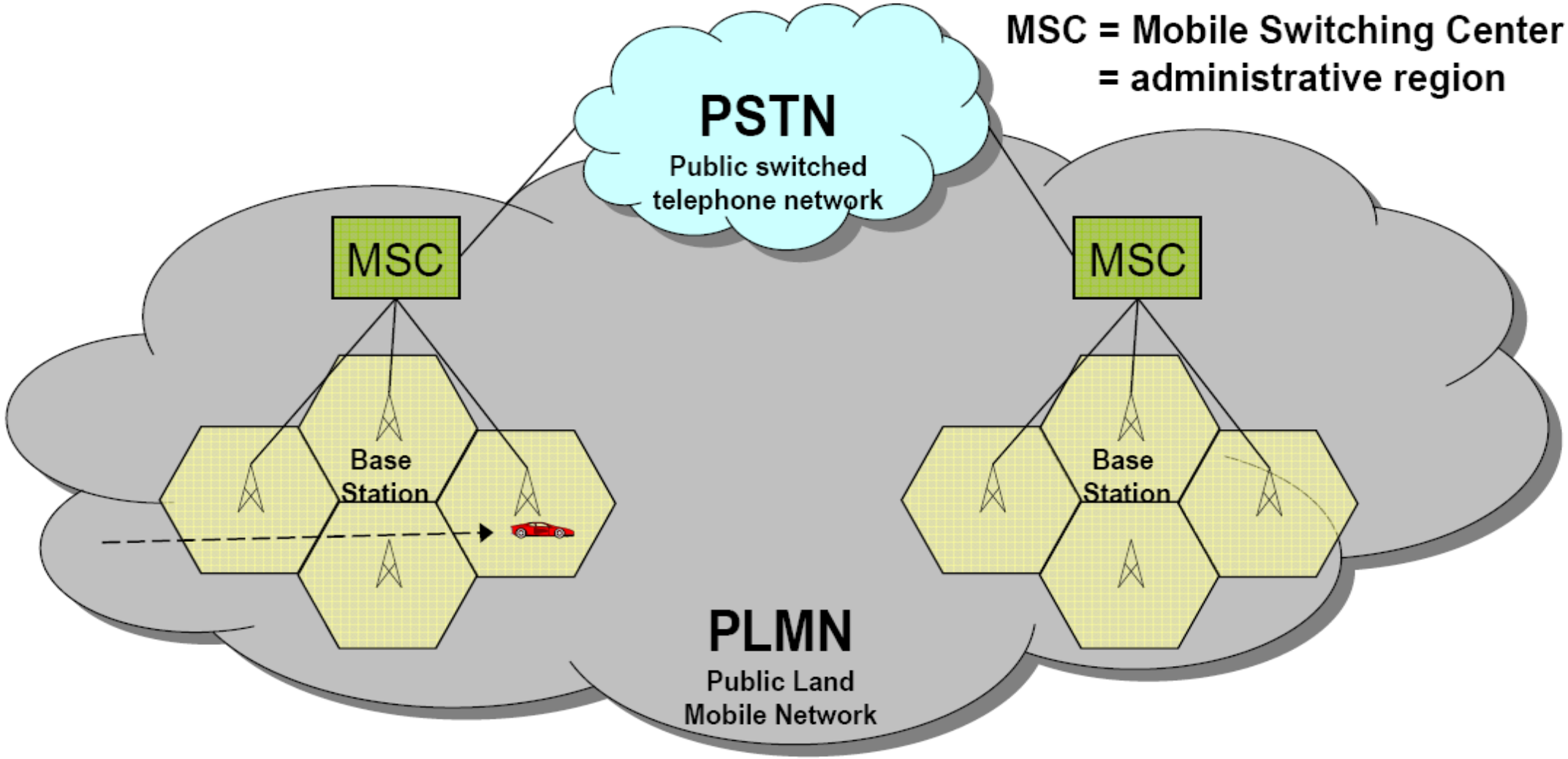
Europe **Mandated** The Technology **By Law**

GSM **Comes** from an Industry **Consortium**

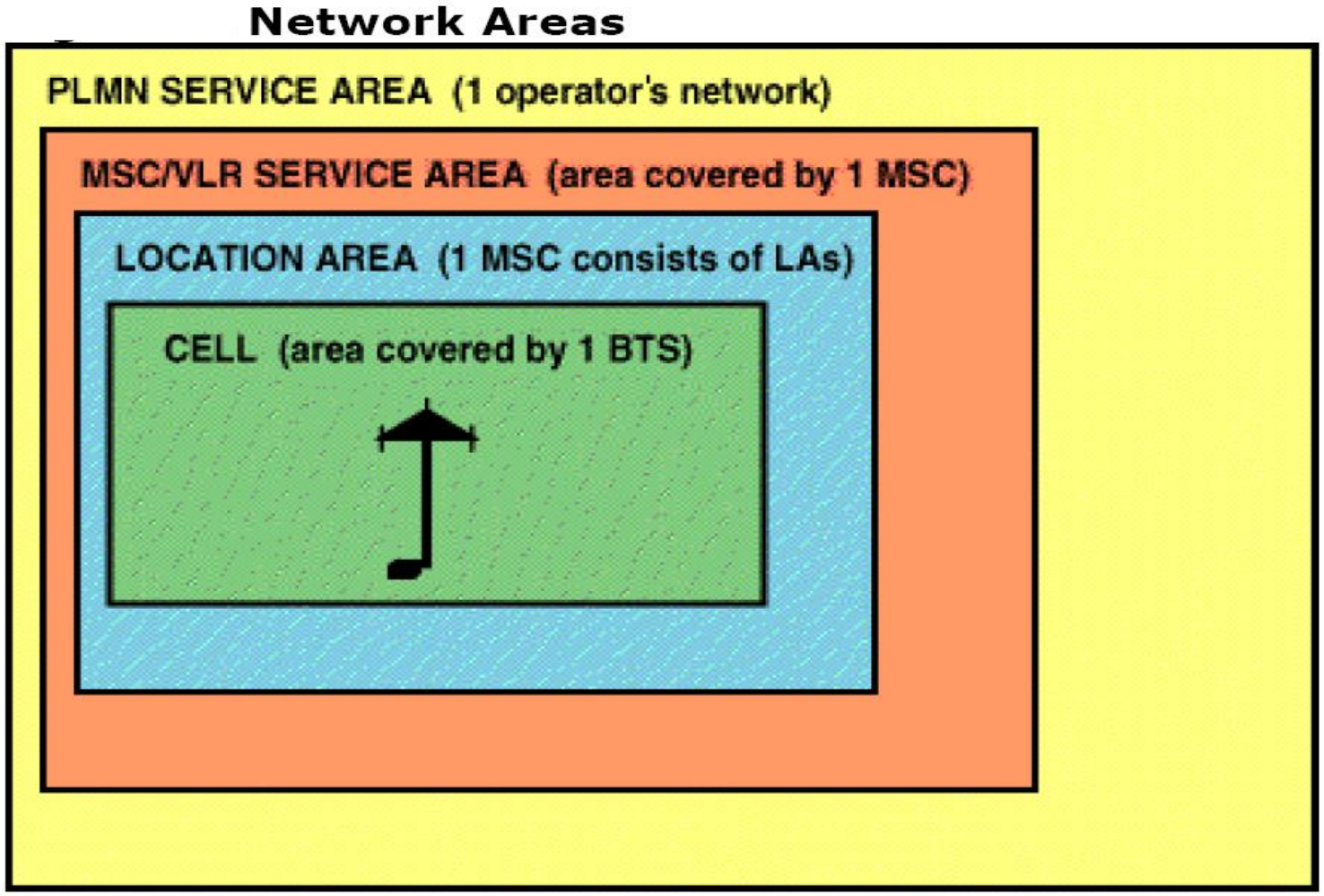
Usage of GSM is Miles **ahead** of **CDMA**

GSM Network

high-level view

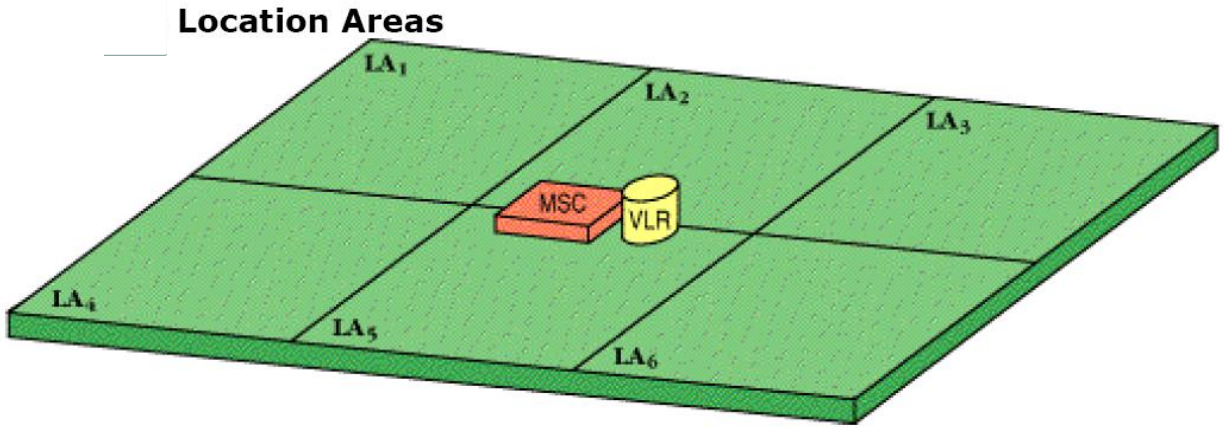


MSC role: telephone switching central with special mobility management capabilities



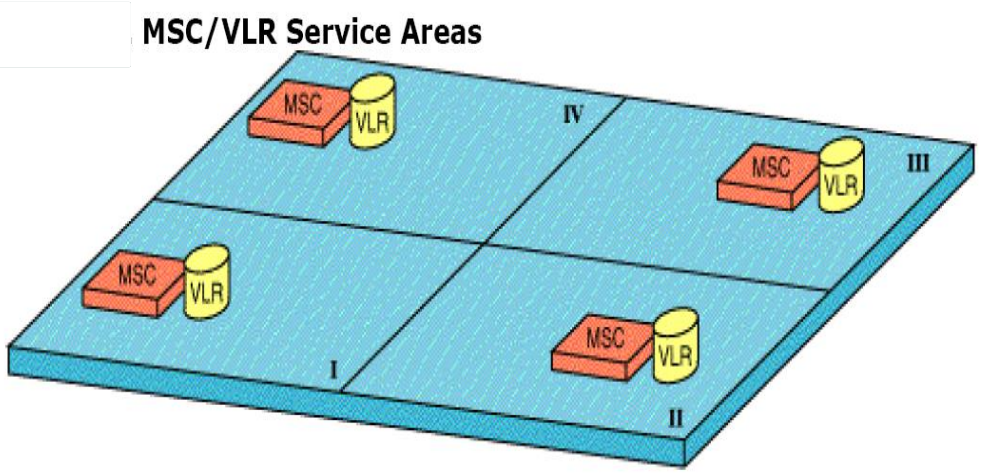
The GSM network identifies each cell via the cell global identity (CGI) number assigned to each cell.

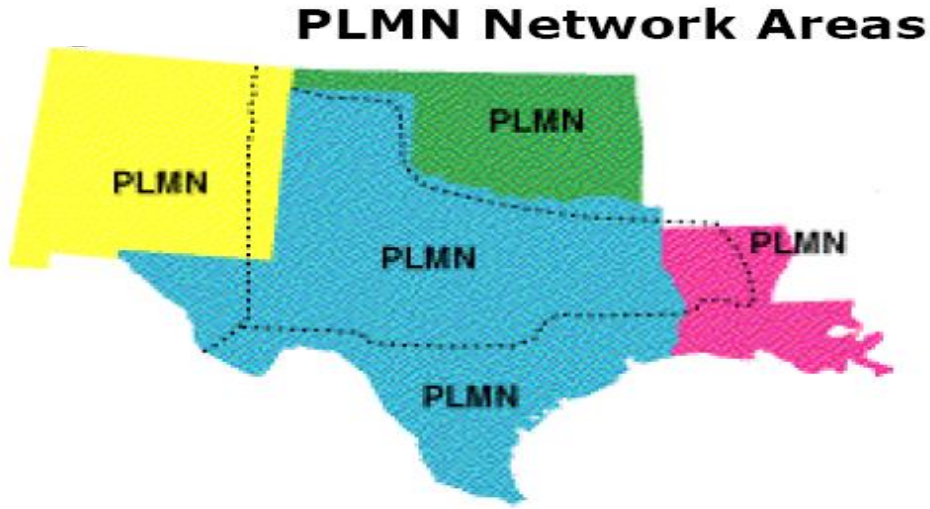
The cell is the area given radio coverage by one base transceiver station.



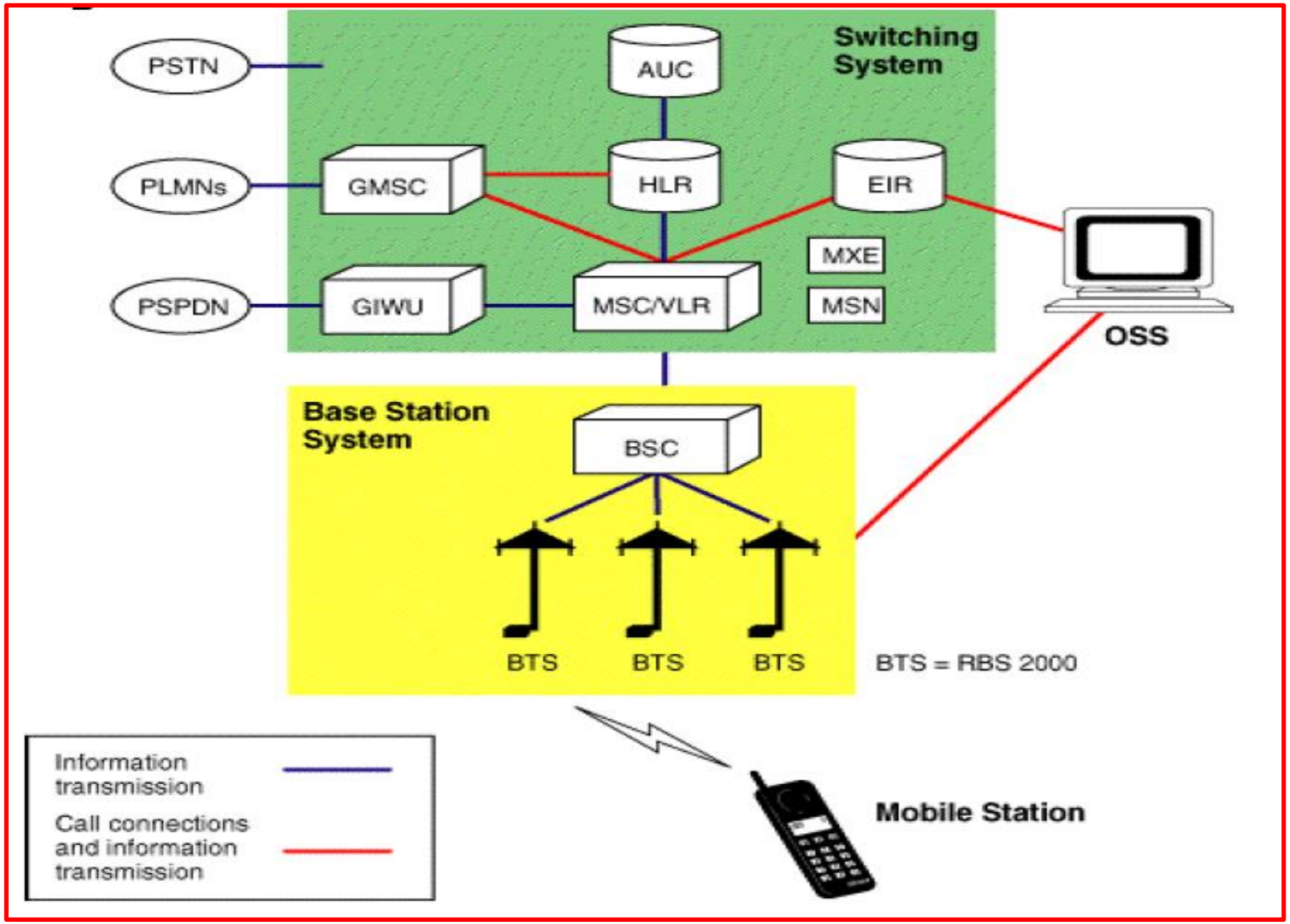
The location area is a group of cells. **It is the area in which the subscriber is paged**. **Each LA is served by one or more base station controllers**, yet only by a **single MSC** Each LA is assigned a location area identity (**LAI**) number

An MSC/VLR service area represents the part of the GSM network that is covered by one MSC and which is reachable, as it is registered in the VLR of the MSC

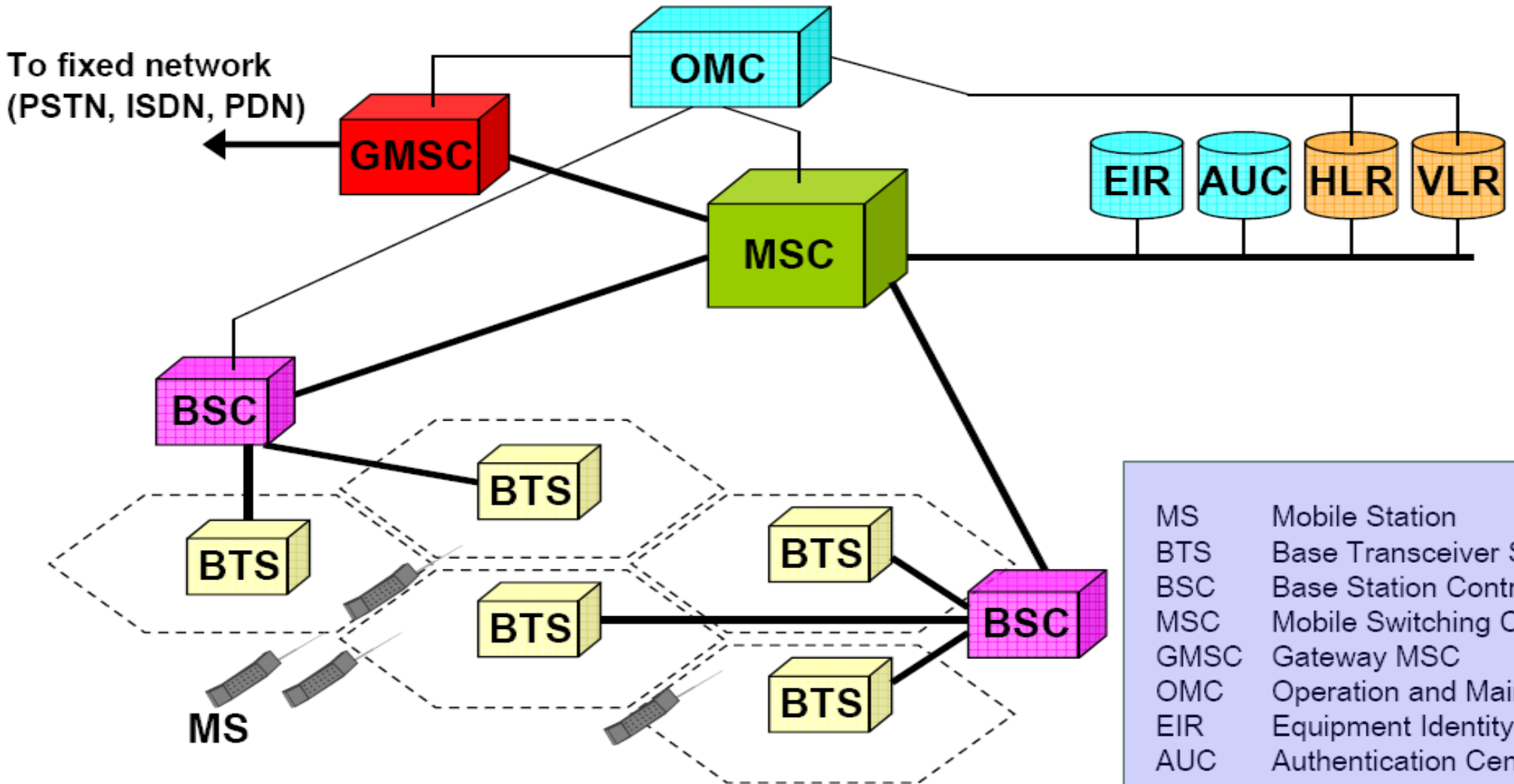




The PLMN service area is an area served by one network operator .


Switching System
PSTN
AUC
PLMNs
GMSC
HLR
EIR
MXE
PSPDN
GIWU
MSC/VLR
MSN
OSS
Base Station System
BSC
BTS
BTS
BTS
BTS = RBS 2000
Information transmission
Call connections and information transmission
Mobile Station

GSM essential components



MS	Mobile Station
BTS	Base Transceiver Station
BSC	Base Station Controller
MSC	Mobile Switching Center
GMSC	Gateway MSC
OMC	Operation and Maintenance Center
EIR	Equipment Identity Register
AUC	Authentication Center
HLR	Home Location Register
VLR	Visitor Location Register

Foundation

- Hybrid frequency-division/time-division multiple access
  - FDMA - division by frequency of the (maximum) 25 MHz allocated bandwidth into 124 carrier frequencies spaced 200 kHz apart.
    - One or more carrier frequencies assigned to each base station
  - Each carrier frequency divided in time, using TDMA
  - Fundamental unit of time in this TDMA scheme is a burst period approx. 0.577 ms long
  - Eight burst periods are grouped into a TDMA frame (approx. 4.615 ms) = basic unit for the definition of logical channels
  - A physical channel is one burst period per TDMA frame
  - Slow frequency at upto 217 times per second
    - hopping algorithm is broadcast on the broadcast control channel
    - helps alleviate multipath fading
    - co-channel interference is effectively randomized
    - Note: broadcast and common control channels are not subject to frequency hopping and are **always** transmitted on the same frequency
- Infrastructure based on Signalling System 7 (SS7)

GSM contributions

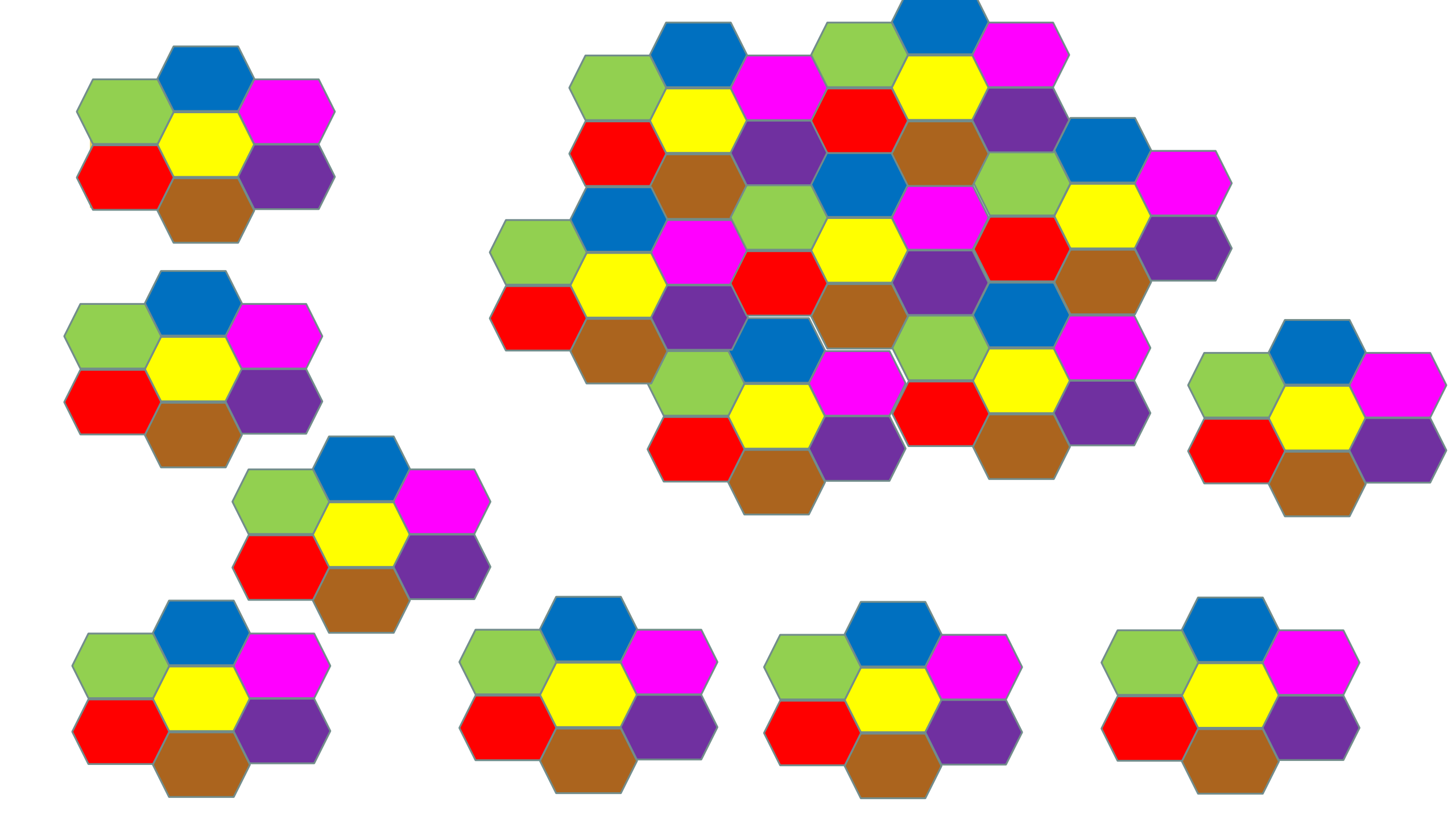
- Location-based mobility management
- Mobile assisted handover
- Temporary Mobile Subscriber ID (TMSI)

Distintive features of GSM

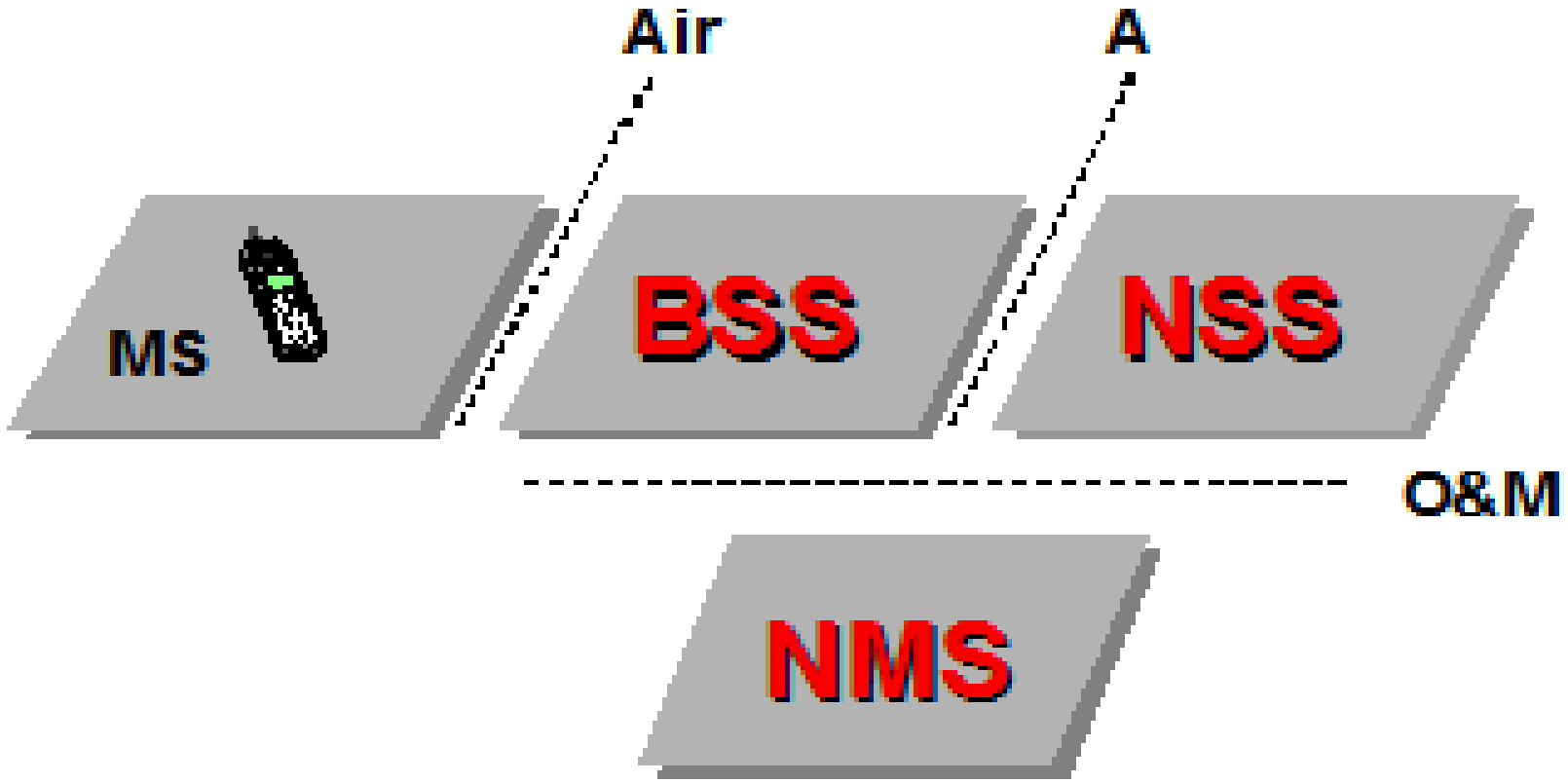
- Cooperative development by many actors from many countries
- preserved open interfaces between the subsystems (especially between infrastructure elements -- particularily between base stations and switches)
- specified a large number of interfaces!
- Phased release - since they could not make all the innovations in time for their targeted 1991 introduction
  - Phase 1 GSM spec. - 100 sections and 5,320 pages!
    - telephony - with some added features
    - emergency calls
    - data transmission at 2.4/4.8/9.6 kbit/s (transparent {the error correction done by a forward error correction (FEC) mechanism}/non-transparent {information is repeated when it has not been correctly received})
    - short message service (SMS)
  - Phase 2
    - non-voice services and enriched telephony

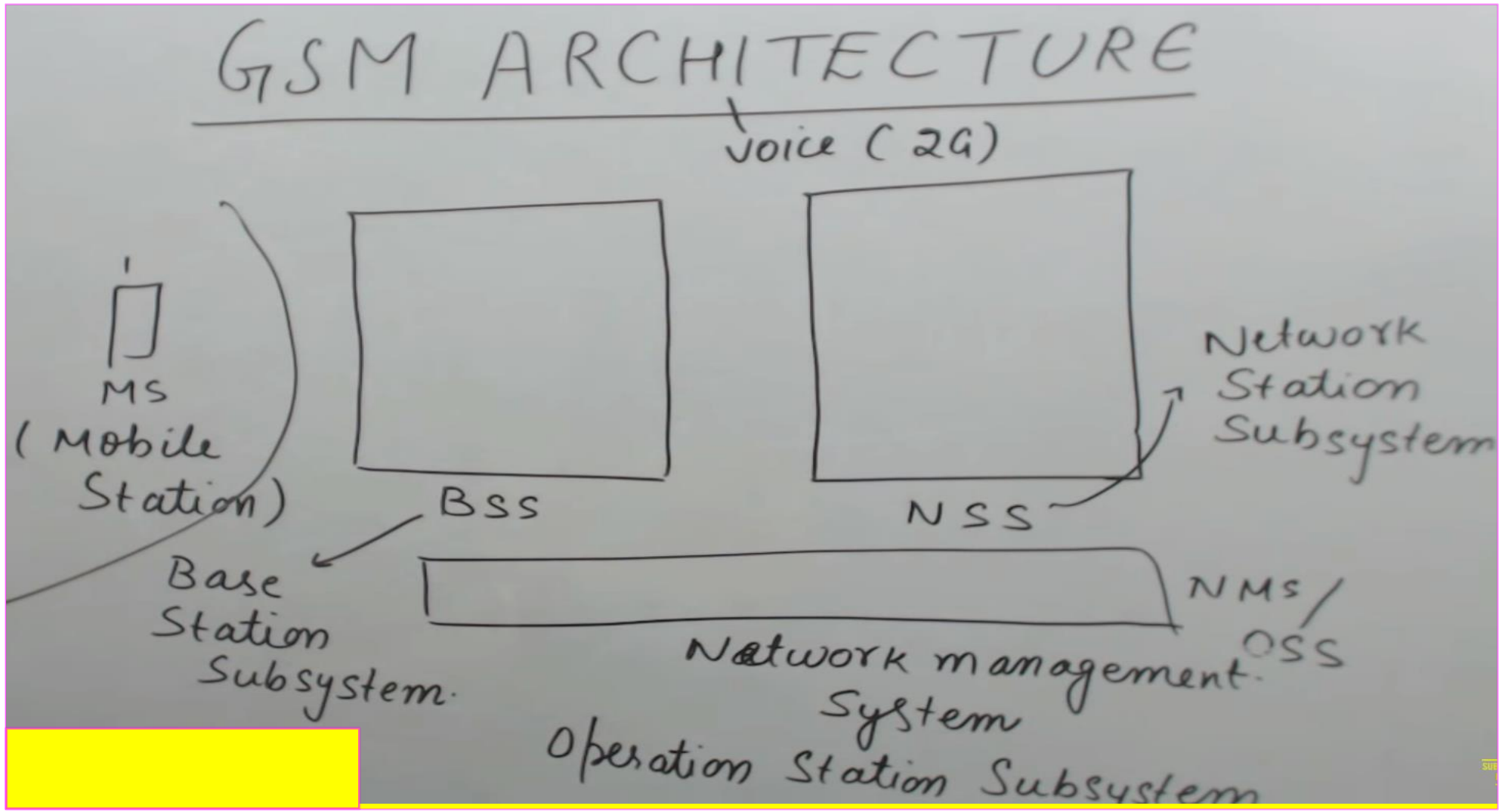
GSM Requirements

- Service portability
  - mobile should be able to be used in any of the participating countries with international roaming and standardized numbering & dialing (but possibly at different rates!)
  - usable for both wireline line services and for mobile service
  - usable when: walking, driving, boating, … (upto 250 km/h)
- Quality of service and Security
  - quality at least as good a previous analog systems
  - capable of offering encryption
- Good radio frequency utilization
  - high spectrum efficiency
  - co-existance with earlier systems in the same bands
- Modern network
  - following ITU recommendations - to allow efficient interoperation with ISDN networks
  - supporting voice and low rate data
  - standardized mobility and switching support
  - standardized interfaces between the subsystems - to allow a mix-and-match system
- System optimized to limit cost of mobiles (and to a lesser extent to limit the cost of the whole system)
  - GSM required higher complexity mobiles than earlier analog systems
  - subscriber cost less than or equal to existing analog systems

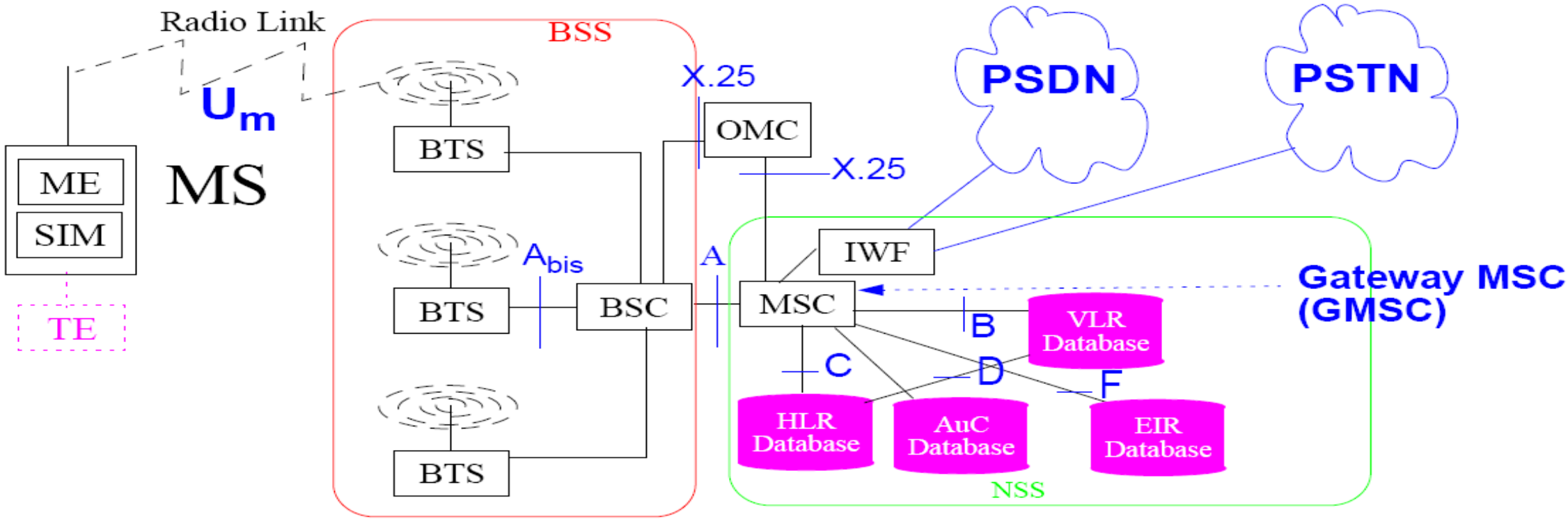


GLOBAL SYSTEMS FOR MOBILE : SYSTEM ARCHITECTURE

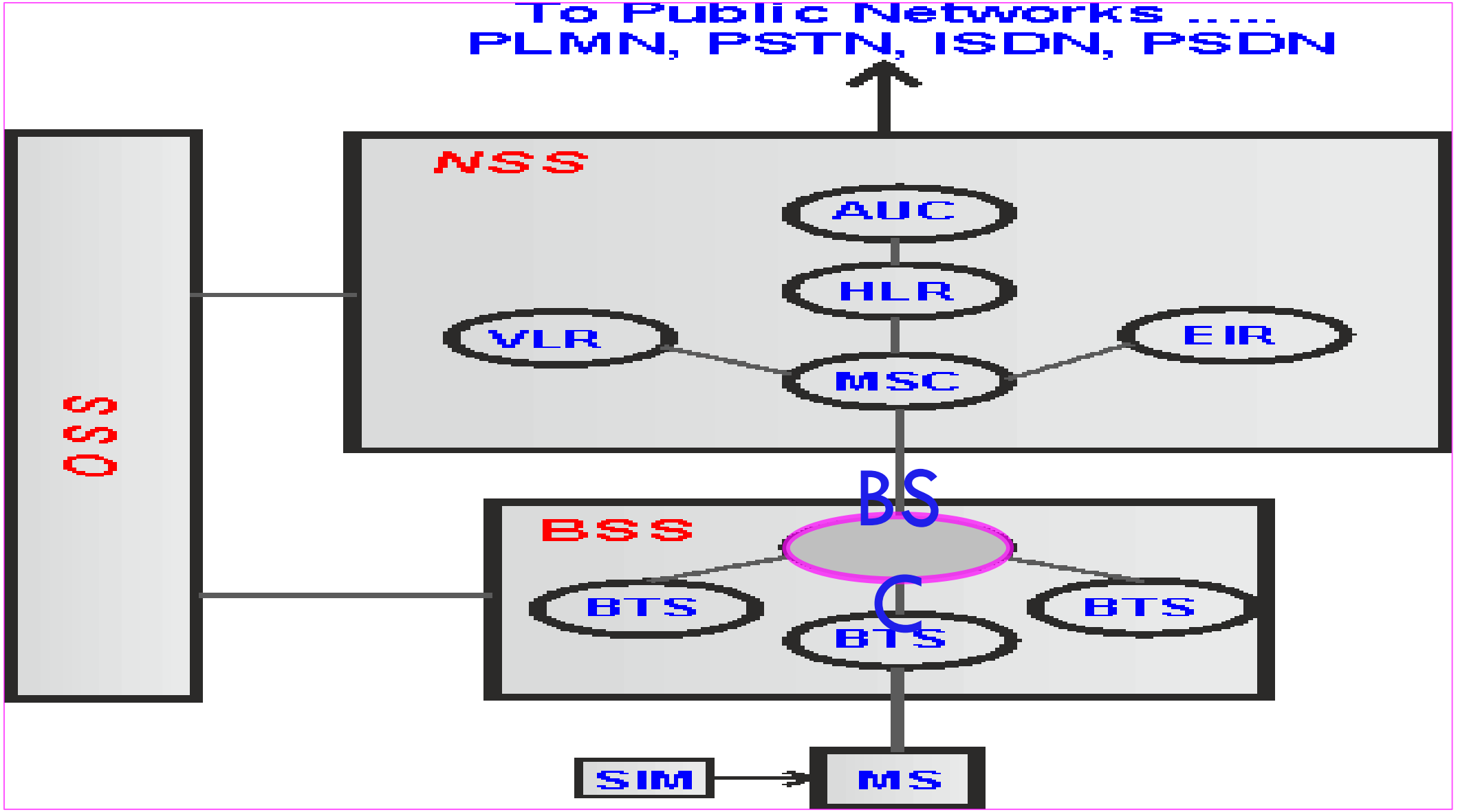

Air
A
MS
BSS
NSS
O&M
NMS


GSM ARCHITECTURE
Voice (2G)
MS
(Mobile Station)
Bss
NSS
Network Station Subsystem
Base Station Subsystem.
NMS/OSS
Network management System
Operation Station Subsystem

GSM Architecture

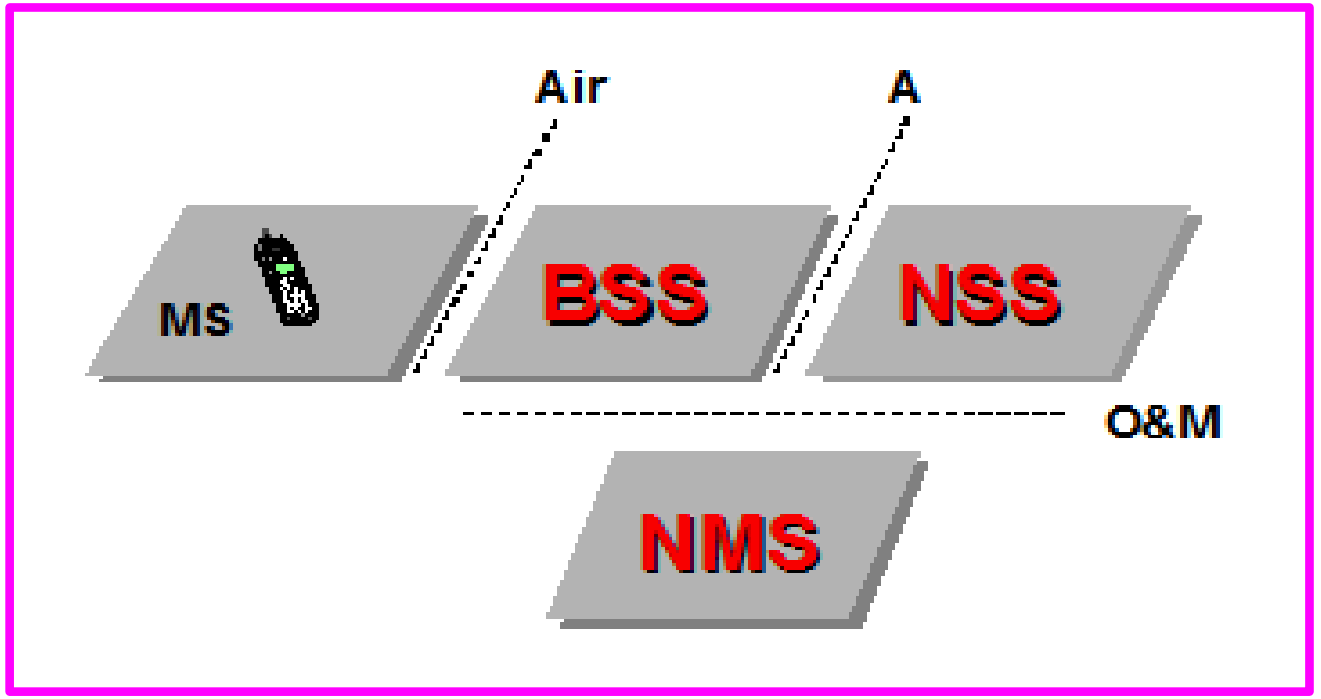


MS	Mobile Station
BSS	base station system
NSS	network and switching subsystem


To Public Networks .....
PLMN, PSTN, ISDN, PSDN
NSS
AUC
HLR
VLR
EIR
MSC
OSS
BSS
BTS
BTS
BTS
SIM
MS

Mobile Station (MS)

- Subscriber Identity Module (SIM)
- Mobile Equipment (ME)
- Mobile Terminal (MT)



Mobile station

Mobile stations (**MS**), mobile equipment (**ME**) or as they are most widely known, cell or mobile phones are the section of a GSM cellular network that the user sees and operates. In recent years their size has fallen dramatically while the level of functionality has greatly increased. A further advantage is that the time between charges has significantly increased.

There are a number of elements to the cell phone, although the two main elements are the main hardware and the SIM.

The hardware itself contains the main elements of the mobile phone including the display, case, battery, and the electronics used to generate the signal, and process the data receiver and to be transmitted. It also contains a number known as the **International Mobile Equipment Identity** (**IMEI**). This is installed in the phone at manufacture and **"cannot"** be changed. It is accessed by the network during registration to **check** whether the equipment has been reported as **stolen**.

The **SIM** or Subscriber Identity Module contains the information that provides the identity of the user to the network. It contains are variety of information including a number known as the **International Mobile Subscriber Identity** (**IMSI**).

Mobile Station (MS)

GSM separates user mobility from equipment mobility, by defining two distinct components

➔ Mobile Equipment

- → The cellular telephone itself (or the vehicular telephone)

⇨ Address / identifier:

- → IMEI (International Mobile Equipment Identity)

➔ Subscriber Identity Module (SIM)

- → Fixed installed chip (plug-in SIM) or
- → exchangeable card (SIM card)

⇨ Addresses / identifiers:

- → IMSI (International Mobile Subscriber Identity)
- → MSISDN (Mobile Subscriber ISDN number)
  - » the telephone number!



15x17mm



Mini UICC
12x15mm
(standardized feb 04)

Mobile Equipment (ME)

"the phone" itself - radio and radio interface, display, keyboard, etc.

performs: radio transmission and reception, authentication, handover, encoding and channel encoding.

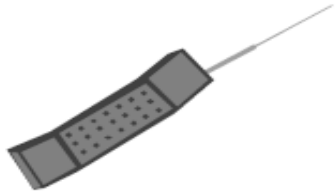
note: ME with SIM can only make emergency (112) calls

Radios operate in one or more of the following bands:

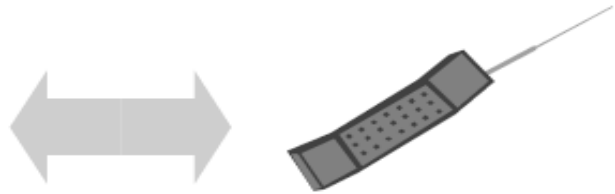
- GSM900 since 900 MHz - the original frequency band
  - Uplink: 890..915 MHz (= mobile station to base station)
  - Downlink: 935..960 MHz (= base station to mobile station)
- GSM1800 (also known as DCS1800)
  - Uplink: 1710..1785 MHz
  - Downlink: 1805..1880 MHz
- GSM1900 (also known as PCS 900)
  - Uplink: 1850..1910 MHz
  - Downlink: 1930..1990 MHz

ME identified by International Mobile Equipment Identity (IMEI)

Mobile Equipment structure



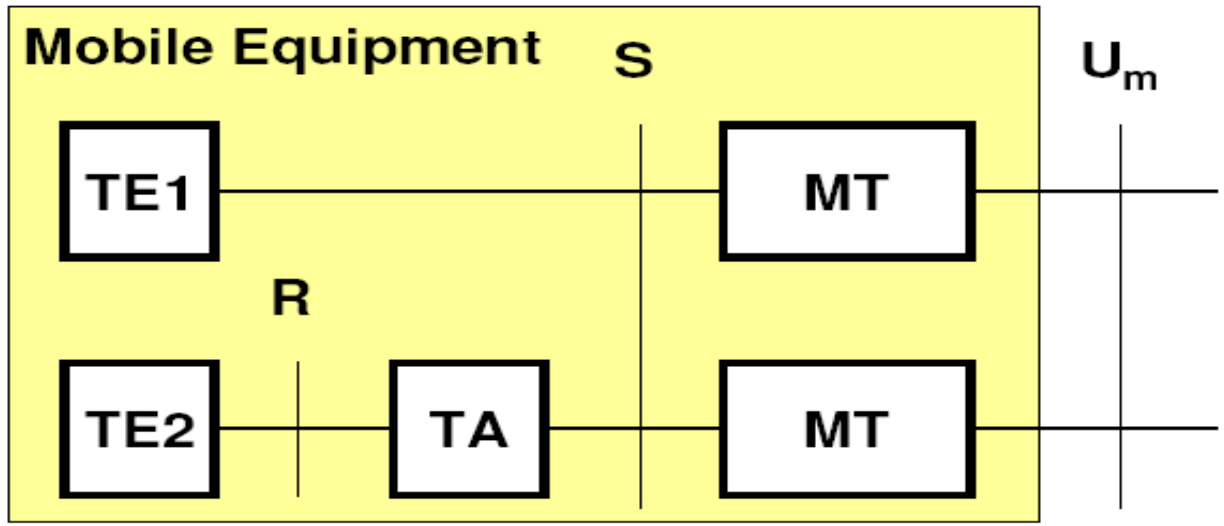








- ➔ **Mobile Termination** functions
  - ⇨ Radio interface (tx, rx, signalling)
- ➔ **Terminal Equipment** functions
  - ⇨ User interface (microphone, keyboard, speakers, etc);
  - ⇨ Functions specific of services (telephony, fax, messaging, etc), independent of GSM
- ➔ **Terminal Adaptor** functions
  - ⇨ Interfaces MT with different types of terminals (PCs, Fax, etc.)



IMEI

International Mobile Equipment Identity

- ➔ **Uniquely identifies the mobile equipment**
- ➔ **15 digits hierarchical address**
- ➔ **assigned to ME during manifacturing and "type approval" testing**
  - ⇨ Type approval procedure: guarantees that the MS meets a minimum standard, regardless of the manifacturer
- ➔ **IMEI structure:**

TAC – 6 digits (Type Allocation Code)	FAC – 2 digits (Final Assembly Code)	SNR – 6 digits (Serial Number)	SP – 1 digit (Spare Digit)
centrally assigned upon type approval **Before april 1, 2004: TAC = Type Approval Code**	assigned by manufacturer Identifies place where ME was assembled or manufactured	assigned by manufacturer Unique for given TAC+FAC combination	Additional digit available >=2: terminal valid for phase 2+

IMEI management

➔ **Protection against stolen and malfunctioning terminals**

⇨ Italian "memorando di intesa" signed on maj 5, 2003

➔ **Equipment Identity Register (EIR): 1 DataBase for each operator; keeps:**

⇨ **WHITE LIST:**

→ valid IMEIs

→ Corresponding MEs may be used in the GSM network

⇨ **BLACK LIST:**

→ IMEIs of all MEs that must be barred from using the GSM network

→ Exception: emergency calls (to a set of emergency numbers)

→ Black list periodically exchanged among different operators

⇨ **GRAY LIST:**

→ IMEIs that correspond to MEs that can be used, but that, for some reason (malfunctioning, obsolete SW, evaluation terminals, etc), need to be tracked by the operator

→ A call from a "gray" IMEI is reported to the operator personnel

Subscriber Identity Module (SIM)

- small form factor - which can be removable and can be moved from one terminal to another
  - smart card (generally too large for handsets!)
  - plug-in SIM (the processor and contact from a smart card)
- **user** authenticated via a Personal Identity Number (PIN)
- if PIN entered incorrectly, N times, then phone is locked for all but emergency calls, until you enter a PIN unblocking key (PUK)
- contains subscriber information:
  - some which is fixed by operator (may include preferred network provider(s))
  - some which is changable by the user (list of short numbers, phone list, SMS messages, …)
- can be updated via:
  - keyboard
  - attached terminal equipment
  - over the air (OTA) via SMS message sent by operator/application/… built using SIM Toolkit
- often the SIM is owned by the operator
- profiles - operator/subscription info; SIMs are required to be able to hold at least two profiles
- contains International Mobile Subscriber Identity (IMSI)

SIM card

Subscriber Identity Module

- **Uniquely associated to a user**
  - Not to an equipment, as in first generation cellular networks
- **Stores user addresses**
  - IMSI
  - MSISDN
  - Temporary addresses for location, roaming, etc
- **authentication and encryption features**
  - All security features of GSM are stored in the SIM for maximum protection
    - subscriber's secret authentication key (*Ki*)
    - Authentication algorithm ("secret" algorithm - A3 – not unique)
    - Cipher key generation algorithm (A8)
- **Personalization**
  - SIM stores user profile (subscribed services)
  - RAM available for SMS, short numbers, user's directory, etc
  - Protection codes
    - PIN (Personal Identification Number, 4-8 digits)
    - PUK (PIN Unblocking Key, 8 digits)

IMSI

International Mobile Subscriber Identity

- **Uniquely identifies the user (SIM card)**
- **GSM-specific address**
  - ⇨ unlike MSISDN - normal phone number
- **15 digits hierarchical address**
- **assigned by operator to SIM card upon subscription**
- **IMSI structure (follows ITU-T E.212 numbering plan):**

MCC – 3 digits (Mobile Country Code)	MNC – 2 digits (Mobile Network Code)	MSIN – max 10 digits (Mobile Subscriber Identification Number)
Internationally standardized; identifies operator country **Italy: 222** **…** **Vatican=225** **San Marino=292**	Identifies operator network (PLMN) within country **TIM=01 VODAFONE=10** **WIND=88 BLU=98** **3ITA=99 IPSE=77 (planned)** **Elsacom=02 (globalstar, satellite)**	Uniquely identifies subscriber in the operator network

MSISDN

Mobile Subscriber ISDN Number

- **MSISDN: the "usual" telephone number**
  - ⇨ Follows international ISDN numbering plan (ITU-T E.164 recommendations)
  - ⇨ Structure:

CC – up to 3 digits (Country Code)	NDC – 3 digits (for PLMN) (National Destination Code)	SN – max 10 digits (Subscriber Number)

- **GSM is the first network to distinguish**
  - ⇨ The user identity (i.e. IMSI)
  - ⇨ From the number to dial (i.e. MSISDN)
- **Separation IMSI-MSISDN protects confidentiality**
  - ⇨ IMSI is the real user address: never public!
  - ⇨ Faking false identity: need signal IMSI to the network; but IMSI hard to obtain!
- **Separation IMSI-MSISDN allows**
  - ⇨ Easy modification of numbering and routing plans
- **single IMSI may be associated to several MSISDN numbers**
  - ⇨ E.g. different services (fax, voice, data, etc) may be associated with different MSISDN numbers

Temporary addresses

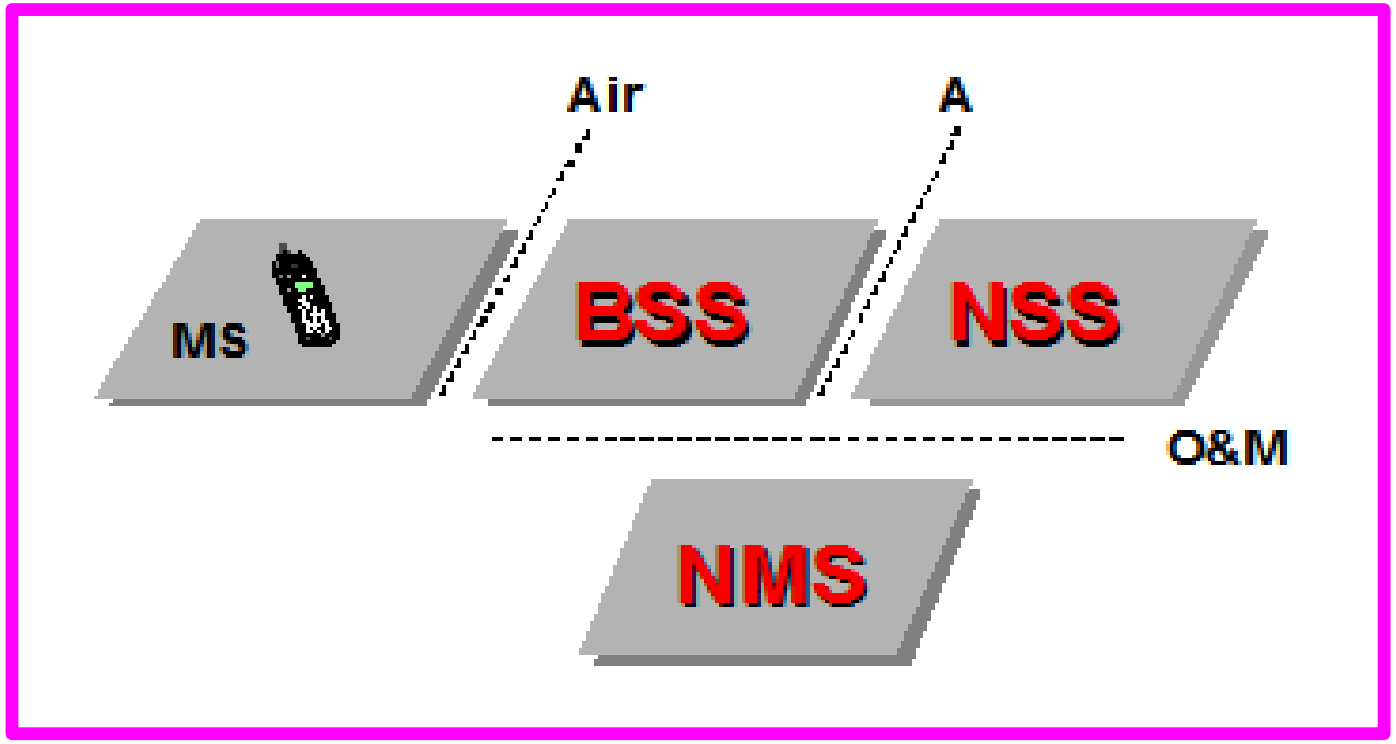
- **TMSI - Temporary Mobile Subscriber Identity**
  - 32 bits
  - assigned by VLR within an administrative area
    - → has significance only in this area
  - transmitted on the radio interface instead of IMSI
    - → reduces problem of "eavesdropping"
- **MSRN - Mobile Station Roaming Number**
  - An MSISDN number
    - → CC, NDC of the visited network
    - → SN assigned by VLR
  - Used to route calls to a roaming MS
    - → Subscriber Number (SN) assigned to provide routing information towards actually responsible MSC

Power saving and interference reduction

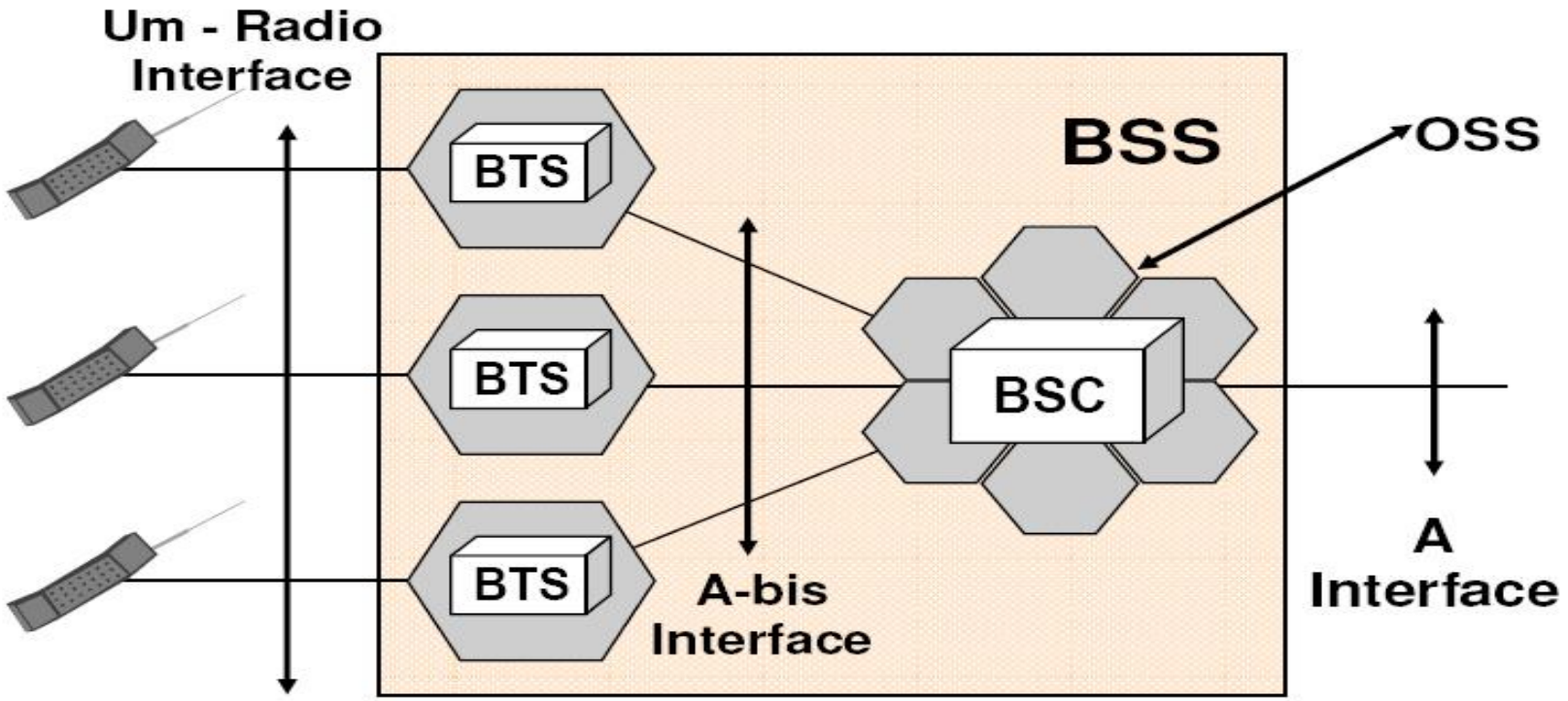
- To reduce the MS's power consumption and minimize interference on the air interface, during pauses in speech the MS does not transmit - this is called: Discontinuous transmission (DTX)
  - "Comfort noise" is artificially generated by the MS
- Discontinuous reception (DRX) - mobile listens to the paging channel, but only needs to wake up for its sub-channel of the paging channel
- To minimize co-channel interference and to conserve power, both the mobiles and the base transceiver stations operate at the lowest power level that will maintain an acceptable signal quality
  - Power levels can be stepped up or down in steps of 2 dBm from the peak power for the class down to a minimum of 13 dBm (20 milliwatts for MS)
  - only one step at a time and each step takes 60ms
  - there are 16 power levels (i.e., 30 db of range)
  - terminal is typically only transmitting in one time slot (i.e., 1/8 of the time - so its radiated power is on average 8db lower than the set power level)
  - Both mobile station and BTS continually measure the signal strength or signal quality (based on the bit error ratio), and pass the information to the base station controller (which manages power levels).

Base Station System (BSS)

- one or more base transceiver station (BTS) and
- base station controller (BSC)



Base Station Sub-System



⇨ **Base Transceiver Station (BTS)**

→ Transmitter and receiver devices, voice coding & decoding, rate adaptation for data

→ Provides signaling channels on the radio interface

→ Limited signal and protocol processing (error protection coding, link layer LAPDm)

⇨ **Base Station Controller (BSC)**

→ performs most important radio interface management functions:

→ Radio channels allocation and deallocation; handover management; …

Base Station Subsystem (BSS)

The Base Station Subsystem (BSS) section of the GSM network architecture that is fundamentally associated with communicating with the mobiles on the network. It consists of two elements:

- ***Base Transceiver Station (BTS):*** The BTS used in a GSM network comprises the radio transmitter receivers, and their associated antennas that transmit and receive to directly communicate with the mobiles. The BTS is the defining element for each cell. The BTS communicates with the mobiles and the interface between the two is known as the Um interface with its associated protocols.
- ***Base Station Controller (BSC):*** The BSC forms the next stage back into the GSM network. It controls a group of BTSs, and is often co-located with one of the BTSs in its group. It manages the radio resources and controls items such as handover within the group of BTSs, allocates channels and the like. It communicates with the BTSs over what is termed the Abis interface.

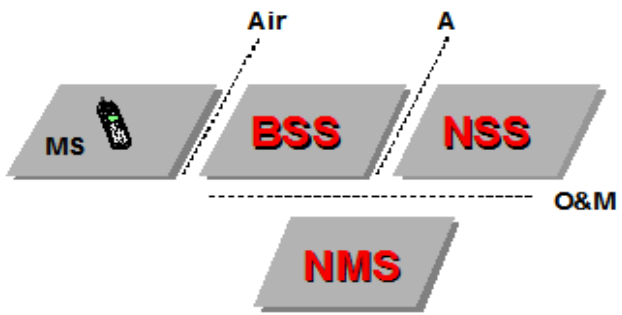
Base station controller (BSC)

BTSs are connected to a BSC which manages the radio resources

- call maintenance using the received signal strength sent by mobile stations normally every 480 ms
- initiate handovers to other cells,
- change BTS transmitter power, …

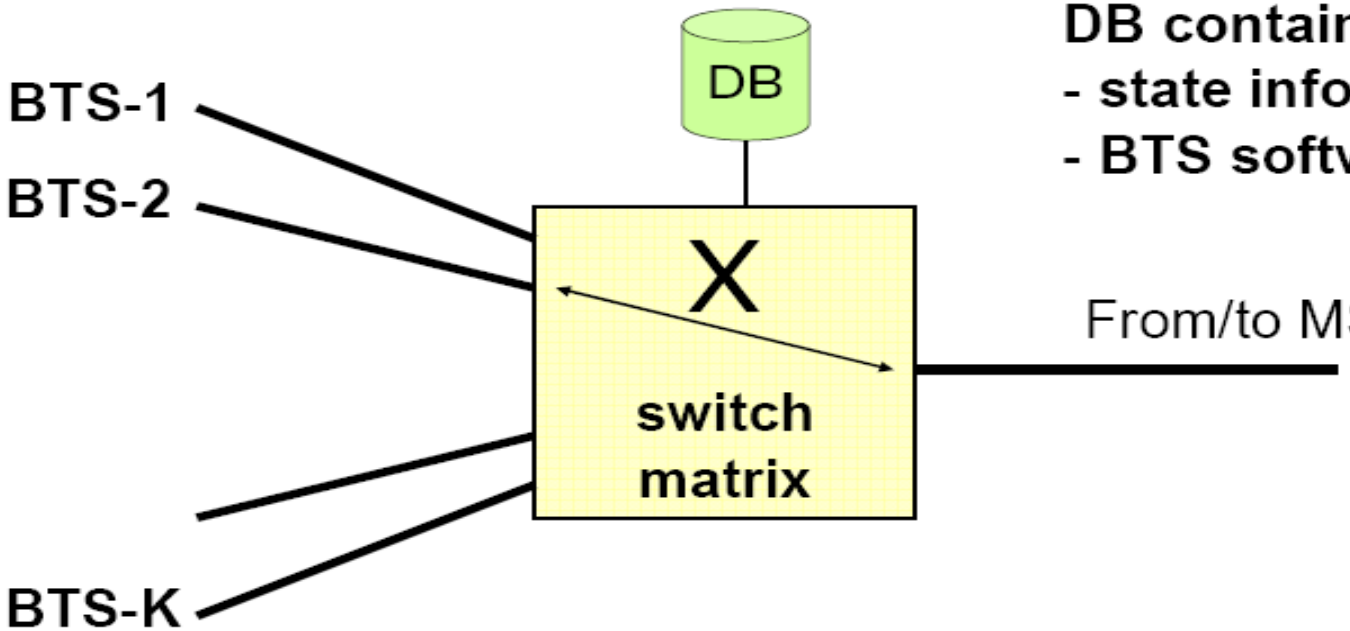
Task breakdown:

call activities	~20-25%
paging and SMS	~10-15%
mobility management	~20-25%
hardware checking/network triggered events	~15-20%



BSCs engineed for about 80% utilization, if overloaded, shed load by: (1) rejecting location updates, (2) rejecting MS originating calls, and (3) ignoring handoffs

Base Station Controller - BSC



DB contains
- **state information for all BSS**
- **BTS software**

1 BSC may control up to 40 BTS

FUNCTIONS:

➔ **switch calls from MSC to correct BTS**

⇨ and conversely

➔ **Protocol and coding conversion**

⇨ for traffic (voice) & signaling (GSM-specific to ISDN-specific)

➔ **Manage MS mobility**

➔ **Enforce power control**

Base Station Controller (BSC)

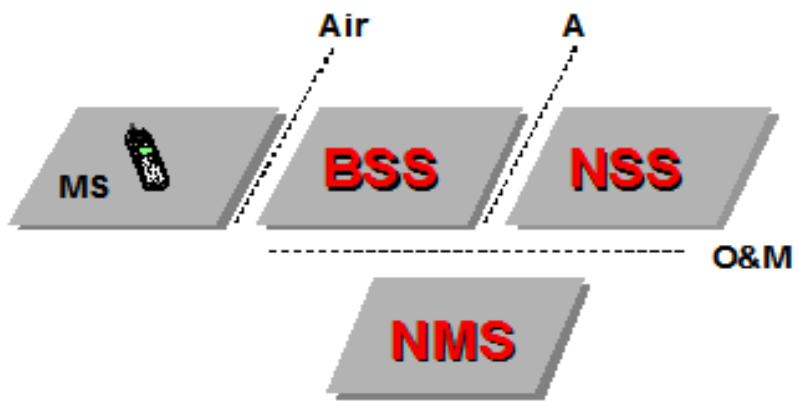
- A base station controller is an automatic coordinator (controller) in a mobile system that allows one or more base transceiver stations (BTS) to communicate with a mobile switching center and/or a packet data communication system.
- A BSC is called a Radio Network Controller (RNC) in the WCDMA system.

Base transceiver station (BTS)

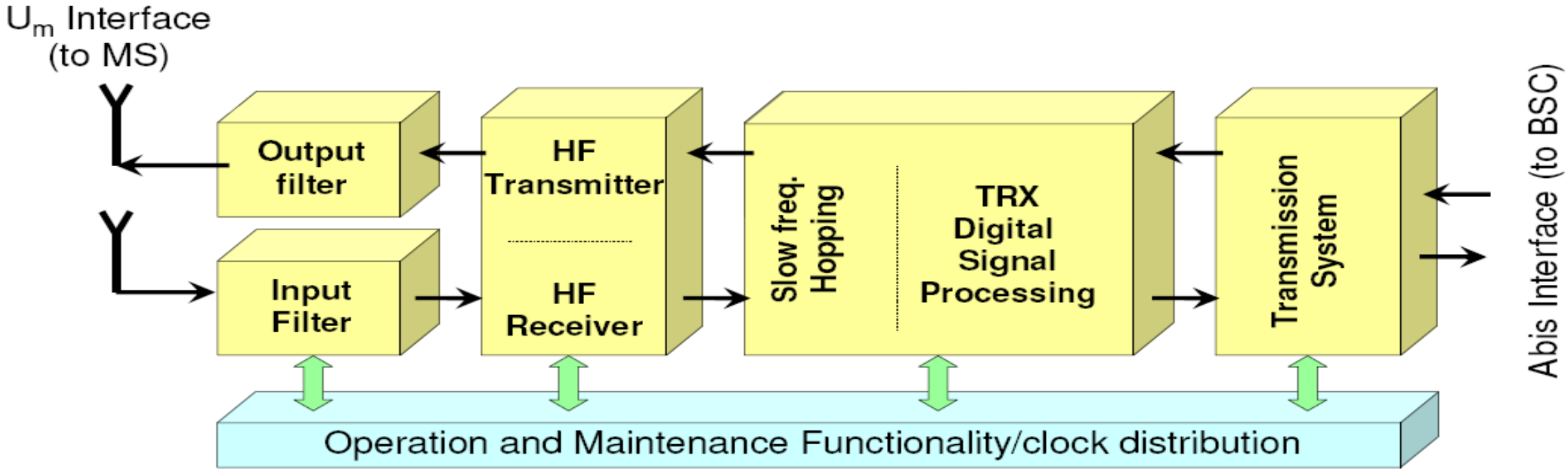
Performs: channel coding/decoding and encryption/decryption

BTS includes: radio transmitters and receivers, antennas, the interface to the PCM facility, …

About 1/2 the processing is associated with transcoding speech channel to/from GSM coding



Base Transceiver Station - BTS



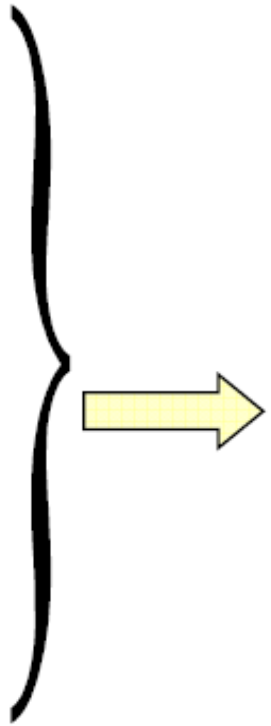
TRX radio interface functions:

- GMSK modulation-demodulation
- channel coding
- encryption/decryption
- burst formatting, interleaving
- signal strength measurements
- interference measurements

In essence, BTS is a complex modem!

Transcoding and Rate Adaptation

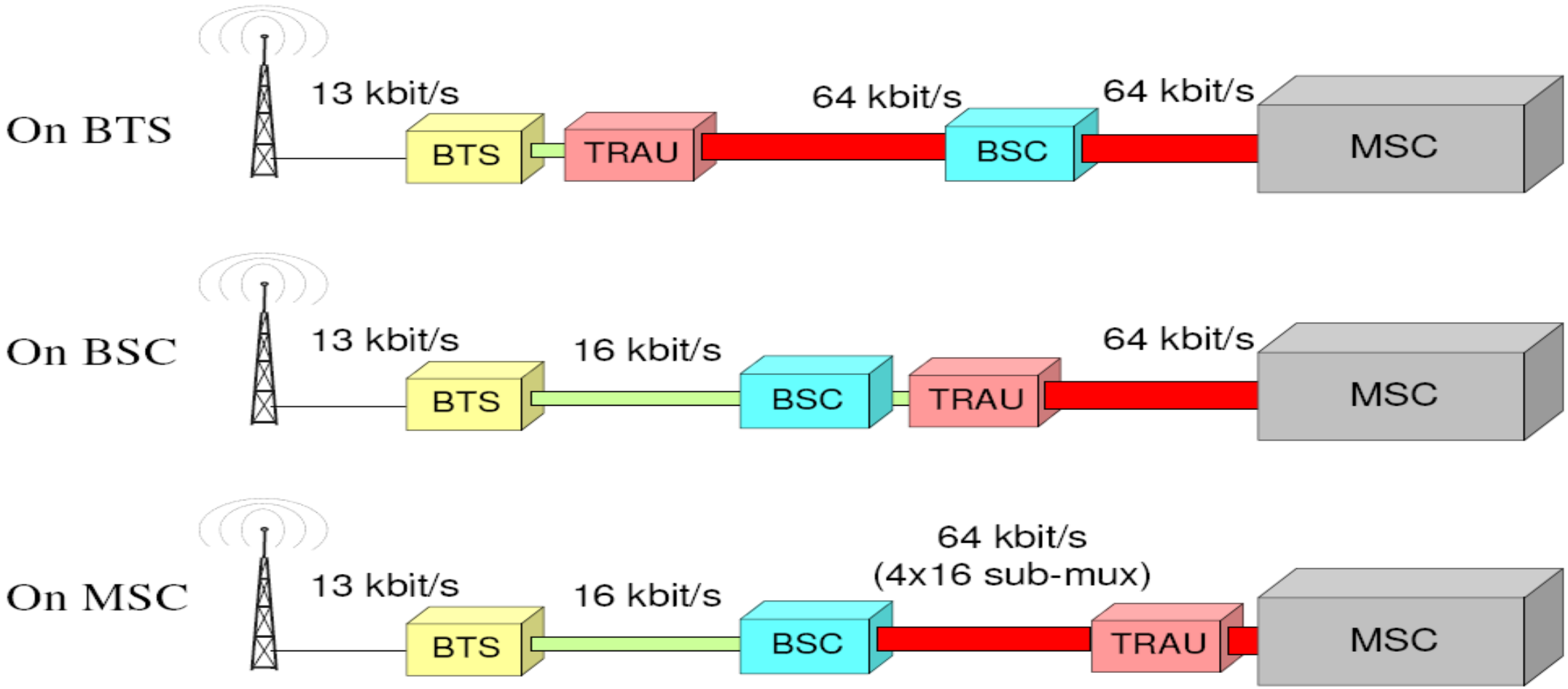
BTS:
-collects speech traffic
-Deciphers and removes error protection
-Result:
-13 kbps air-interface GSM speech-coded signal
MSC:
-A modified ISDN switch
-Needs to receive ISDN-coded speech
-64 kbps PCM format (A-law)



Transcoding and Rate Adaptation Unit (TRAU) needed!

Rationale: re-use existing ISDN switches & protocols

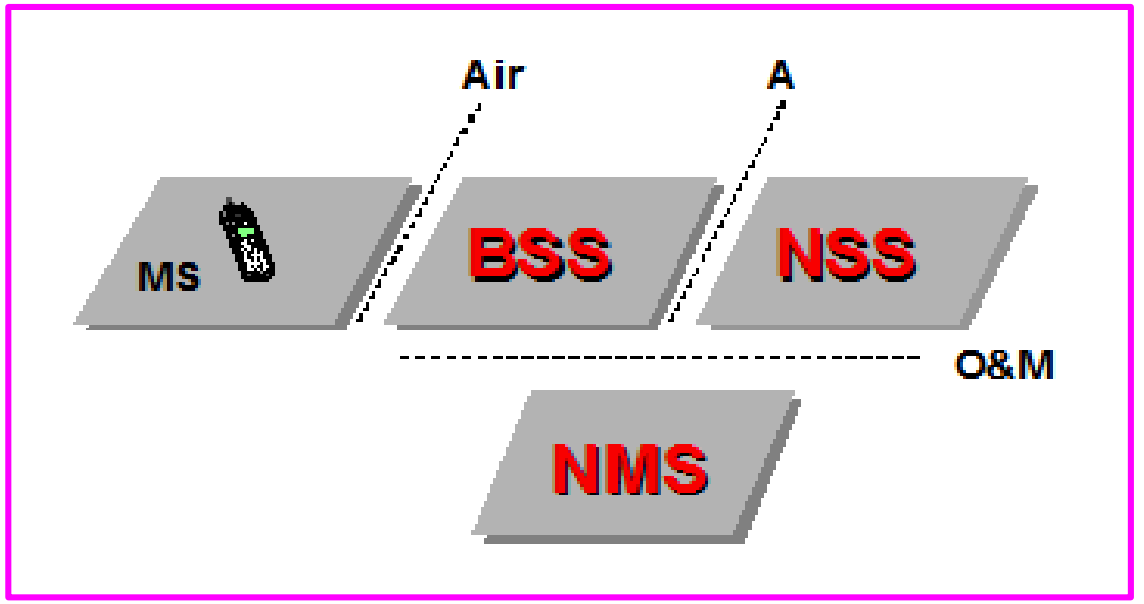
TRAU possible placements



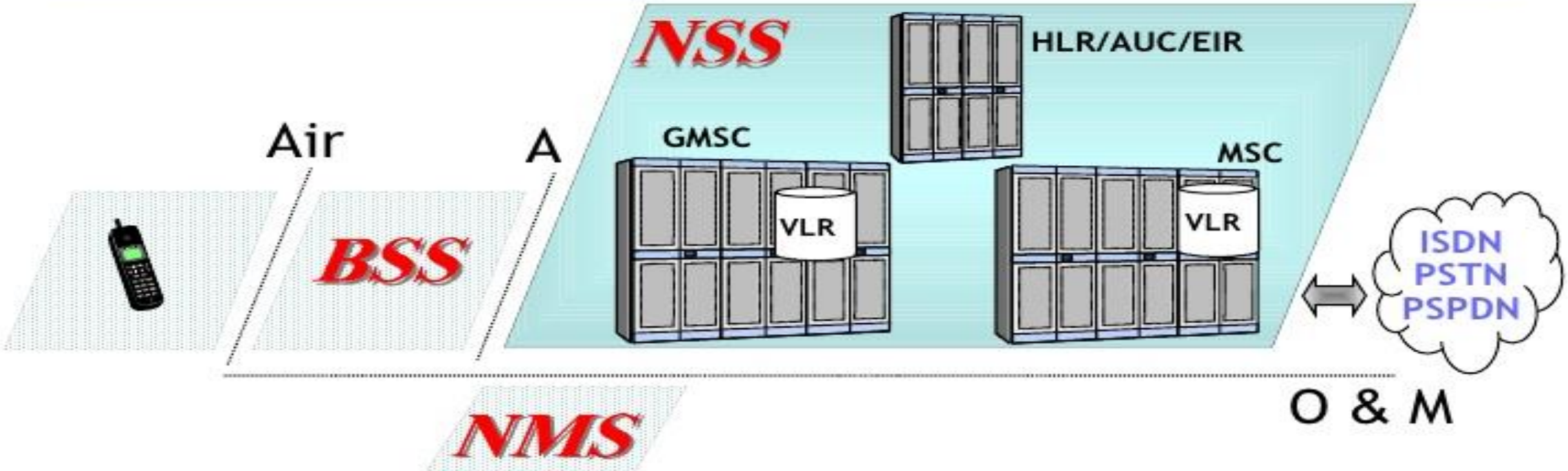
Why 16 kbps instead of 13? Inband signalling needed for BTS control of TRAU (TRAU needs to receive synchro & decoding information from BTS)

Network and Switching Subsystem (NSS)

- MSCs
  - Gateway MSC (GMSC) has interconnections to other networks
- Databases
- Gateways



NSS Elements



- Mobile services Switching Centre (MSC) or Gateway MSC
- Visitors Location Register (VLR)
- Home Location Register (HLR)
- AUthentication Centre (AUC)
- Equipment Identity Register (EIR)

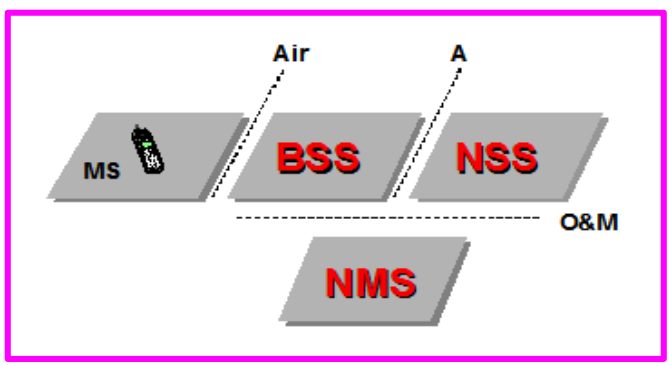
Network Switching Subsystem (NSS)

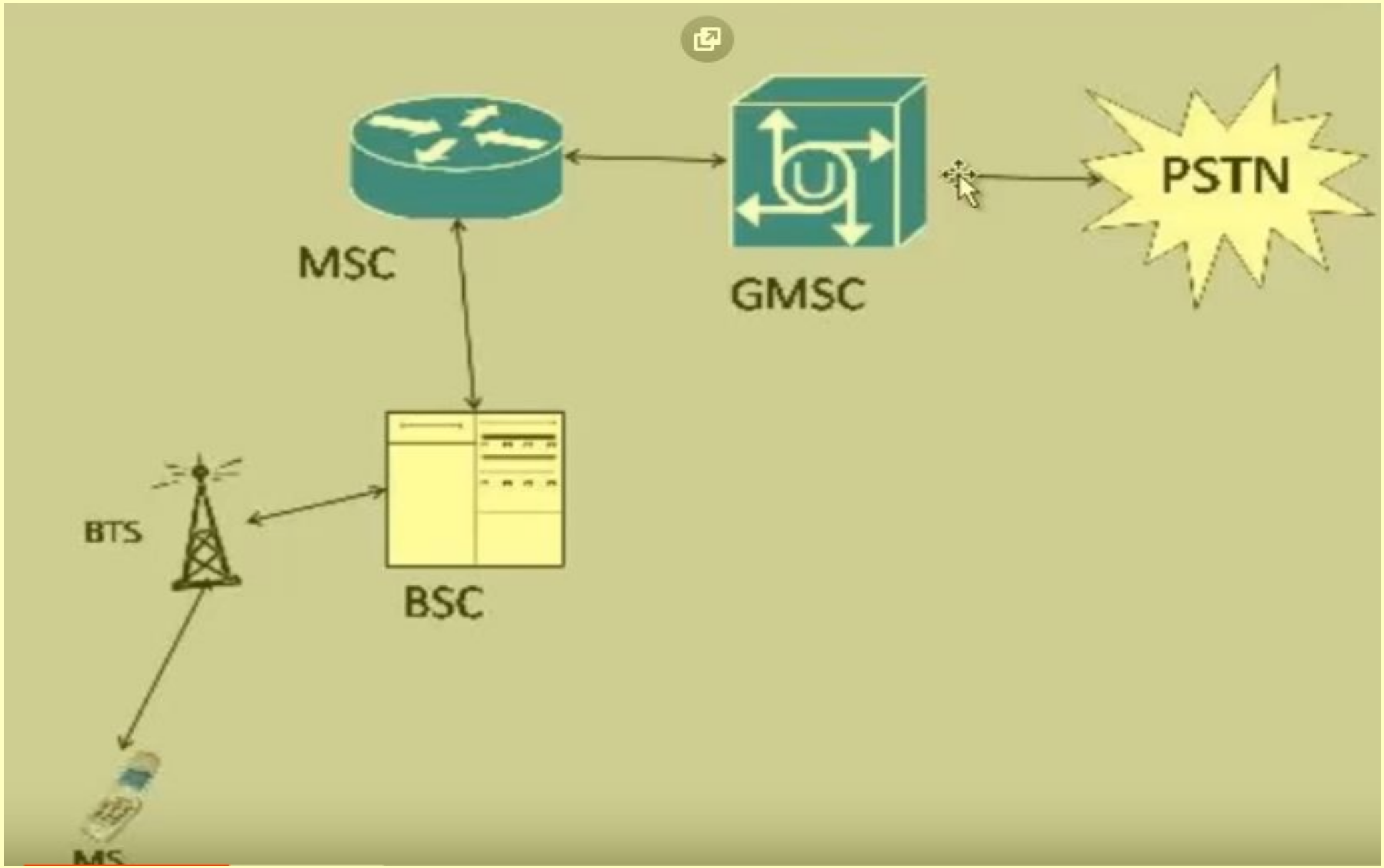
The GSM system architecture contains a variety of different elements, and is often termed the core network. It provides the **main control** and **interfacing** for the **whole** mobile network. The major elements within the core network include:

- ***Mobile Services Switching Centre (MSC):*** The main element within the core network area of the overall GSM network architecture is the Mobile switching Services Centre (MSC). The MSC acts like a **normal switching node within a PSTN** or **ISDN**, but also provides additional functionality to enable the requirements of a mobile user to be supported. These include registration, authentication, call location, inter-MSC handovers and call routing to a mobile subscriber. **It also provides an interface to the PSTN so that calls can be routed from the mobile network to a phone connected to a landline**. **Interfaces to other MSCs are provided to enable calls to be made to mobiles on different networks**.

Mobile Switching Center - MSC

- **An ISDN switch (64 kbps channels)**
- **Performs all the switching and routing functions of a fixed network switching node**
- **PLUS specific mobility-related functions:**
  - ⇨ Allocation and administration of radio resources
  - ⇨ Management of mobile users
    - → registration, authentication
    - → handover execution and control
    - → paging
- **A PLMN (operator network) has, in general, many MSC**
  - ⇨ Each MSC is responsible of a set of BSS
    - → (note: a BSS refers to just 1 MSC, not many)




MSC
GMSC
PSTN
BTS
BSC
MS

MSC

- The mobile switching center (MSC) processes requests for service from mobile devices and land line callers, and routes calls between the base stations and the public switched telephone network (PSTN).
- The MSC receives the dialed digits, creates and interprets call-processing tones, and routes the call paths.
- Mobile Switching Center (MSC) - The MSC is the heart of the GSM netowrk. It handles call routing, call setup, and basic switching functions. An MSC handles multiple BSCs and also interfaces with other MSC's and registers. It also handles iner-BSC handoffs as well as coordinates with other MSC's for inter-MSC handoffs.

Databases

Home Location Register (HLR)	database for management of mobile subscribers, stores the international mobile subscriber identity(IMSI), mobile station ISDN number (MSISDN) and current visitor location register (VLR) address keeps track of the services associated with each MS an HLR may be used by multiple MSCs
Visitor Location Register (VLR)	caches some information from the HLR as necessary for call control and service provisioning for each mobile currently located in the geographical area controlled by this VLR connected to one MSC and is often integrated into the MSC
Authentication Center (AuC)	a protected database which has a copy of the secret key stored in each subscriber's SIM card this secret is used for authentication and encryption over the radio channel normally located close to HLR
Equipment Identity Register (EIR)	contains a list of all valid mobile station equipment within the network, where each mobile station is identified by its international mobile equipment identity (IMEI) - split into 3 databases:

- White list: all known, good IMEIs
- Black list: bad or stolen handsets
- Grey list: handsets/IMEIs that are uncertain

Home Location Register - HLR

➔ **1 database per operator (PLMN)**

⇨ In principle; in practice may be

→N-plicated for reliability reasons

→In large operator networks, there may be 2+ HLR with distinct information, although MSISDN-HLR association needs to be introduced (e.g. first two digits of the Subscriber Number)

➔ **HLR entries:**

⇨ Every user / MSISDN that has subscribed to the operator

➔ **Stores:**

➔ **Permanent information associated to the user**

⇨ IMSI, MSISDN
⇨ Services subscribed
⇨ Service restrictions (e.g. roaming restrictions)
⇨ Parameters for additional services
⇨ info about user equipment (IMEI)
⇨ Authentication data

➔ **Temporary information associated to the user**

⇨ Link to current location of the user:

→Current VLR address (if avail)
→Current MSC address (if avail)
→MSRN (if user outside PLMN)

- ***Home Location Register (HLR):*** This database contains all the administrative information about each subscriber along with their last known location. In this way, the GSM network is able to route calls to the relevant base station for the MS. When a user switches on their phone, the phone registers with the network and from this it is possible to determine which BTS it communicates with so that incoming calls can be routed appropriately. Even when the phone is not active (**but switched on**) it ***re-registers periodically to ensure*** that the network (**HLR**) is aware of **its latest position**. There is one HLR per network, although it may be distributed across various sub-centres to for operational reasons.

Home Location Register (HLR)

- The home location register (HLR) is a subscriber database containing each
  - Customer's international mobile subscriber identity (IMSI)
  - International mobile equipment identifier (IMEI) to uniquely identify each customer.

- There is usually only one HLR for each carrier even though each carrier may have many MSCs.

HLR Functions

- The HLR holds each customer's user profile which includes the selected long distance carrier, calling restrictions, service fee charge rates, and other selected network options.
- The subscriber can change and store the changes for some feature options in the HLR (such as call forwarding.)
- The MSC system controller uses this information to authorize system access and process individual call billing.

Fault Tolerance System

- The HLR is a magnetic storage device for a computer (commonly called a hard disk).
- Subscriber databases are critical, so they are usually regularly backed up, typically on tape or CDROM, to restore the information if the HLR system fails.

Visitor Location Register - VLR

- **At most 1 database per MSC**
  - ⇨ Generally, joint MSC-VLR implementation
    - → No need to carry heavy MSC-VRL signalling load on network links
  - ⇨ but 1 VLR may serve many MSCs
- **VLR entries:**
  - ⇨ Every user / MSISDN actually staying in the administrative area of the associated MSC
    - → Entry created when an MS enters the MSC area (registration)
  - ⇨ NOTE: may store data for roaming users (subscribed to different operators)
- **Stores:**

Subscriber and subscription data

- ⇨ IMSI, MSISDN
- ⇨ Parameters for additional services
- ⇨ info about user equipment (IMEI)
- ⇨ Authentication data

Tracking and routing information

- ⇨ Mobile Station Roaming Number (MSRN)
- ⇨ Temporary Mobile Station Identity (TMSI)
- ⇨ Location Area Identity (LAI) where MS has registered
  - → Used for paging and call setup

- ***Visitor Location Register (VLR):*** This contains selected information from the HLR that enables the selected services for the individual subscriber to be provided. The VLR can be implemented as a separate entity, but it is commonly realised as an integral part of the MSC, rather than a separate entity. In this way access is made faster and more convenient.

Visitor Location Register (VLR)

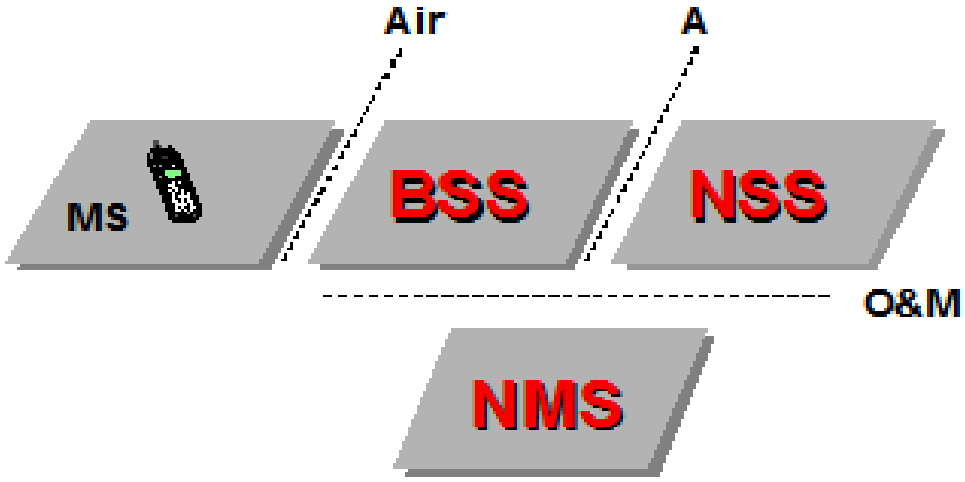
- The VLR is a database that contains a subset of the information located on the HLR.
- It contains similar information as the HLR, but only for subscribers currently in its Location Area.
- There is a VLR for every Location Area.
- The VLR reduces the overall number of queries to the HLR and thus reduces network traffic.
- VLRs are often identified by the Location Area Code (LAC) for the area they service.

Equipment Identity Register (EIR)

Optional in a GSM network, i.e., not required

EIR block (bars) calls from a MS, **not** from a subscriber.

Sometimes the AuC and EIR are combined.



• ***Equipment Identity Register (EIR):*** The EIR is the entity that decides whether a given mobile equipment may be allowed onto the network. **Each mobile equipment has a number known as the International Mobile Equipment Identity.** This number, as mentioned above, is installed in the equipment and is checked by the network during registration. Dependent upon the information held in the EIR, the mobile may be allocated one of three states - allowed onto the network, barred access, or monitored in case its problems.

Equipment Identity Register (EIR)

- The equipment identity register is a database that contains the identity of telecommunicatio devices (such as wireless telephones) and the status of these devices in the network (such as authorized or not-authorized).
- The EIR is primarily used to identify wireless telephones that may have been stolen or have questionable usage patterns that may indicate fraudulent use.
- The EIR is a database that keeps tracks of handsets on the network using the IMEI

EIR Types

- There is only one EIR per network. It is composed of three lists; white, black and gray.
- The white list holds known good mobile devices.
- The black list is a list if IMEIs that are to be denied service by the network for some reason. Reasons include the IMEI being listed as stolen or clonedor if the handset is malfunctioning or doesn't have the technical capabilities to operate on the network.
- The gray list holds mobile devices that may be suspect for fraud or are being tested for validation.

Authentication Center - AUC

- **Associated to HLR**
  - Eventually integrated with HLR
- **Search key: IMSI**
- **Responsible of storing security-relevant subscriber data**
  - Subscriber's secret key Ki (for authentication)
  - Encryption key user on the radio channel (Kc)
  - Algorithms to compute volatile keys used during authentication process

- ***Authentication Centre (AuC):*** The AuC is a protected database that contains the secret key also contained in the user's SIM card. It is used for authentication and for ciphering on the radio channel.

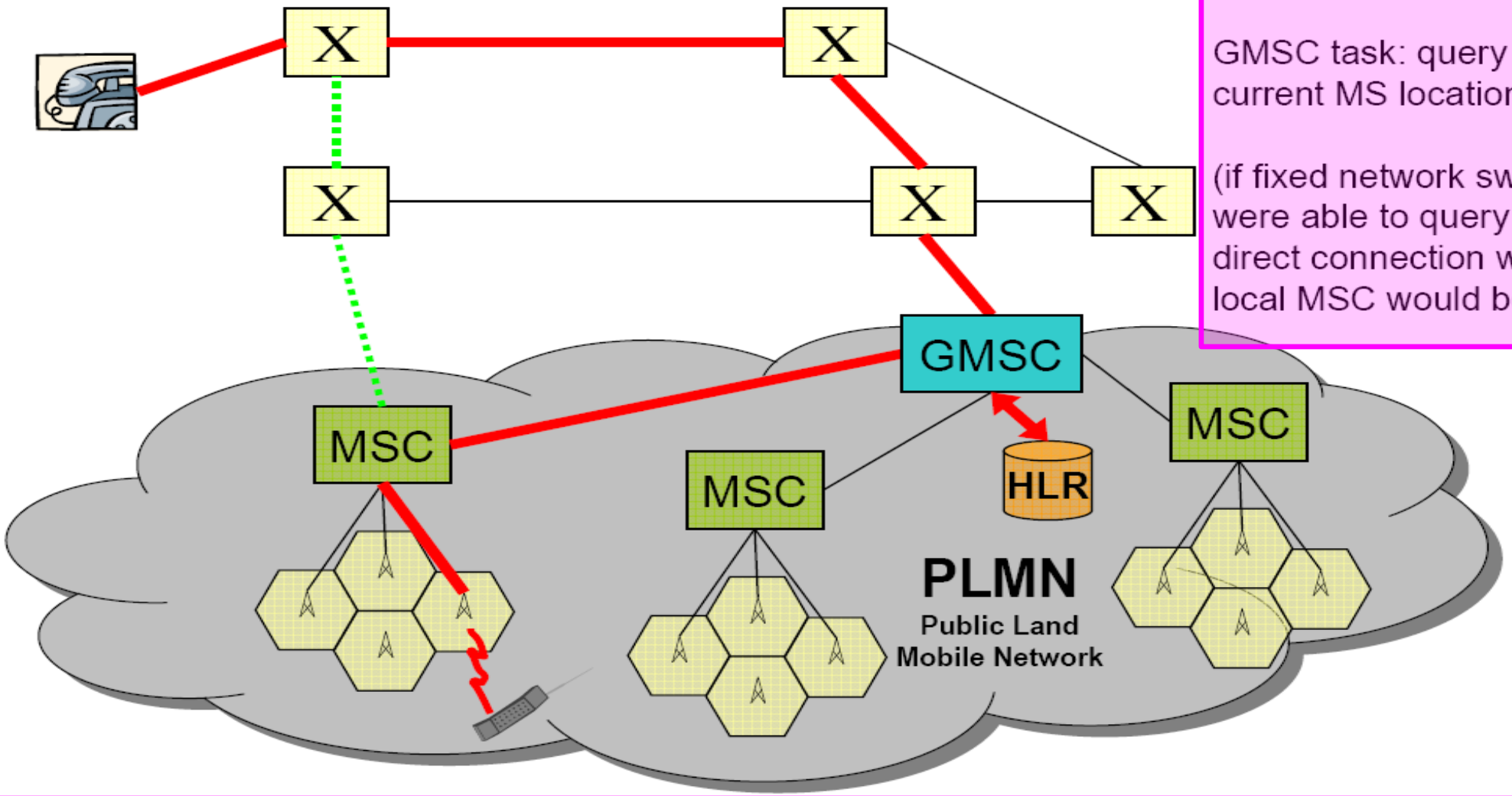
Authentication Centre (AuC)

- The Authentication Centre stores and processes information that is required to validate (authenticate) the identity of a wireless device before service is provided.
- During the authentication procedure, the AuC provides information to the system to allow it to validate the mobile device
- AuC handles the authentication and encryption tasks for the network.

Gateway Mobile Switching Center (GMSC)

- There is another important type of MSC, called a Gateway Mobile Switching Center (GMSC).
- The GMSC functions as a gateway between two networks.
- If a mobile subscriber wants to place a call to a regular landline, then the call would have to go through a GMSC in order to switch to the Public Switched Telephone Network (PSTN).

Gateway MSC – GMSC

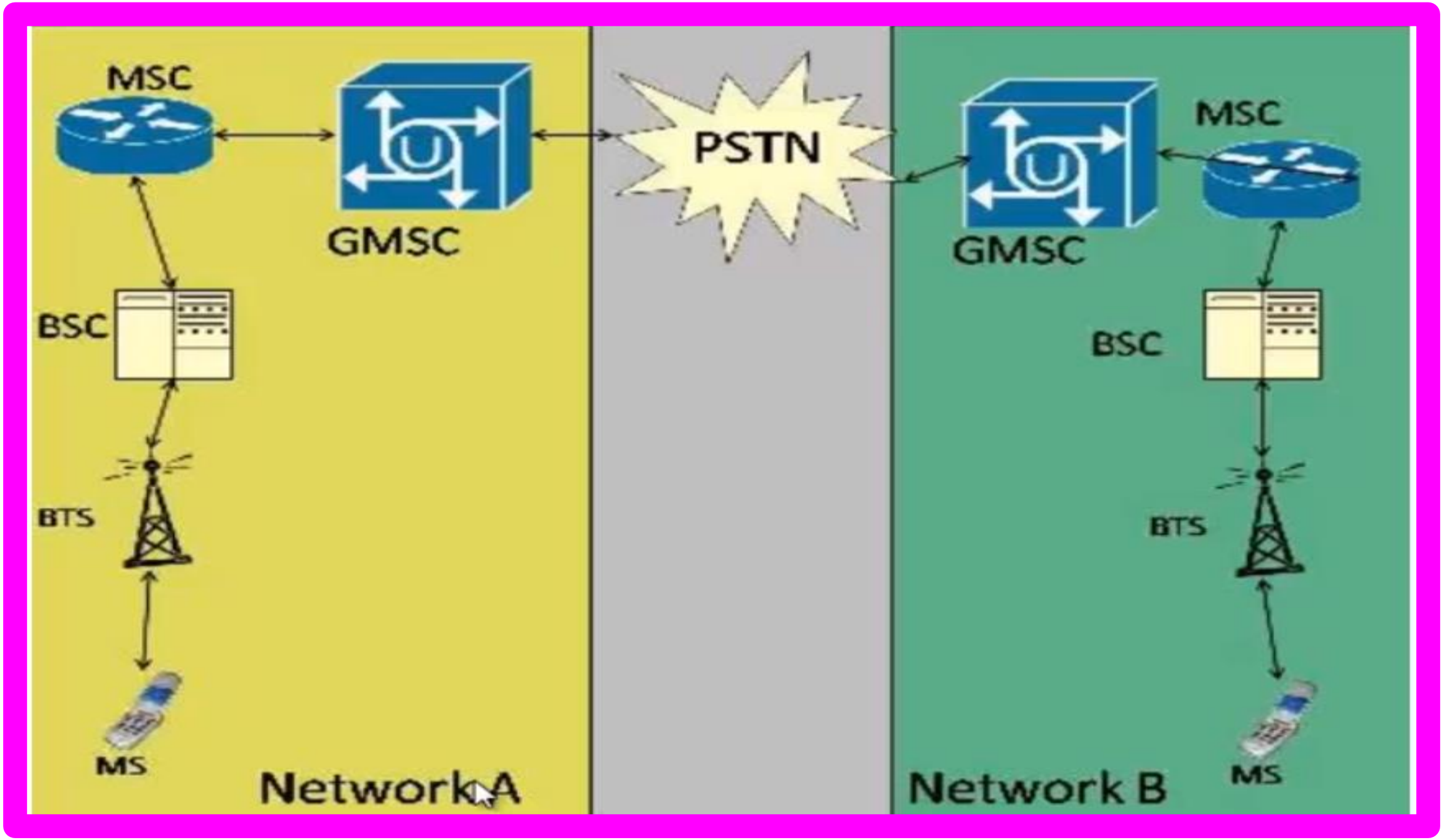


Needed, as fixed network switches are not mobile capable!!

GMSC task: query HLR for current MS location

(if fixed network switches were able to query HLR, direct connection with local MSC would be available)

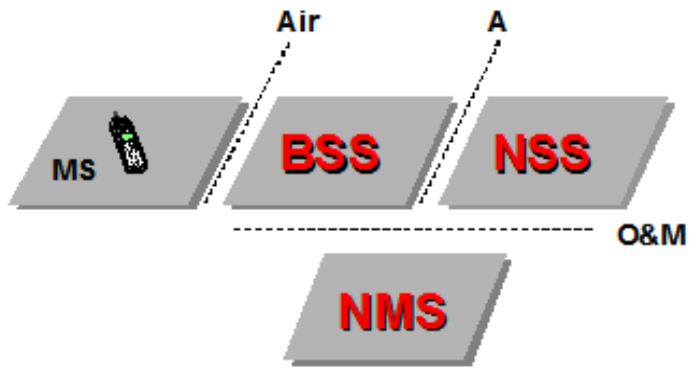
- ***Gateway Mobile Switching Centre (GMSC):*** The GMSC is the point to which a ME terminating call is initially routed, without any knowledge of the MS's location. The GMSC is thus in charge of obtaining the MSRN (Mobile Station Roaming Number) from the HLR based on the MSISDN (Mobile Station ISDN number, the "directory number" of a MS) and routing the call to the correct visited MSC. The "MSC" part of the term GMSC is misleading, since the gateway operation does not require any linking to an MSC.


MSC
GMSC
PSTN
GMSC
MSC
BSC
BSC
BTS
BTS
MS
MS
Network A
Network B

MS Gateway (SMS-G): The SMS-G or SMS gateway is the term that is used to collectively describe the two Short Message Services Gateways defined in the GSM standards. The two gateways handle messages directed in different directions. The SMS-GMSC (Short Message Service Gateway Mobile Switching Centre) is for short messages being sent to an ME. The SMS-IWMSC (Short Message Service Inter-Working Mobile Switching Centre) is used for short messages originated with a mobile on that network. The SMS-GMSC role is similar to that of the GMSC, whereas the SMS-IWMSC provides a fixed access point to the Short Message Service Centre.

Operation Sub-System (OSS)

- Operation and Maintenance Center
- Service management
  - subscription management for registering new subscriptions, modifying and removing subscriptions, as well as billing information
  - billing
  - fraud detection
  - …



Operation and Support Subsystem (OSS)

The **OSS** or **operation support subsystem** is an element within the overall GSM network architecture that is **connected** to components of the **NSS** and the **BSC**. It is used to **control** and **monitor** the **overall GSM** network and it is also used to control the traffic load of the BSS. It **must be** noted that as the number of BS increases with the scaling of the **subscriber population** some of the maintenance tasks are **transferred** to the **BTS**, allowing savings in the cost of ownership of the system.

Operation and Maintenance Center (OMC)

Manages the GSM functional blocks: MSC, BSC (and indirectly the BTSs)

Task: to maintain satisfactory operation of the GSM network

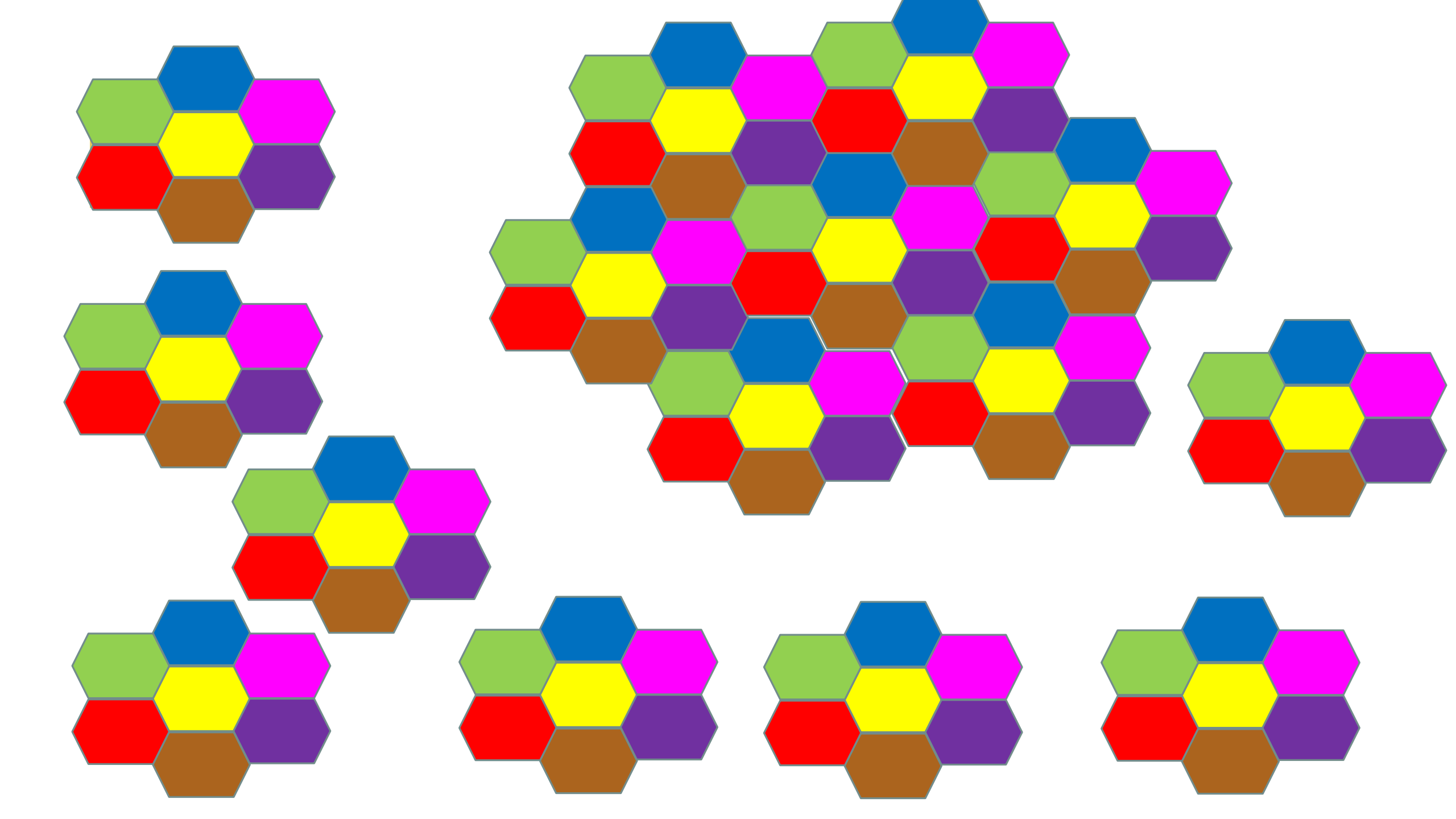
Based on observing system load, blocking rates, handovers,…

Activities:

- Network Management System (NMS)
  - modify network configuration
- equipment maintenance aiming at detecting,locating, and correcting faults

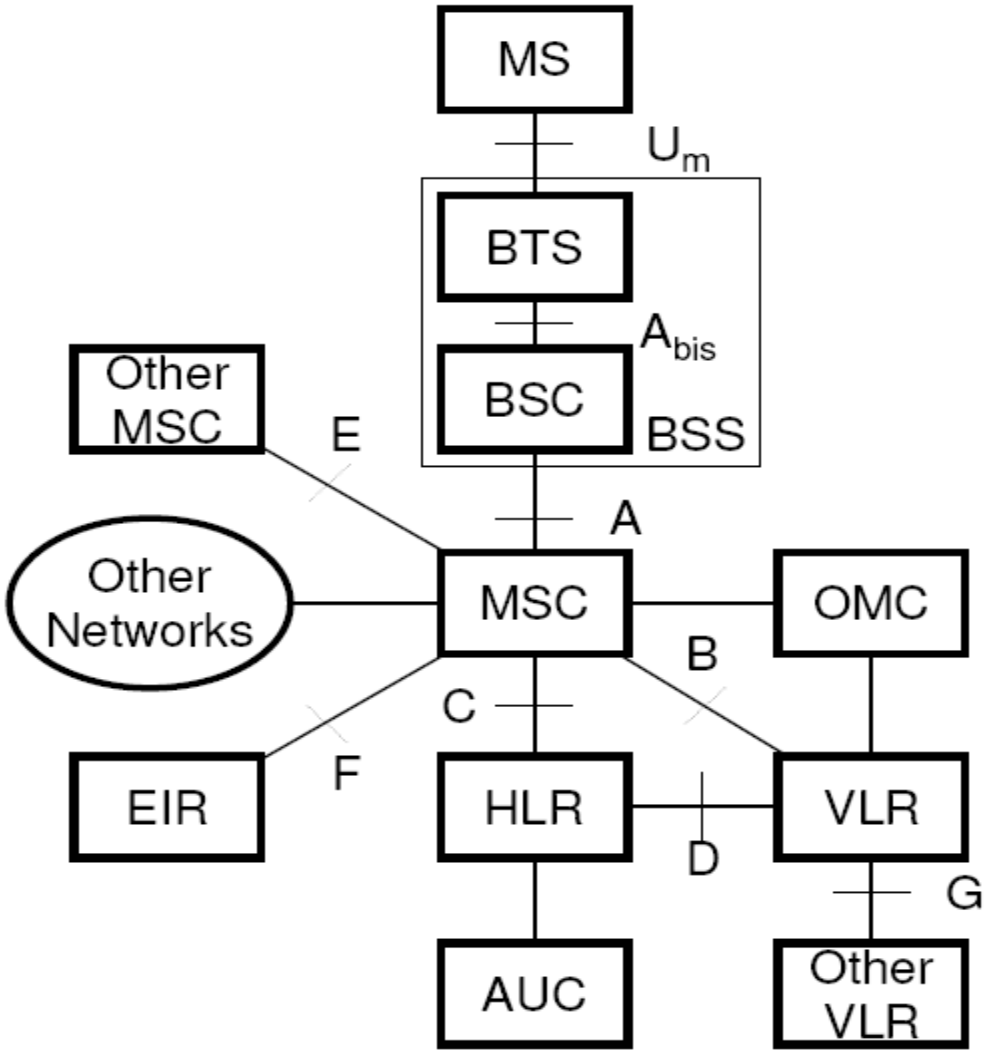
Public Switched Telephone Network (PSTN)

- Public switched telephone networks are communication systems that are available to the public to allow users to interconnect communication devices.
- Public telephone networks within countries and regions are standard integrated systems of transmission and switching facilities, signaling processors, and associated operations support systems that allow communication devices to communicate with each other when they operate.



GSM INTERFACE

Components and interfaces



Components

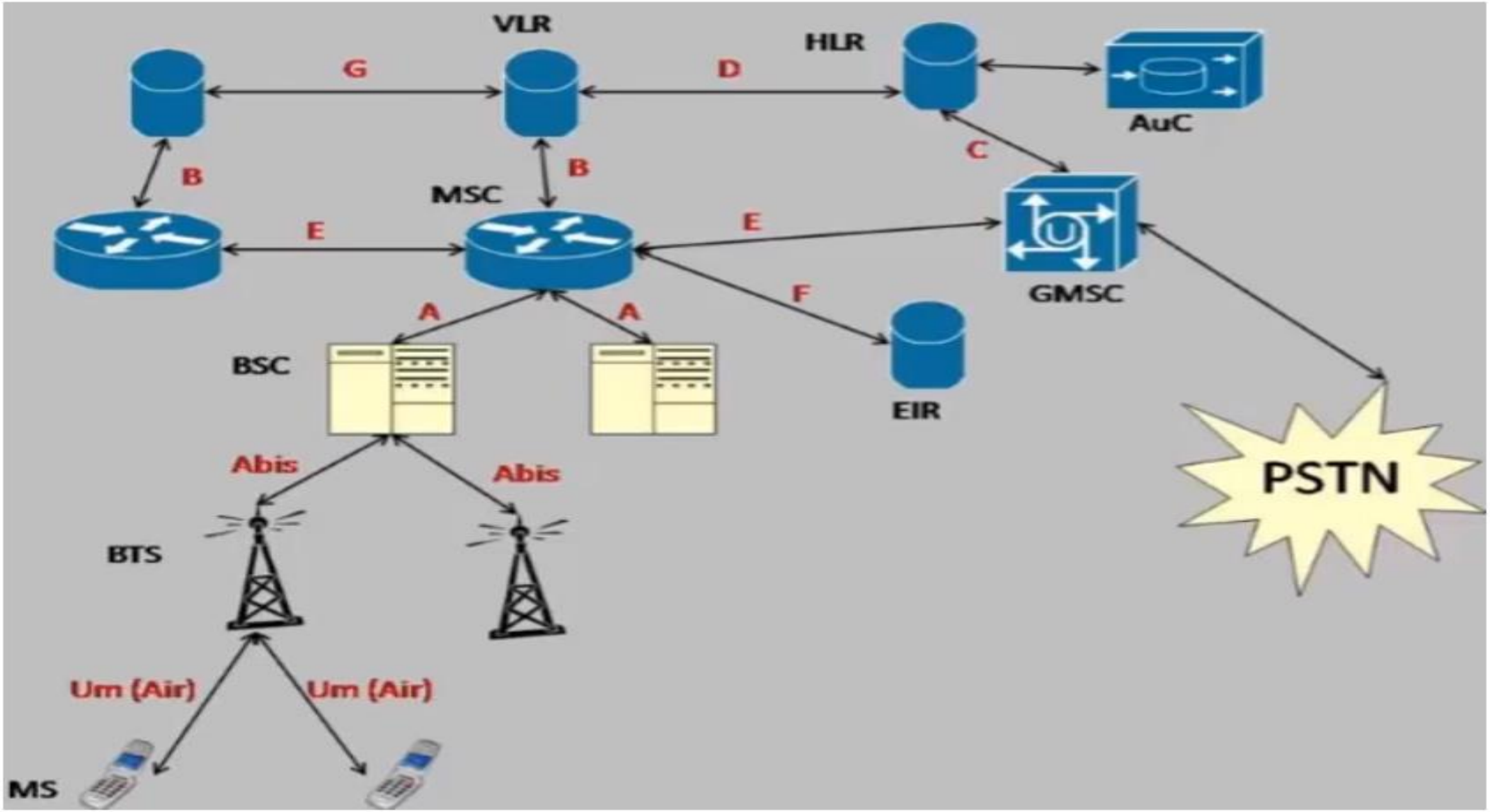
MS	Mobile Station
BTS	Base Transceiver Station
BSC	Base Station Controller
MSC	Mobile Switching Center
OMC	Operation and Maintenance Center
EIR	Equipment Identity Register
AUC	Authentication Center
HLR	Home Location Register
VLR	Visitor Location Register

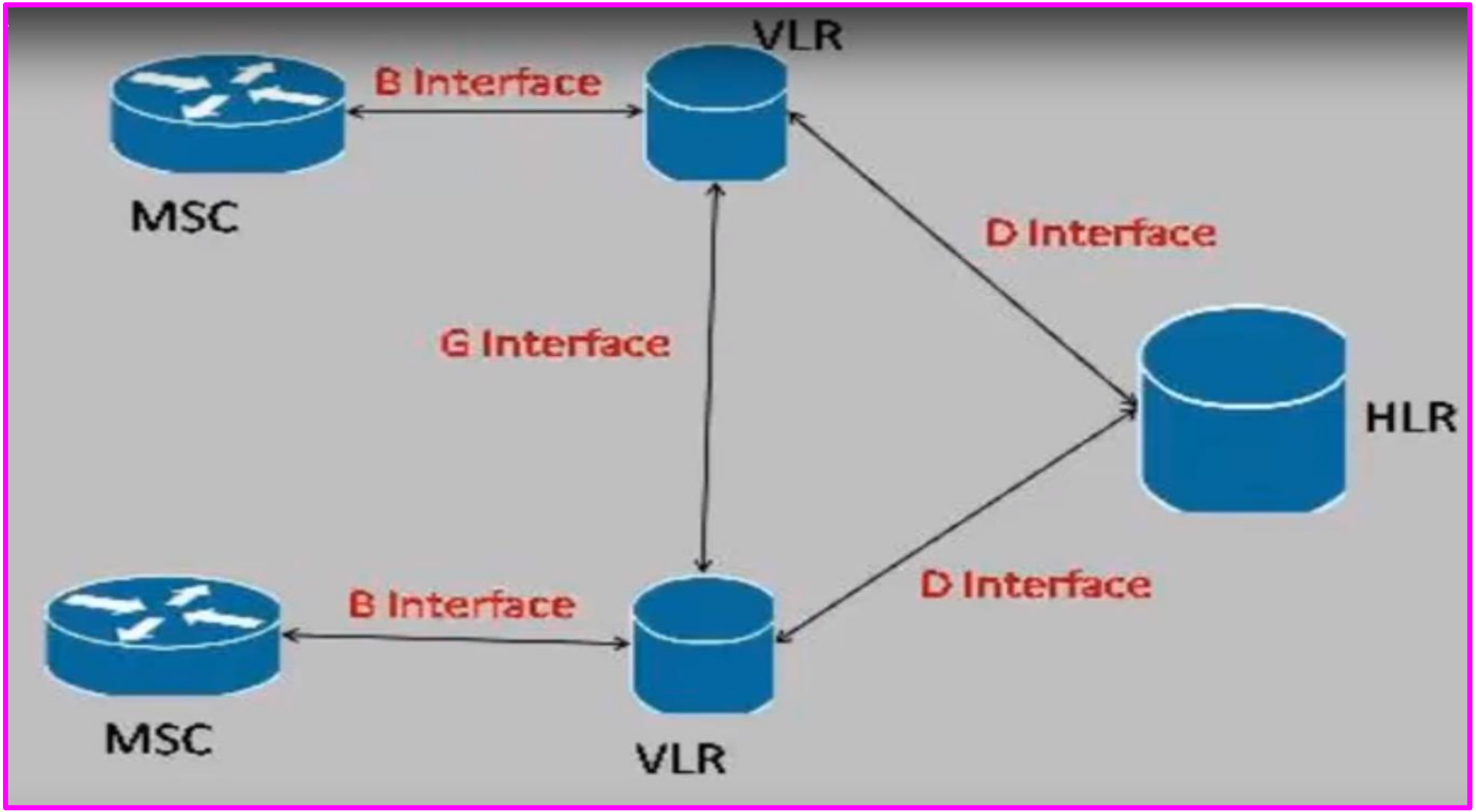
Interfaces

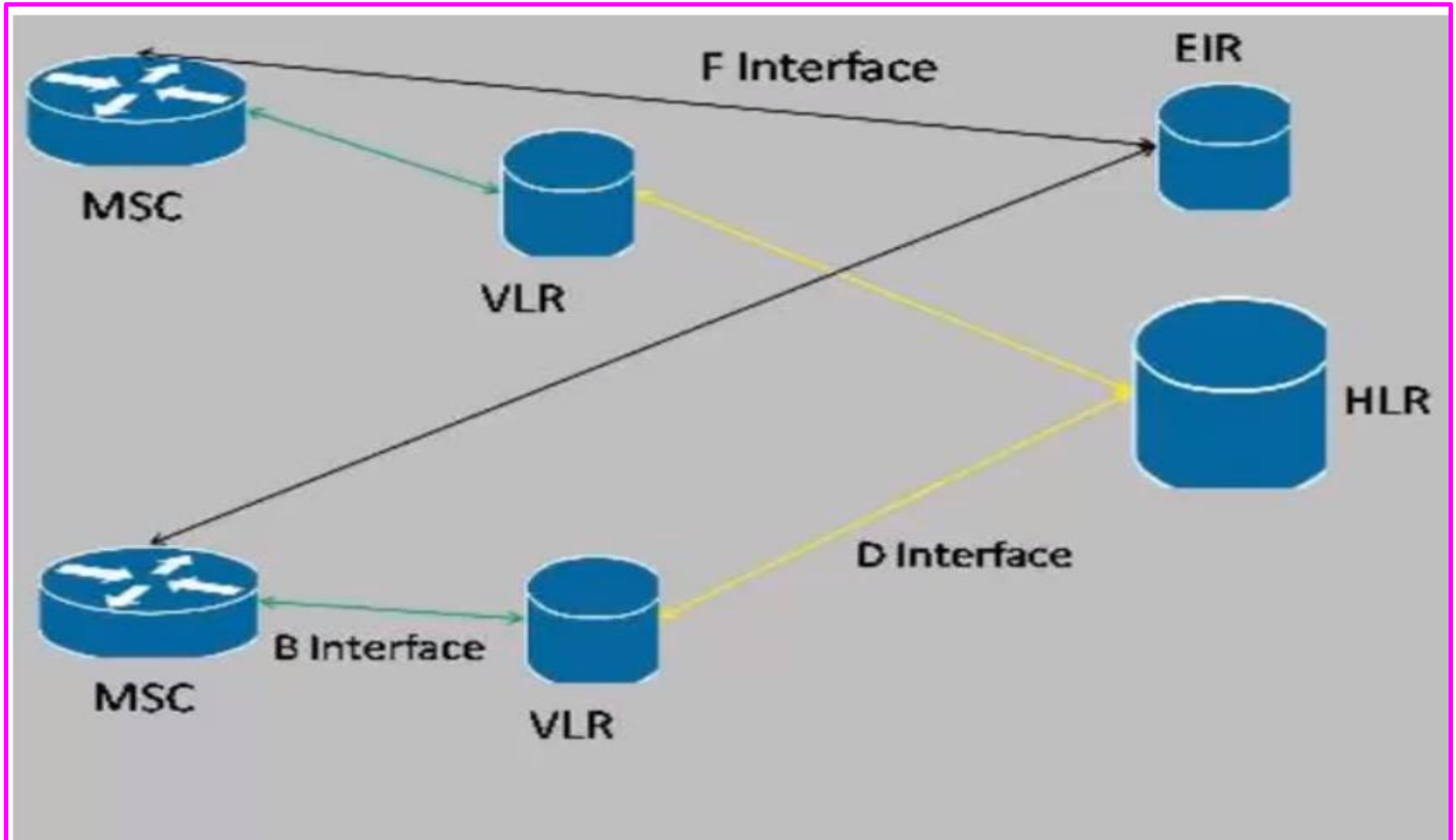
Um	Radio Interface
Abis	BTS-BSC
A	BSS-MSC
B	MSC-VLR
C	**MSC-HLR**
D	HLR-VLR
E	MSC-MSC
F	MSC-EIR
G	VLR-VLR

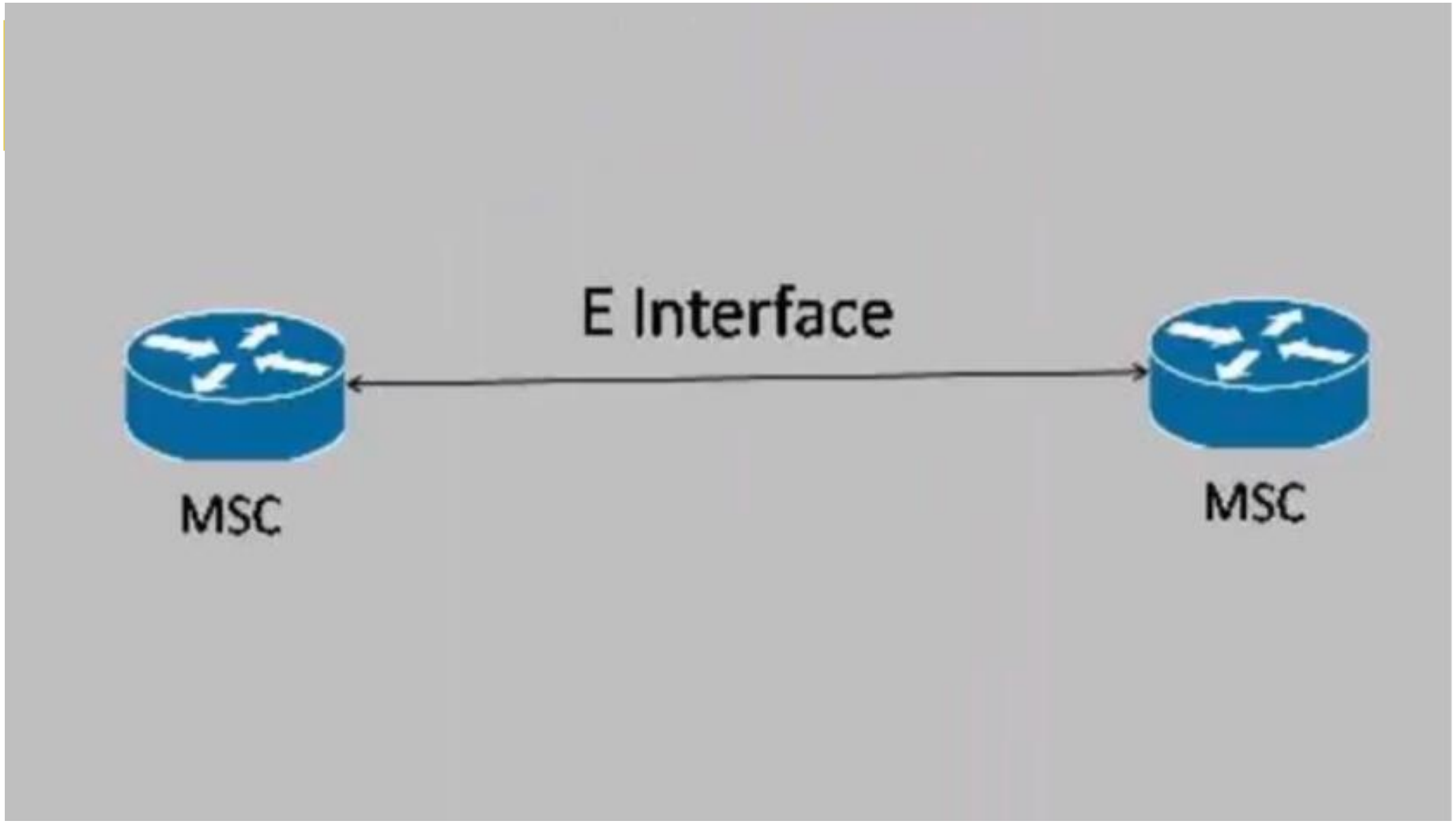
GSM Interfaces

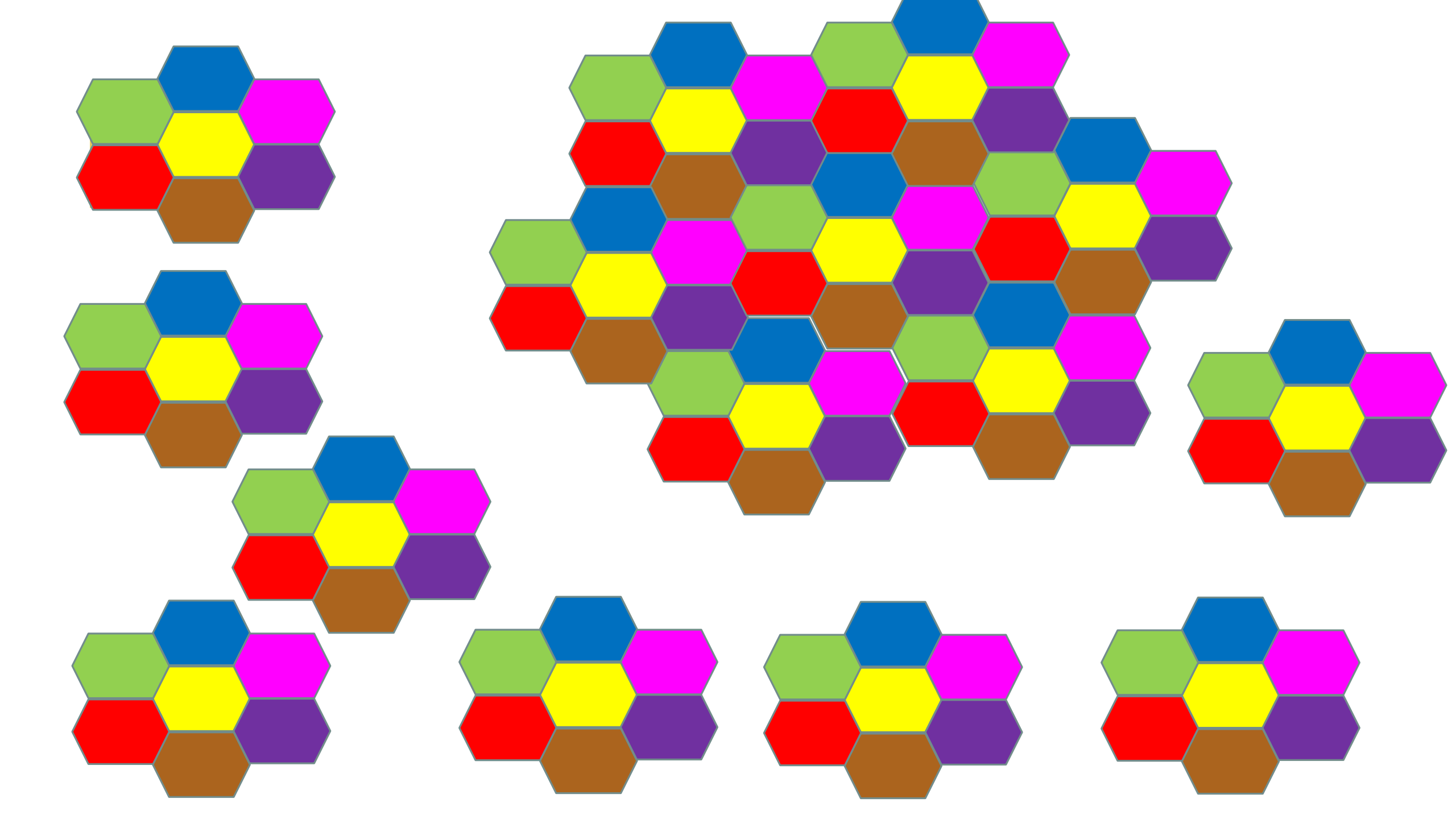
Interface	Description
U_m	Radio link between MS and BTS
A_{bis}	between BTS and BSC, PCM 2 Mbit/s, G. 703
A	between BSC and MSC, PCM 2 Mbit/s, G. 703
B	between MSC and VLR (use MAP/TCAP protocols)
C	between MSC and HLR (MAP/TCAP)
D	between HLR and VLR (MAP/TCAP)
E	between two MSCs (MAP/TCAP + ISUP/TUP)
F	between MSC and EIR (MAP/TCAP)
G	between VLRs (MAP/TCAP)


VLR
HLR
G
D
AuC
B
B
C
MSC
E
E
GMSC
A
A
F
BSC
EIR
PSTN
Abis
Abis
BTS
Um (Air)
Um (Air)
MS


VLR
B Interface
MSC
D Interface
G Interface
HLR
B Interface
D Interface
MSC
VLR


F Interface
EIR
MSC
VLR
HLR
D Interface
B Interface
MSC
VLR

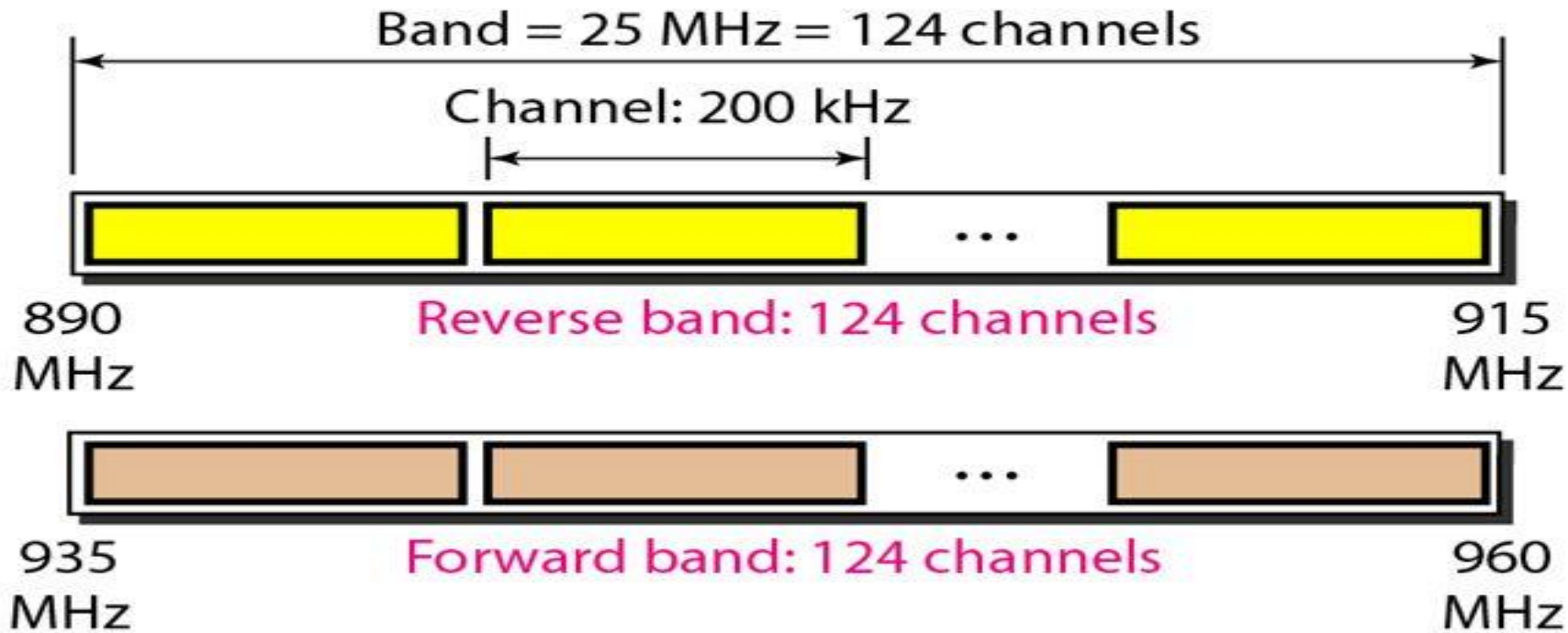

E Interface
MSC
MSC



GSM FREQUENCY BANDS

GSM bands

(digital cellular phone system using TDMA and FDMA)



In some countries the GSM-900 band has been extended to cover a larger frequency range. This 'extended GSM', E-GSM, uses frequency range **880 - 915 MHz** (uplink) and **925 - 960 MHz** (downlink), adding 50 channels (channel numbers 975 to 1023 and 0) to the original GSM-900 band.

GSM Band Allocations (MHz)

GSM systems	Uplink	Downlink	Band	Duplex Spacing	Duplex channels
GSM 450	*450.4-457.6*	*460.4-467.6*	*2x7.2*	*10*	*35*
GSM 480	*478.8-486*	*488.8-496*	*2x7.2*	*10*	*35*
GSM 850	824-849	869-894	2x25	45	124
GSM 900	890-915	935-960	2x25	45	124
E-GSM (900)	880-915	925-960	2x35	45	174
R-GSM (900)	876-880	921-925	2x04	41	40
GSM 1800	1710-1785	1805-1880	2x75	95	374
GSM 1900	1850-1910	1930-1990	2x60	80	299

Frequencies are in MHz

Carrier frequency = ARFCN = Absolute Radio Frequency Channel Number

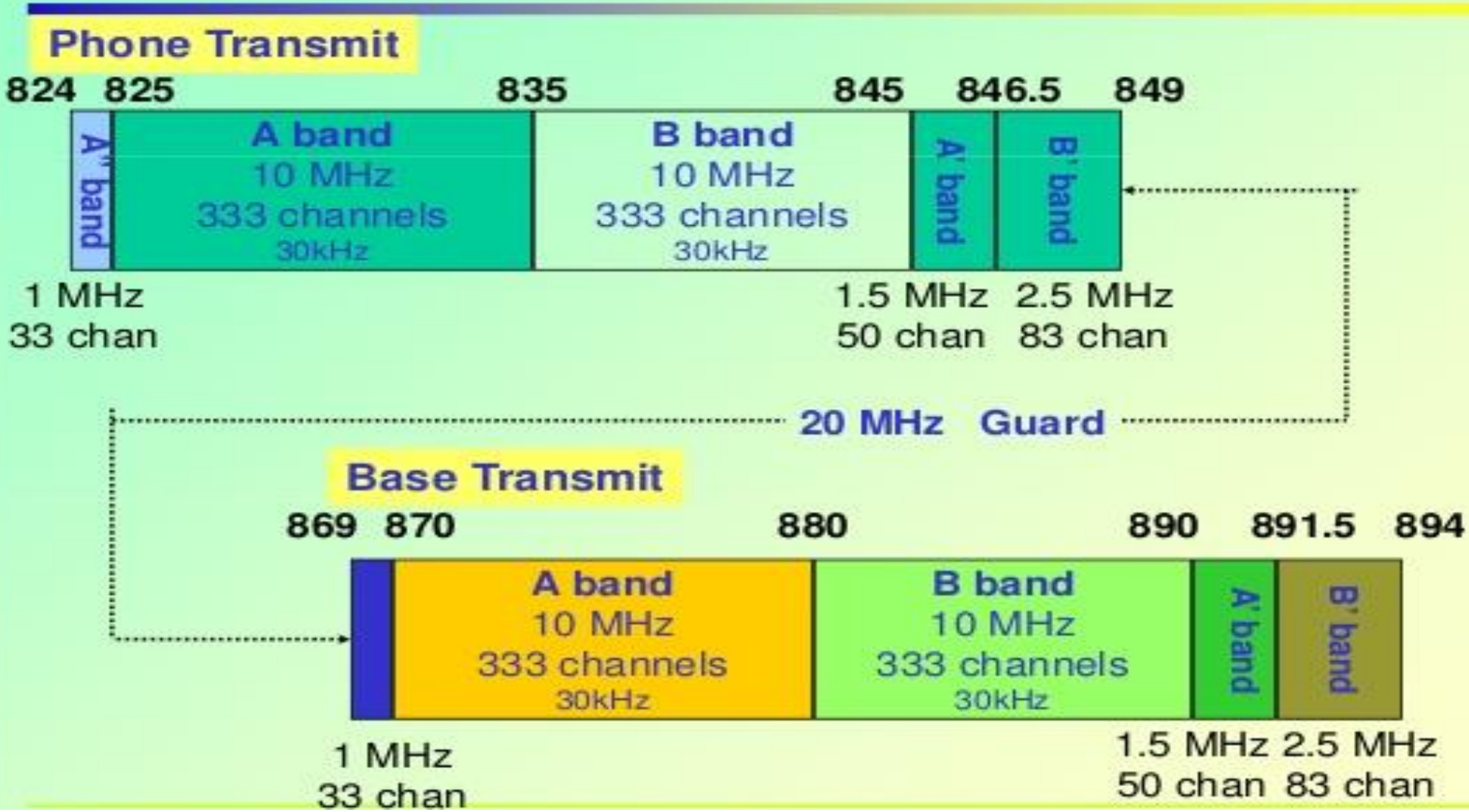
AIR INTERFACE

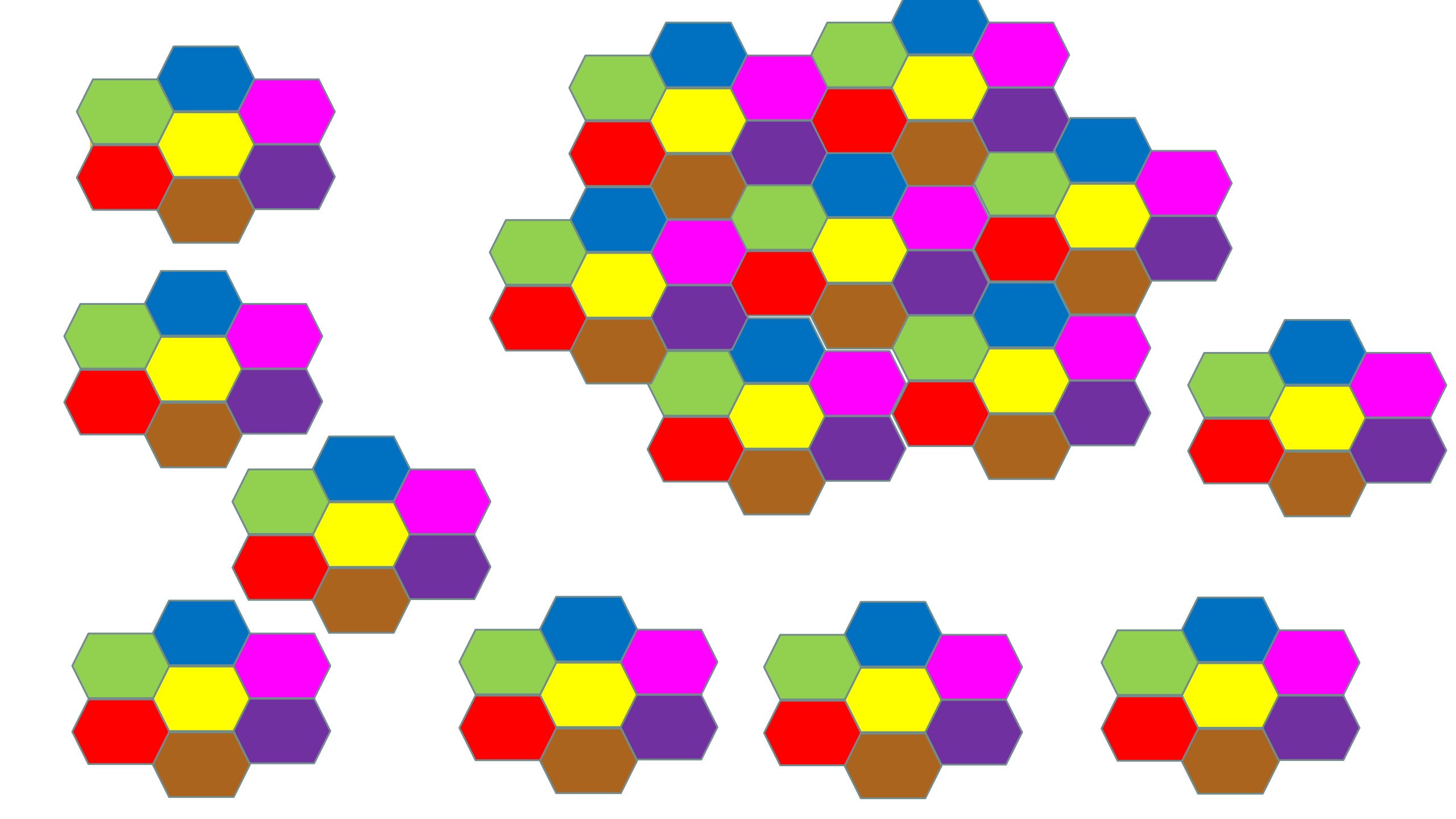
- GSM Frequency Bands

System	P-GSM 900	E-GSM 900	GSM(DCS) 1800	GSM(PCS) 1900
Uplink (MS → BS) Downlink(BS → MS)	890 - 915 MHz 935 - 960 MHz	880 - 915 MHz 925 - 960 MHz	1710 - 1785 MHz 1805 - 1880 MHz	1850 - 1910 MHz 1930 - 1990 MHz
Wavelength	≅ 33 cm	≅ 33 cm	≅ 17 cm	≅ 16 cm
Bandwidth	25 MHz	35 MHz	75 MHz	60 MHz
Duplex distance	45 MHz	45 MHz	95 MHz	80 MHz
Carrier separation	200 kHz	200 kHz	200 kHz	200 kHz
No. of carriers	124	174	374	299
Channel rate	270.8 kbps	270.8 kbps	270.8 kbps	270.8 kbps

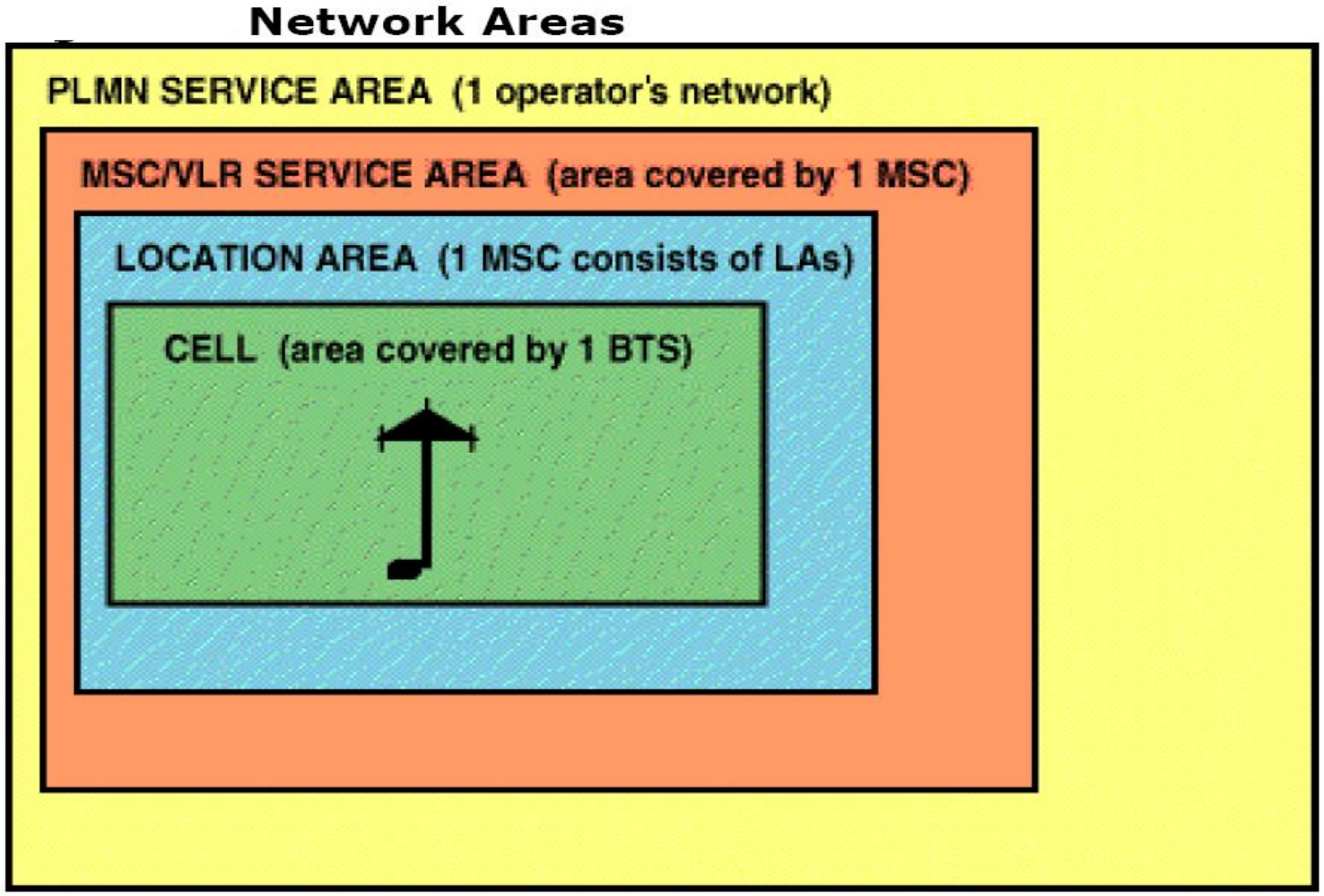
GSM Band	ARFCN(N)	Uplink Frequency equation(F_{UL})	Downlink Frequency equation
GSM 450	259-293	450.6 + 0.2*(N-259)	$F_{UL}(N) + 10$
GSM 480	306-340	479+0.2*(N-306)	$F_{UL}(N) + 10$
GSM 750	438-511	747.2 + 0.2*(N-438)	$F_{UL}(N) + 30$
GSM 850	128-251	824.2+0.2*(N-128)	$F_{UL}(N) + 45$
P-GSM	1-124	890+0.2*N	$F_{UL}(N) + 45$
E-GSM	975-1023	890+0.2*(N-1024)	$F_{UL}(N) + 45$
GSM-R	940- 974	890+0.2*(N-1024)	$F_{UL}(N) + 45$
DCS 1800	512-885	1710.2+0.2*(N-512)	$F_{UL}(N) + 95$
PCS 1900	512-810	1850.2 + 0.2*(N-512)	$F_{UL}(N) + 80$

Mobile Communs. - Cellular Spectrum



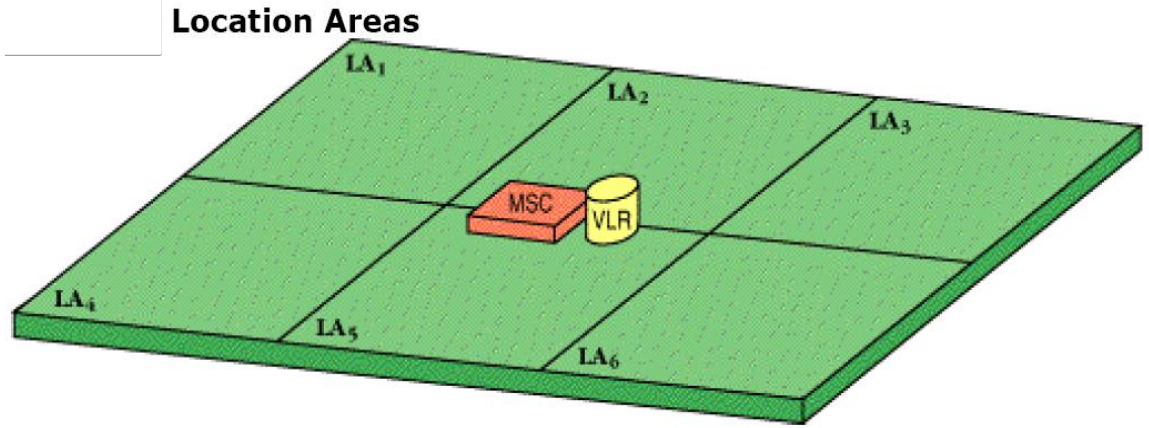


GSM PLMN



The GSM network identifies each cell via the cell global identity (CGI) number assigned to each cell.

The cell is the area given radio coverage by one base transceiver station.



The location area is a group of cells. **It is the area in which the subscriber is paged**. **Each LA is served by one or more base station controllers**, yet only by a **single MSC** Each LA is assigned a location area identity (**LAI**) number

An MSC/VLR service area represents the part of the GSM network that is covered by one MSC and which is reachable, as it is registered in the VLR of the MSC

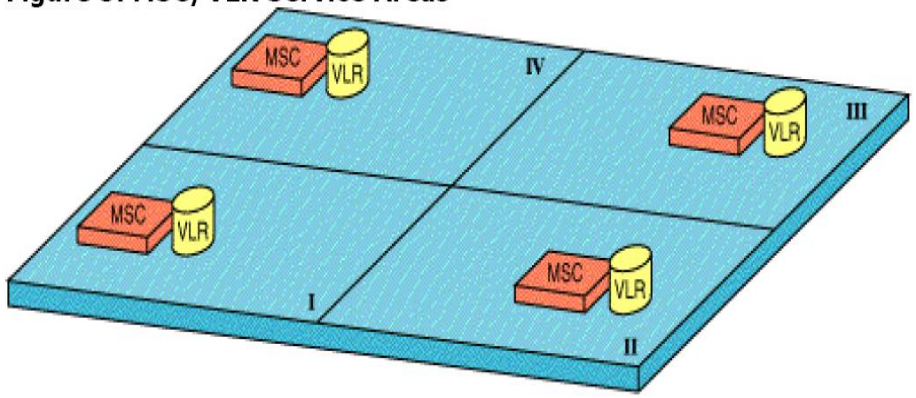
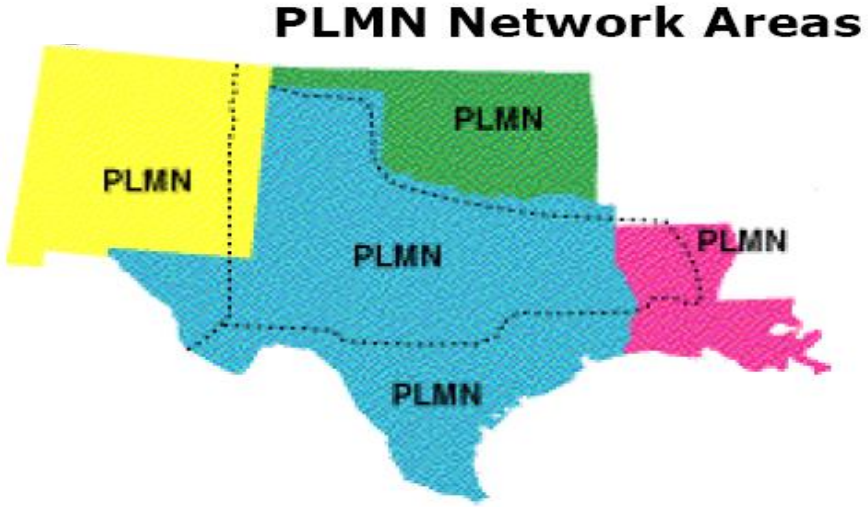


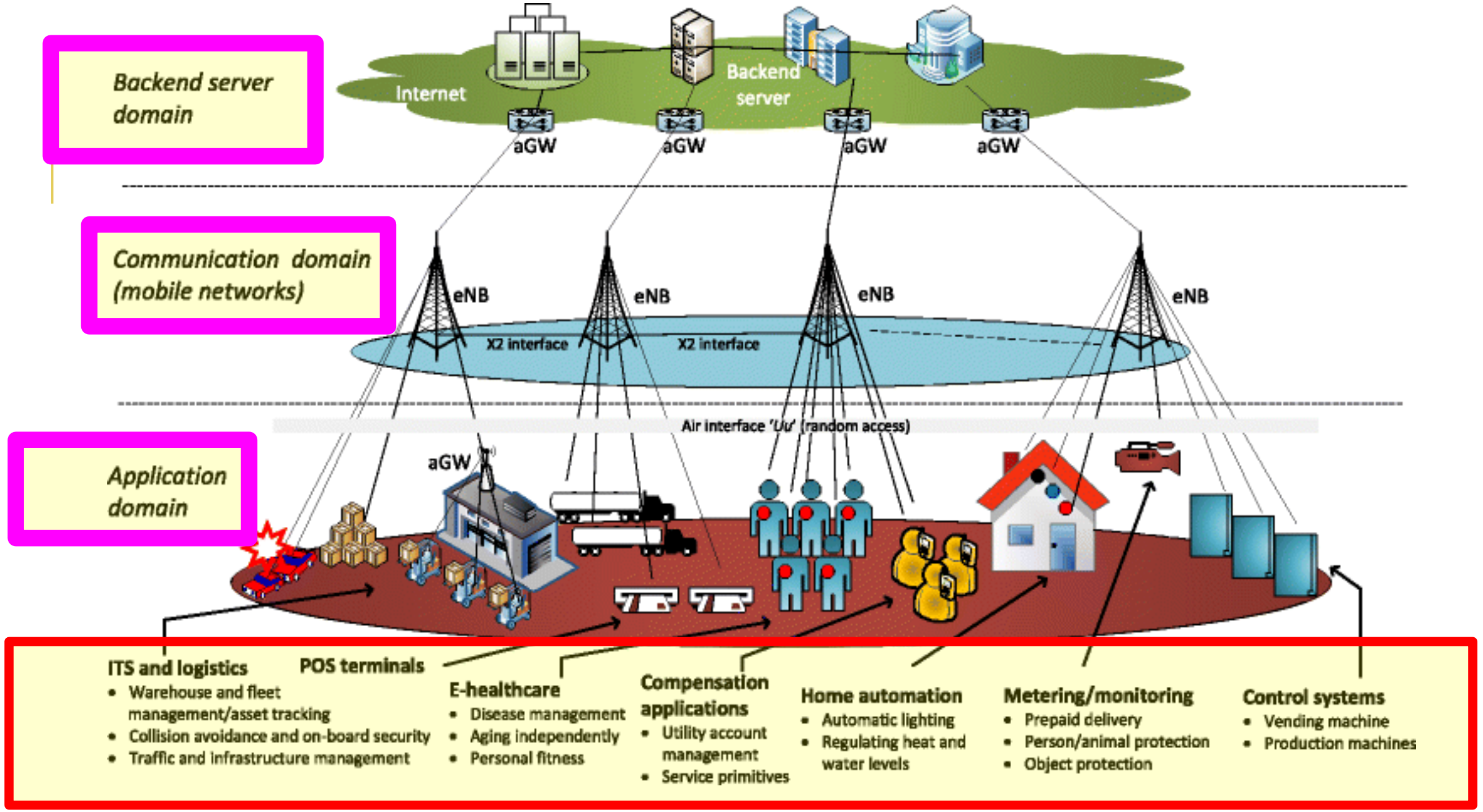
Figure 5: MSC/VLR Service Areas

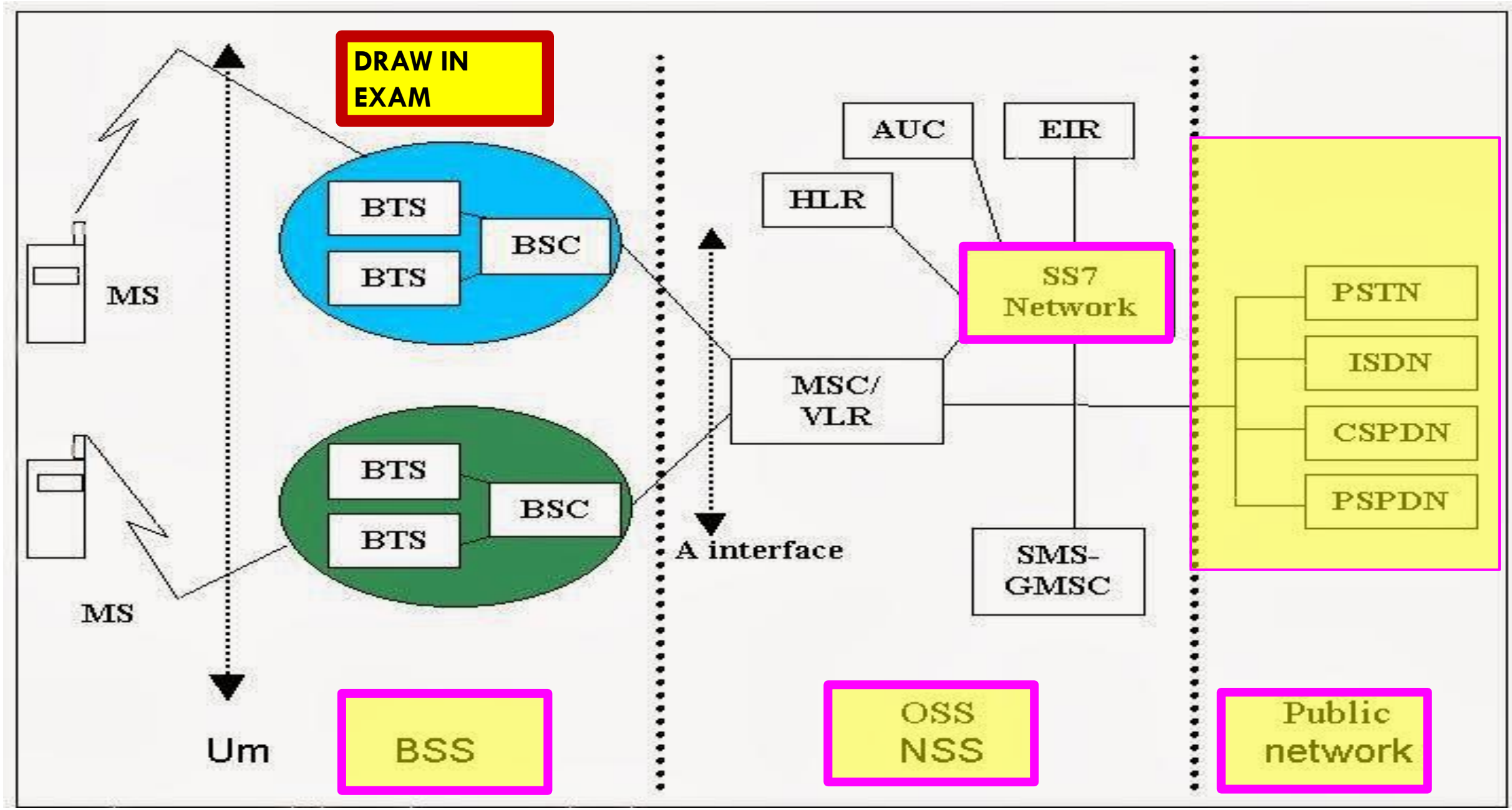


The PLMN service area is an area served by one network operator .

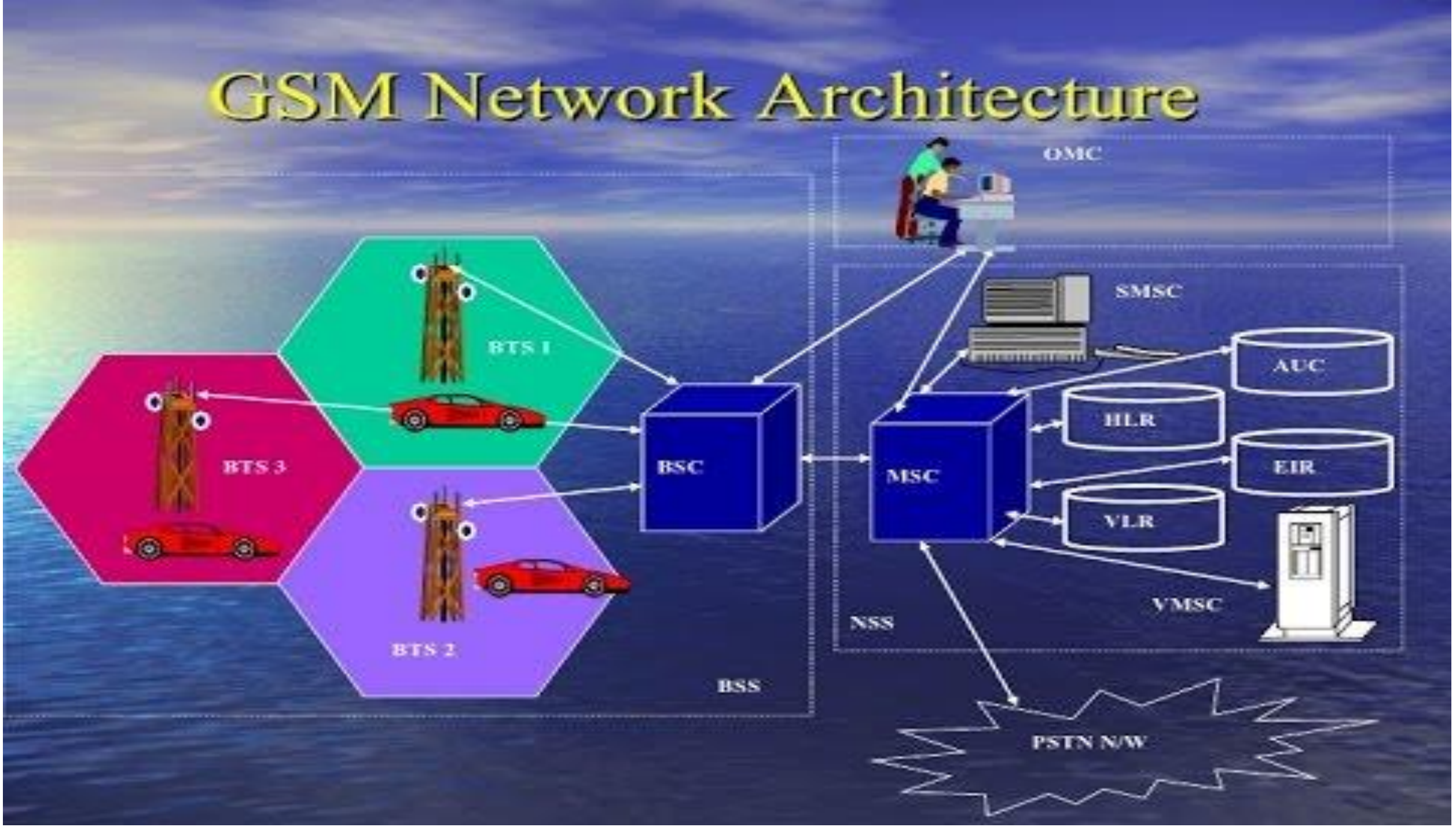
A public land mobile network (PLMN) is any wireless communications system intended for use by terrestrial subscribers in vehicles or on foot. Such a system can stand alone, but often it is interconnected with a fixed system such as the public switched telephone network (PSTN).

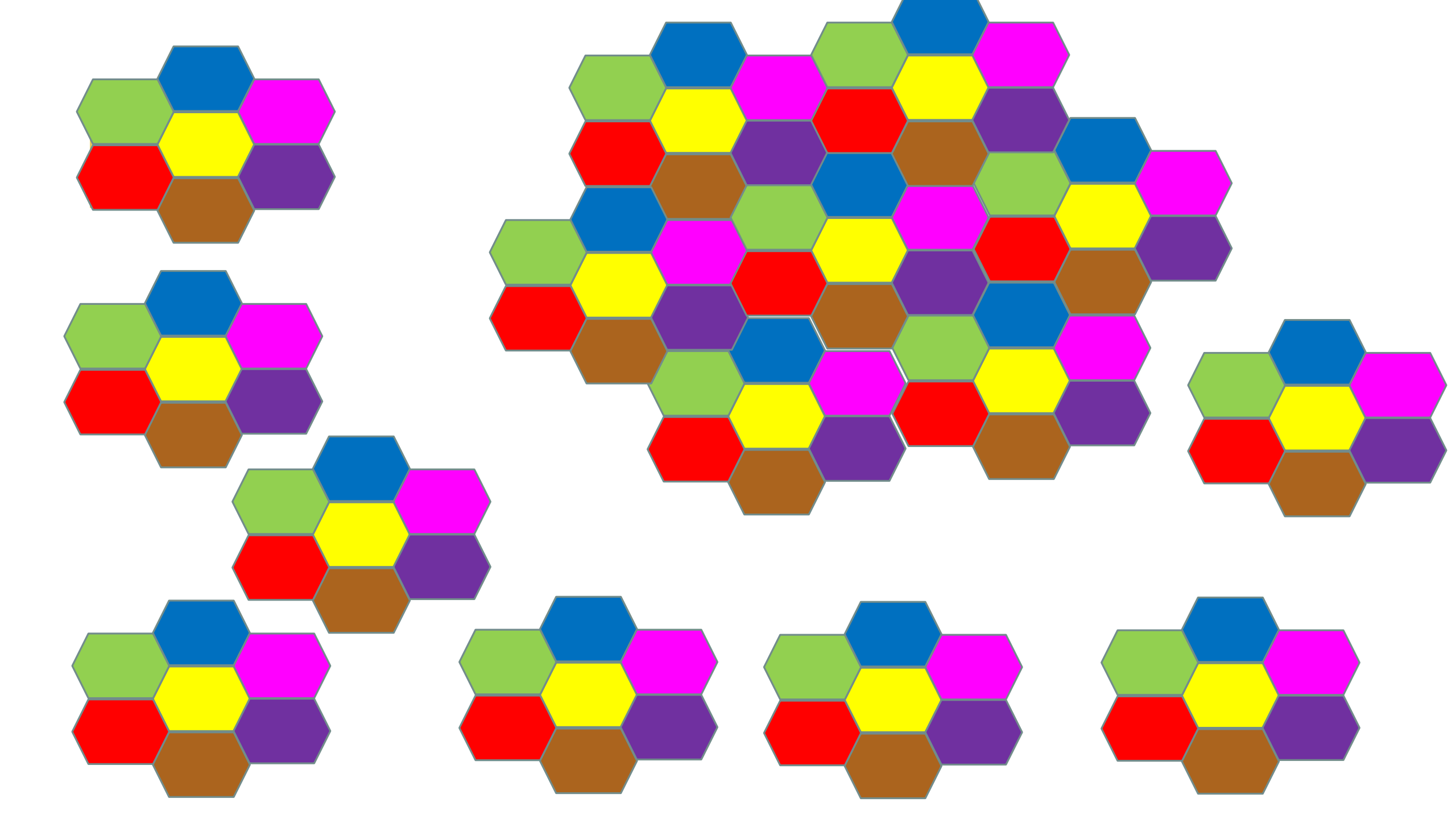
A public land mobile network (PLMN), as defined in telecommunications regulation, is a network that is established and operated by an administration or by a recognized operating agency (ROA) for the specific purpose of providing land mobile telecommunications services to the public.


Backend server domain
Communication domain (mobile networks)
Application domain
Internet
Backend server
aGW
aGW
aGW
aGW
eNB
eNB
eNB
eNB
X2 interface
X2 interface
Air interface 'Uu' (random access)
aGW
ITS and logistics
• Warehouse and fleet management/asset tracking
• Collision avoidance and on-board security
• Traffic and infrastructure management
POS terminals
E-healthcare
• Disease management
• Aging independently
• Personal fitness
Compensation applications
• Utility account management
• Service primitives
Home automation
• Automatic lighting
• Regulating heat and water levels
Metering/monitoring
• Prepaid delivery
• Person/animal protection
• Object protection
Control systems
• Vending machine
• Production machines


DRAW IN EXAM
MS
MS
BTS
BTS
BSC
BTS
BTS
BSC
Um
BSS
A interface
AUC
EIR
HLR
SS7 Network
MSC/ VLR
SMS-GMSC
OSS NSS
PSTN
ISDN
CSPDN
PSPDN
Public network

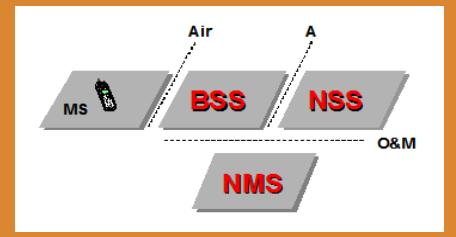
GSM Network Architecture

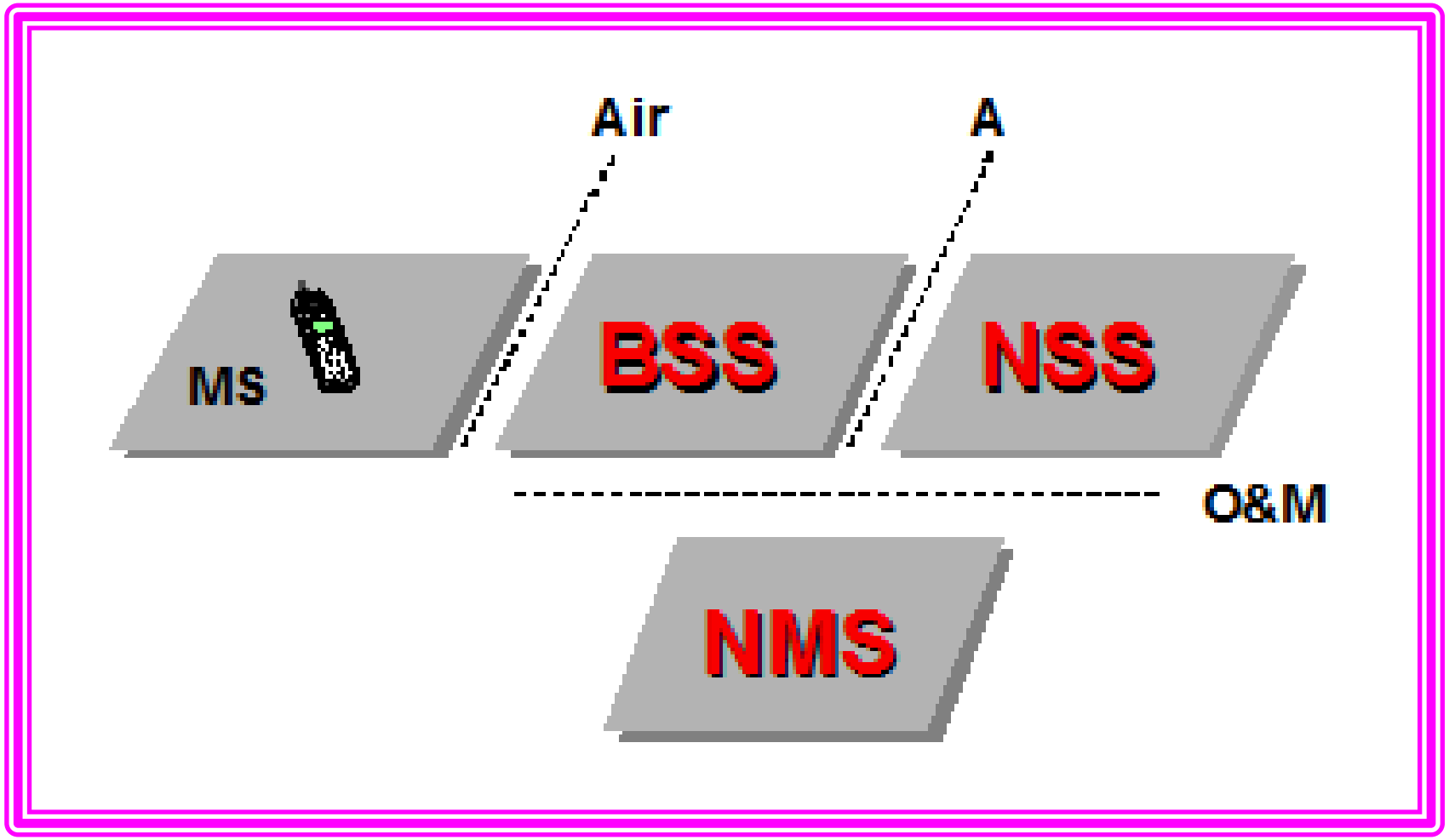




GSM

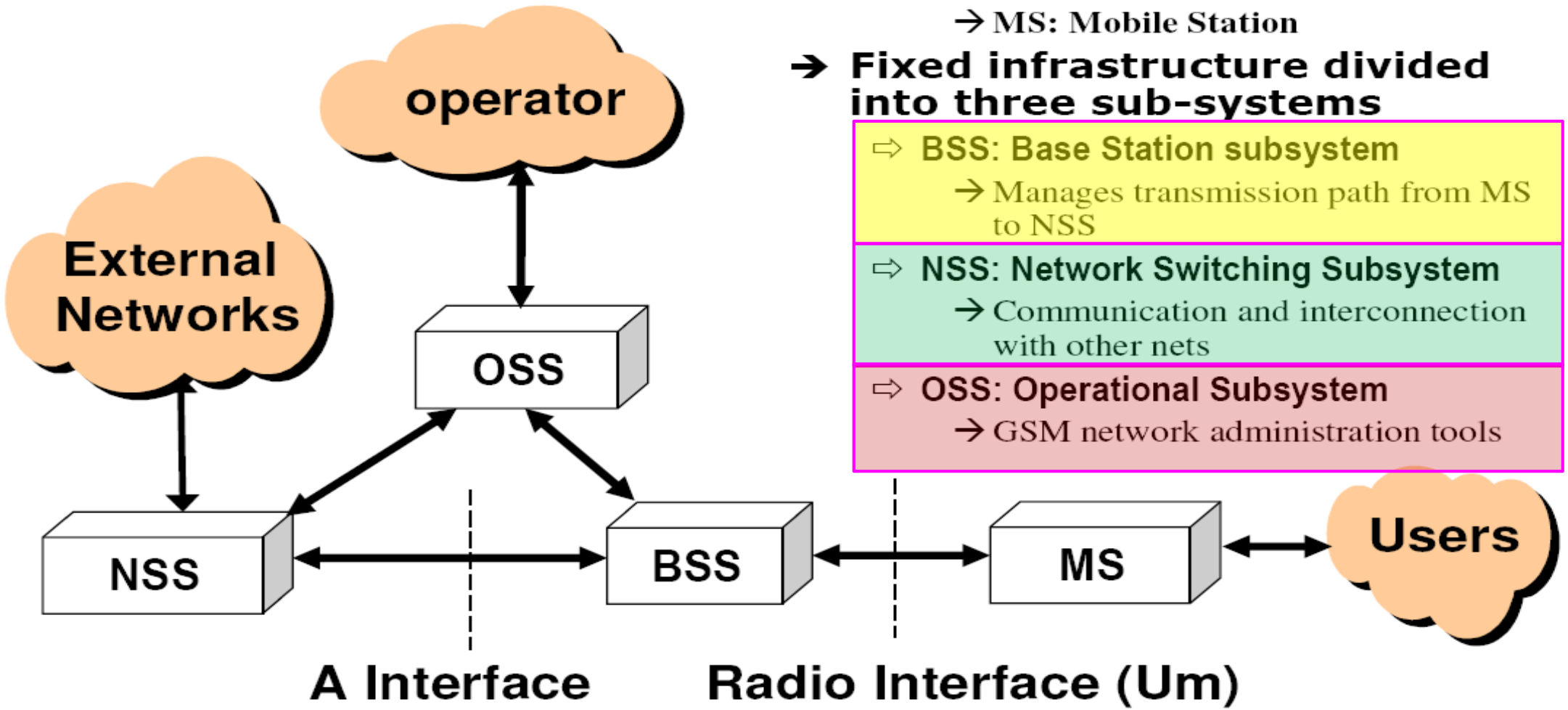
SUB-SYSTEM




Air
A
MS
BSS
NSS
O&M
NMS

GSM Sub-Systems

- **Two components:**
  - ***Fixed installed infrastructure***
    - The network in the proper sense
  - ***Mobile subscribers***
    - **MS: Mobile Station**
- **Fixed infrastructure divided into three sub-systems**
  - **BSS: Base Station subsystem**
    - Manages transmission path from MS to NSS
  - **NSS: Network Switching Subsystem**
    - Communication and interconnection with other nets
  - **OSS: Operational Subsystem**
    - GSM network administration tools



Network Switching Sub-System

➔Elements:

- ⇨Mobile Switching Center (MSC) / Gateway MSC (GMSC)
- ⇨Home Location Register (HLR) / Authentication Center (AuC)
- ⇨Visitor Location Register (VLR)
- ⇨Equipment Identity Register (EIR)

➔Functions:

- ⇨Call control
- ⇨User management

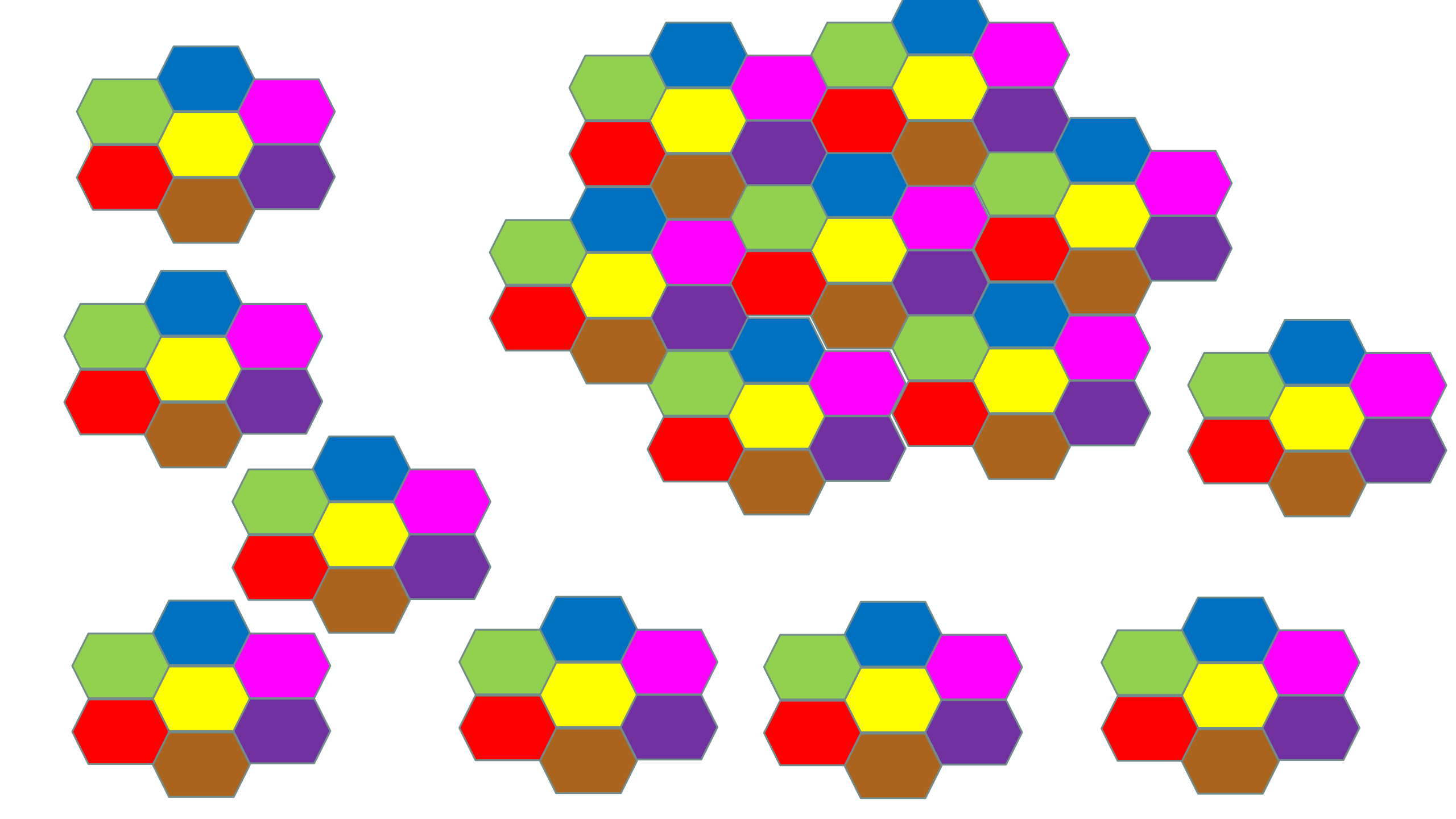
➔Inter-component communication

- ⇨Via SS7 signalling network
- ⇨With suitable extensions (e.g. MAP – Mobile Application Part)

Operation & Maintenance Sub-system (OSS)

- **Network measurement and control functions**
- **Monitored and initiated from the OMC (Operation and Maintenance Center)**
- **Basic functions**
  - ⇨ Network Administration
    - → configuration, operation, performance management, statistics collection and analysis, network maintenance
  - ⇨ Commercial operation & charging
    - → Accounting & billing
  - ⇨ Security Management
    - → E.g. Equipment Identity Register (EIR) management

O&M functions based on ITU-T TMN standards (Telecommunication Network Management) – complex topic out of the scopes of this course



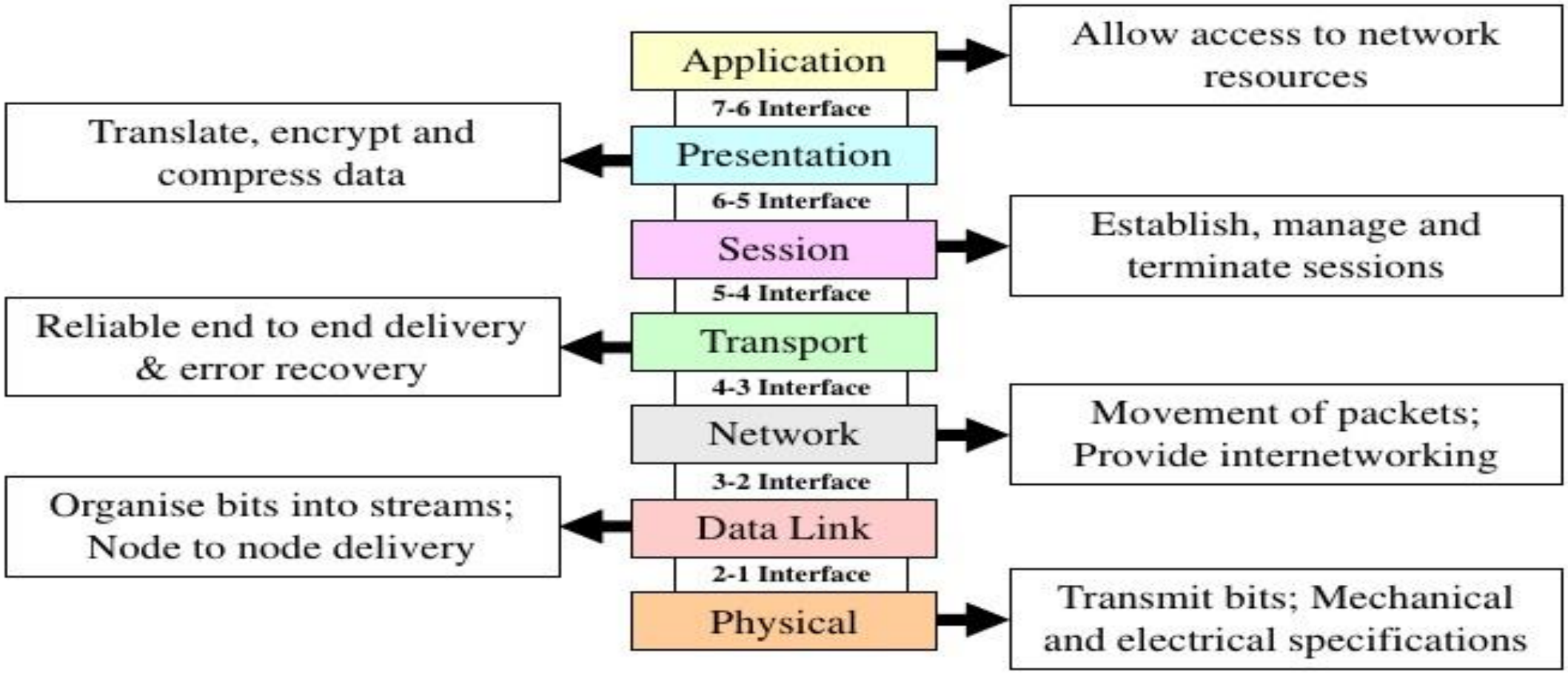
MAPPING OF GSM LAYER ON TO OSI LAYERS

OSI Model

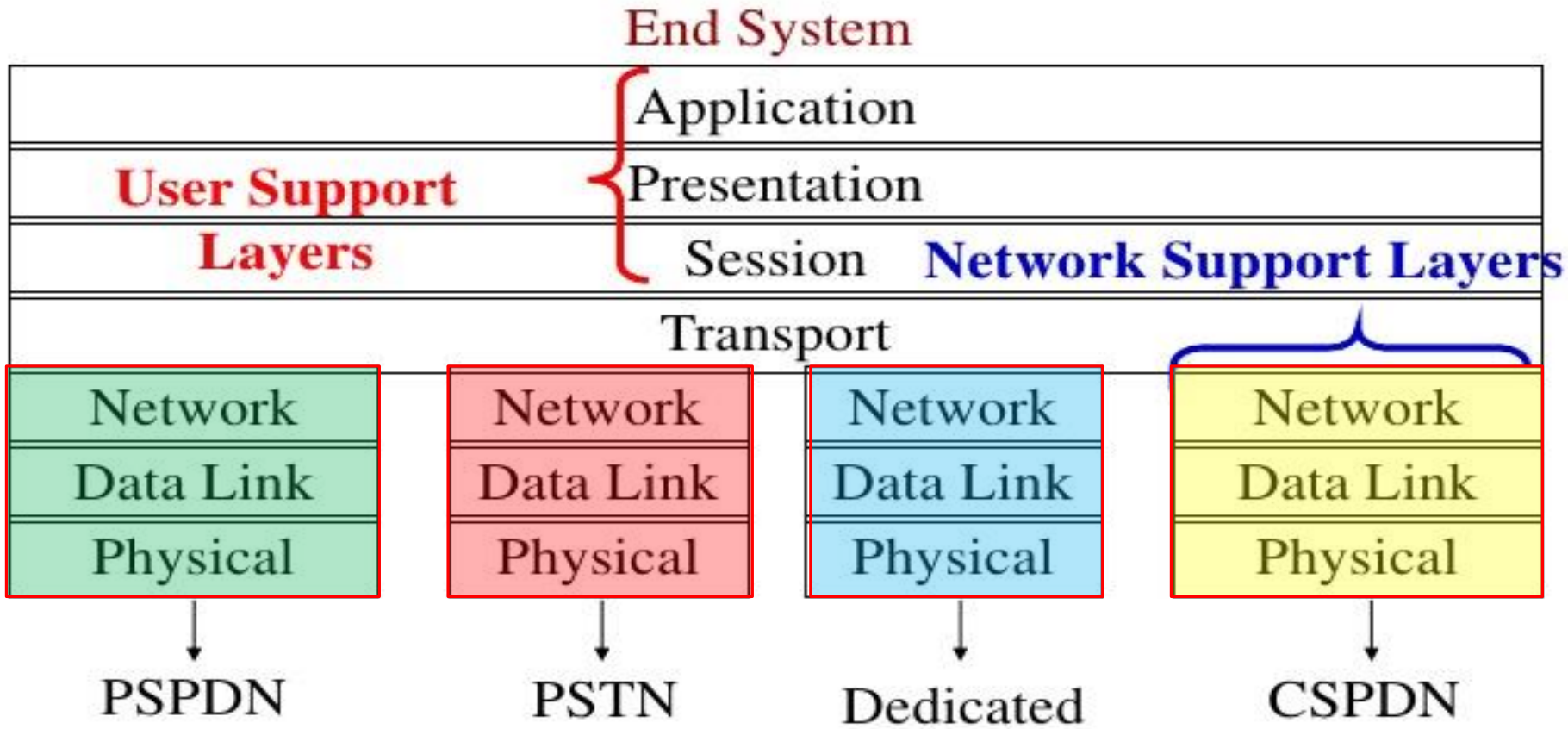


7	Application	Data Generation
6	Presentation	Encryption and Formatting
5	Session	Establish Connection
4	Transport	Delivery and Sequencing
3	Network	Routing to Destination
2	Data Link	Local Network Host Delivery
1	Physical	Access to Media

Summary of OSI Layers Functions



OSI Layers

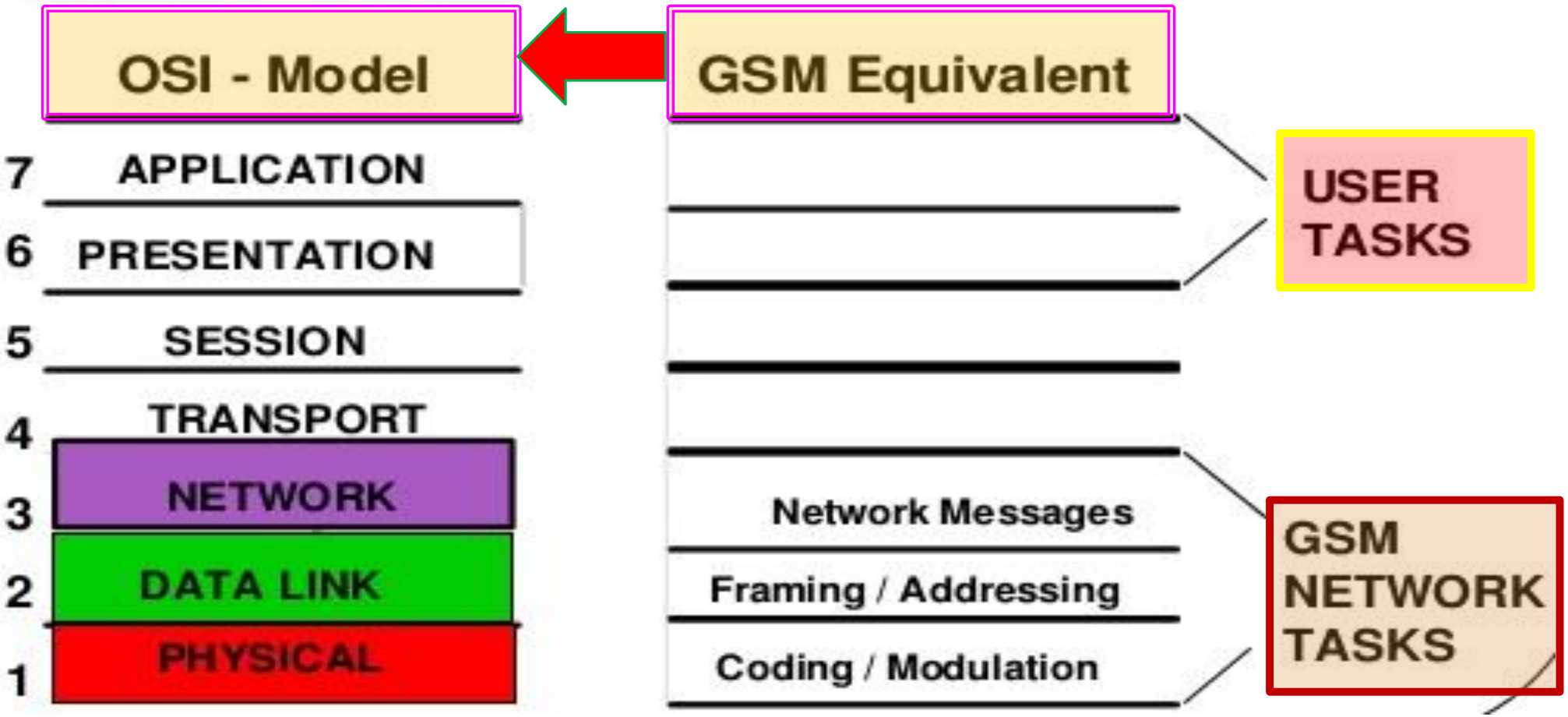

End System
Application
Presentation
Session
Transport
User Support Layers
Network Support Layers
Network
Data Link
Physical
Network
Data Link
Physical
Network
Data Link
Physical
Network
Data Link
Physical
PSPDN
PSTN
Dedicated
CSPDN

GSM Layers

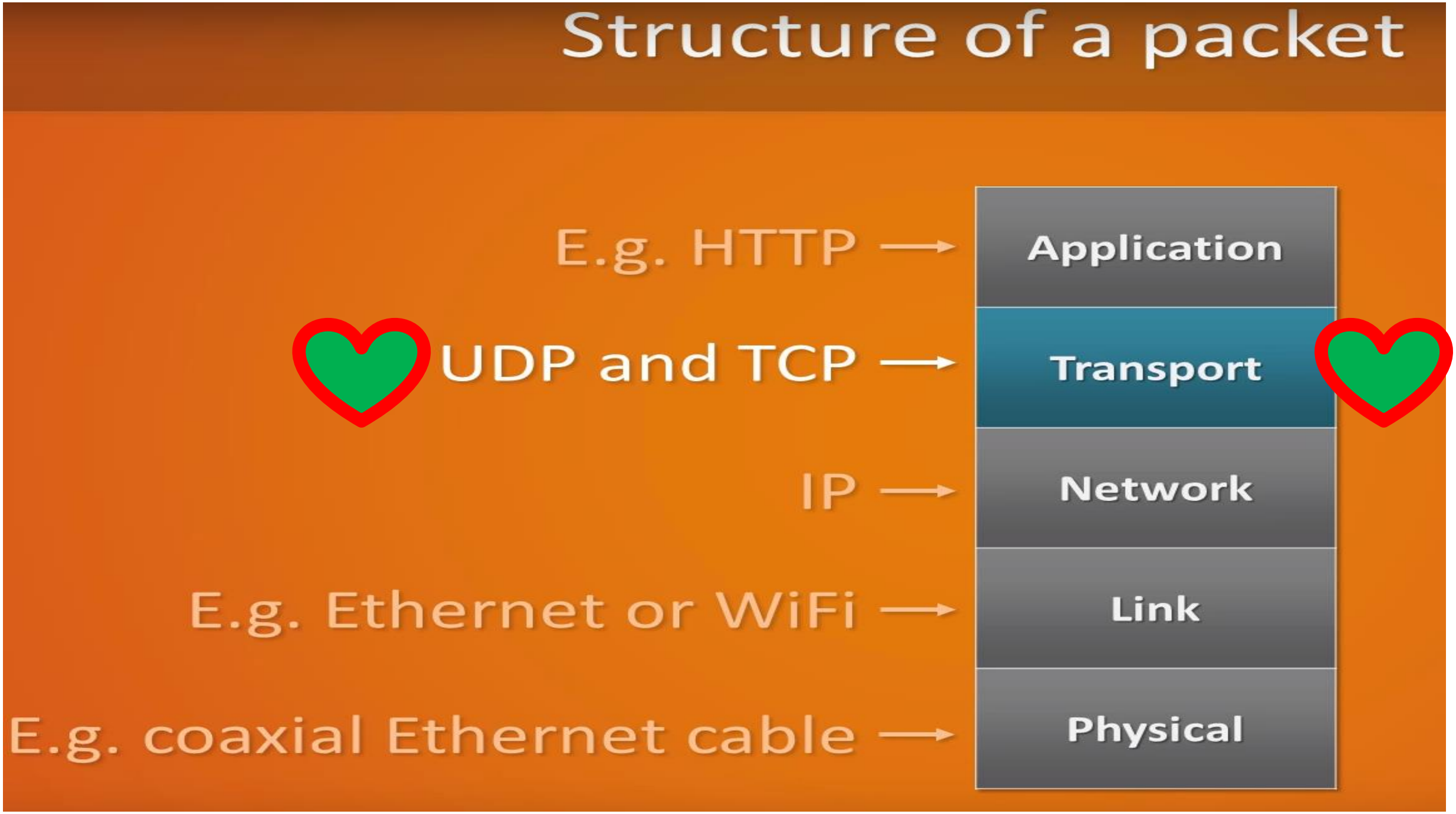
- Layer 1: Physical layer
  - physical transmission
  - channel quality measurements
  - GSM Rec. 04.04, PCM 30 or ISDN links are used (GSM Rec. 08.54 on A bis interface and 08.04 on A to F interfaces)
- Layer 2: Data link layer
  - Multiplexing of layer 2 connections on control/signaling channels
  - Error detection (based on HDLC)
  - Flow control
  - Transmission quality assurance
  - Routing
- Layer 3: Network layer
  - Connection management (air interface)
  - Management of location data
  - Subscriber identification
  - Management of added services (SMS, call forwarding, conference calls, etc.)

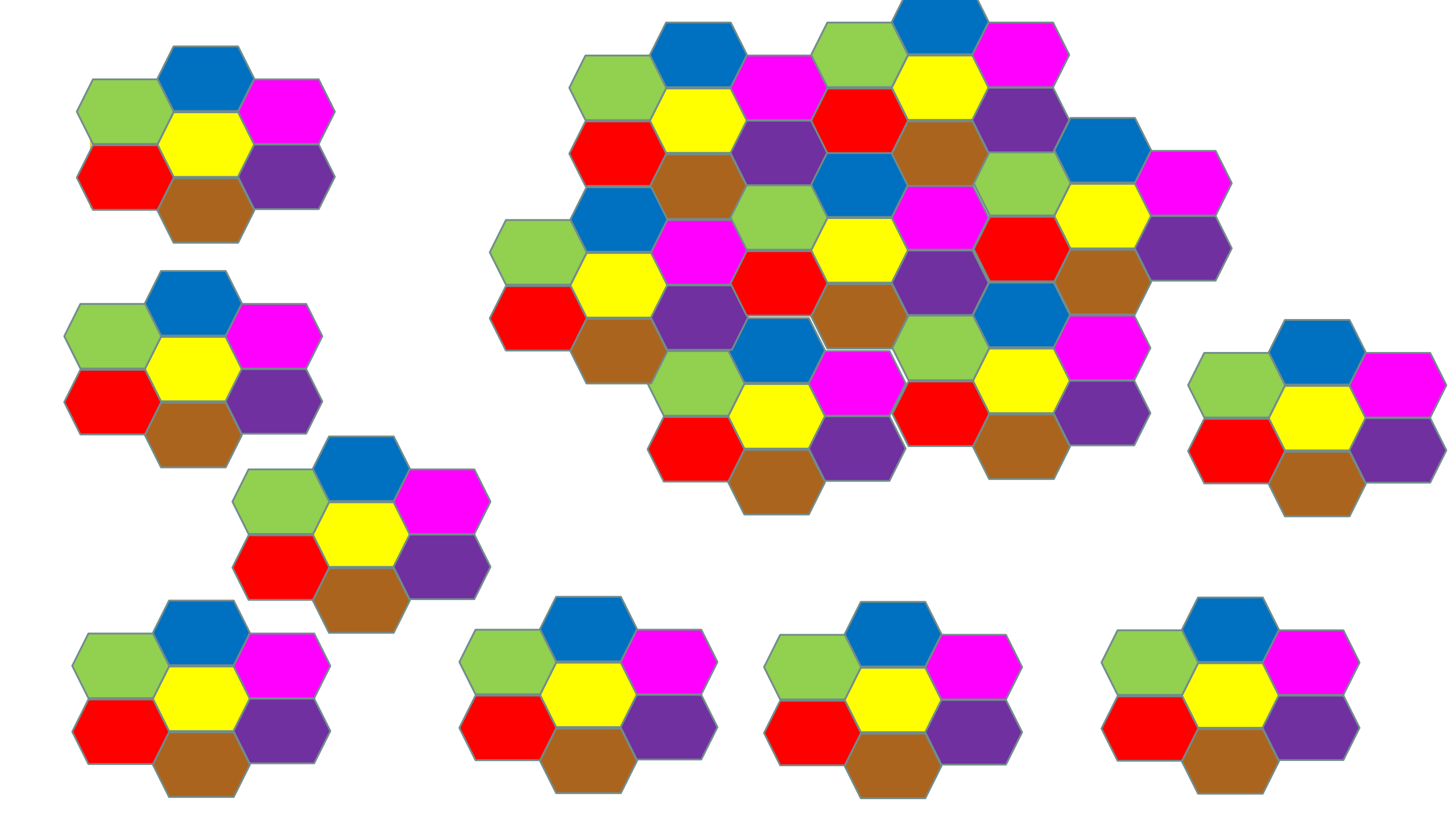
GSM Signaling

Open Systems Interconnect uses “layers” where only layer 1, the physical layer is actually physically connected between devices. Messages are sent between devices by handing them down to lower layers from transmission across the physical interface.



Structure of a packet





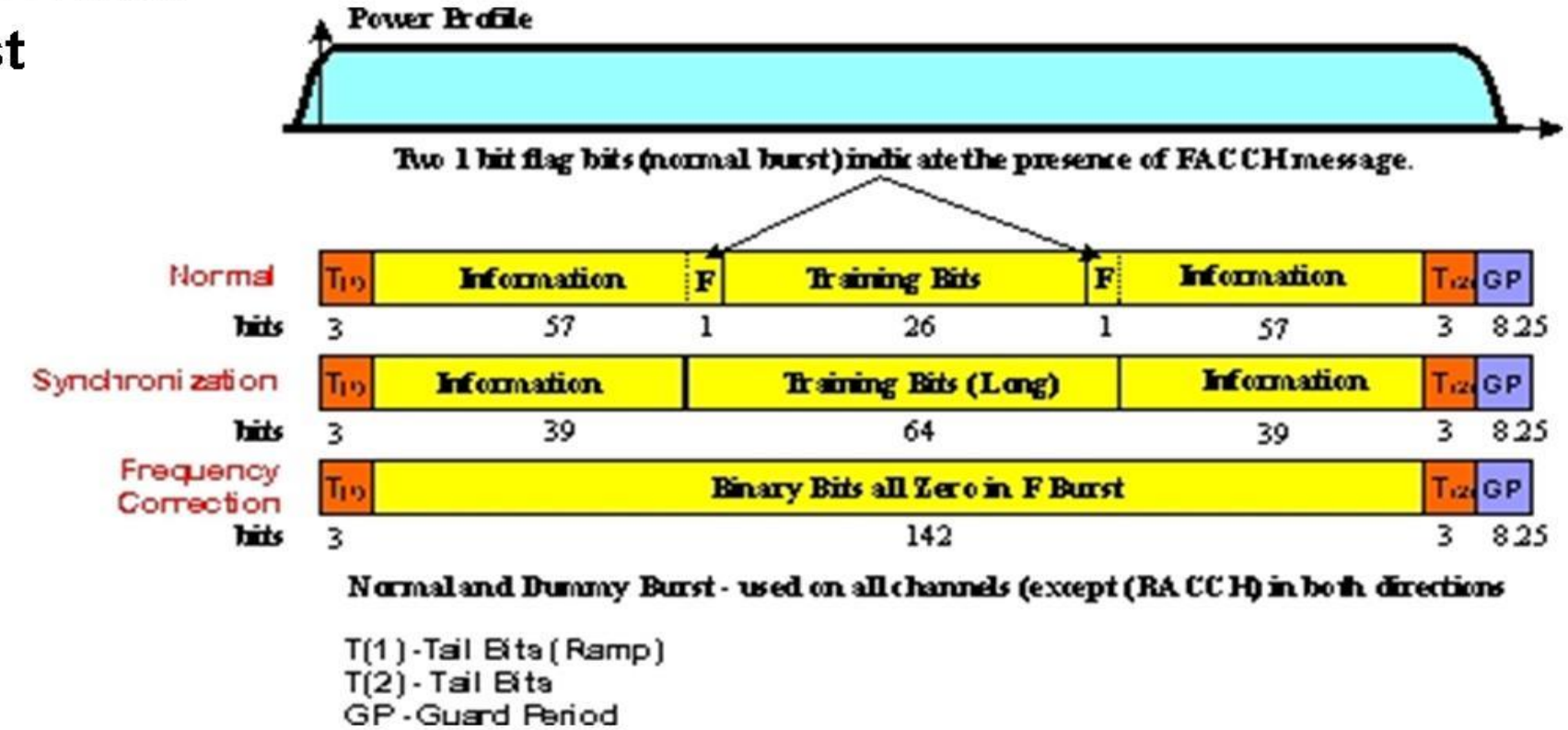
GSM
LOGICAL CHANNEL
AND FRAME

GSM Radio Channels

- Radio Channel Structure
- Time Slot Structure
- Frame Structure
- Dynamic Time Alignment

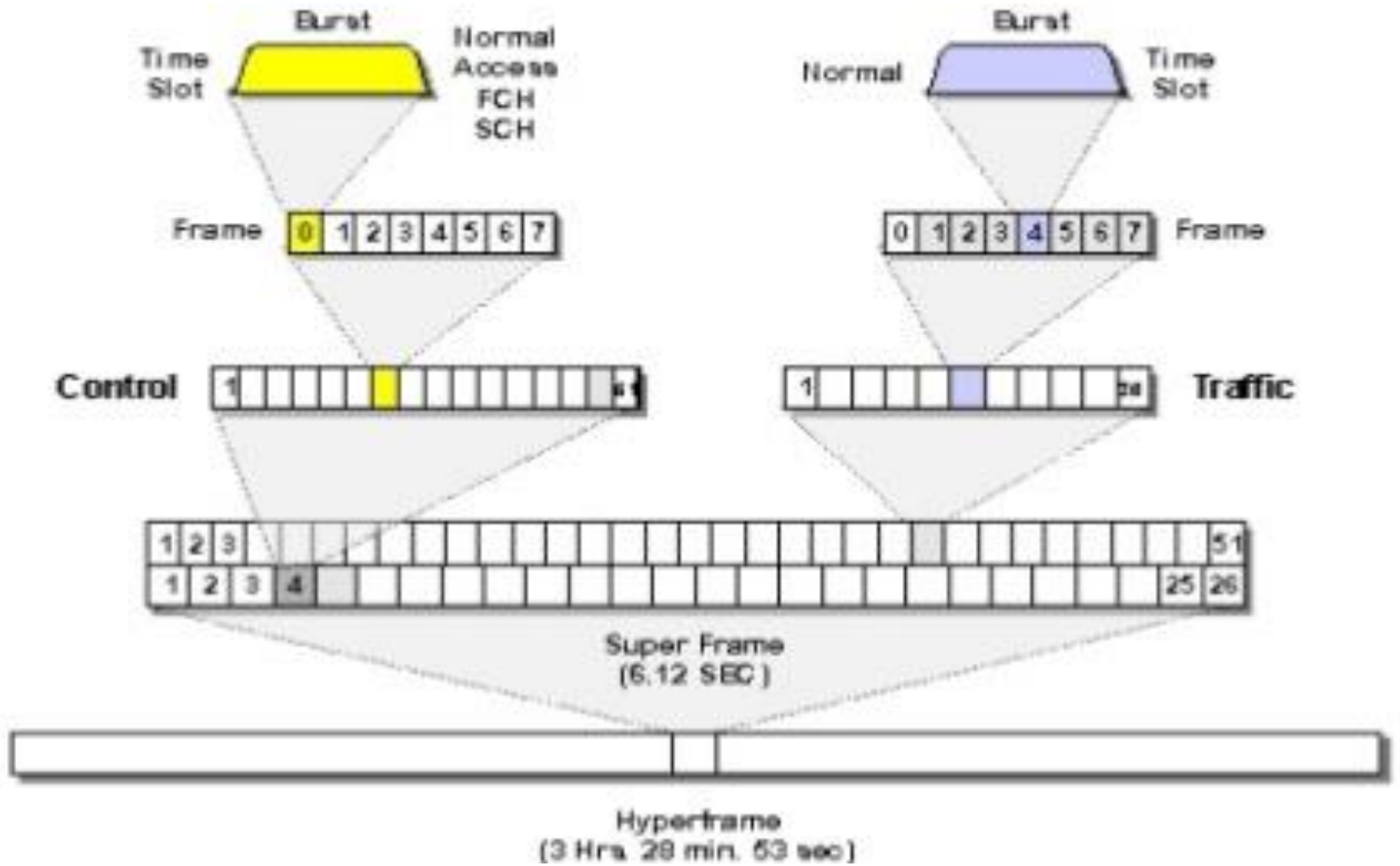
Time Slot Structure

- Normal Burst
- Random Access Burst (Shortened Burst)
- Frequency Correction Burst
- Synchronization Burst



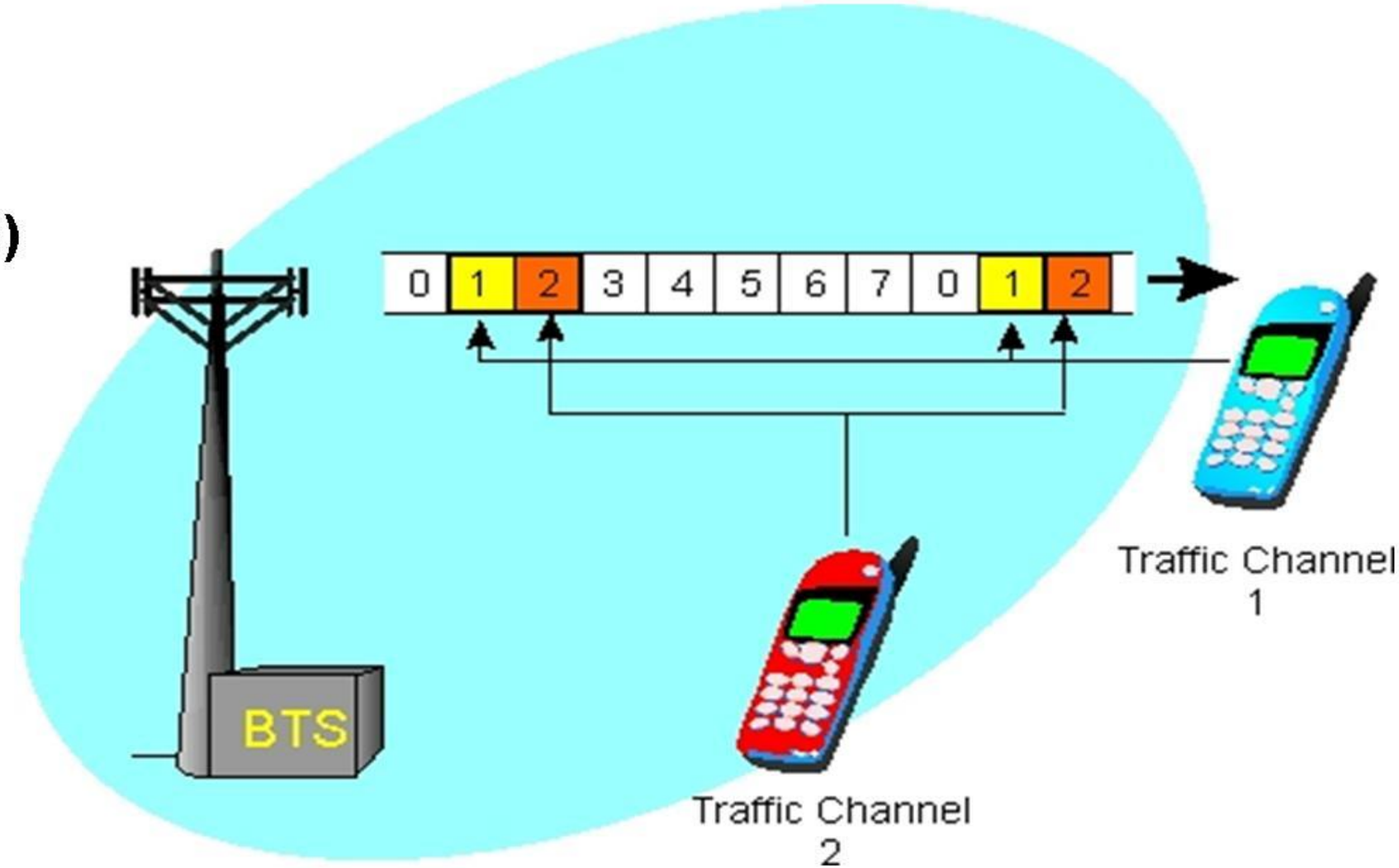
Frame Structure

- **GSM Frame**
  - 4.615 msec
- **Multiframes**
  - Traffic
  - Control
- **Superframes**



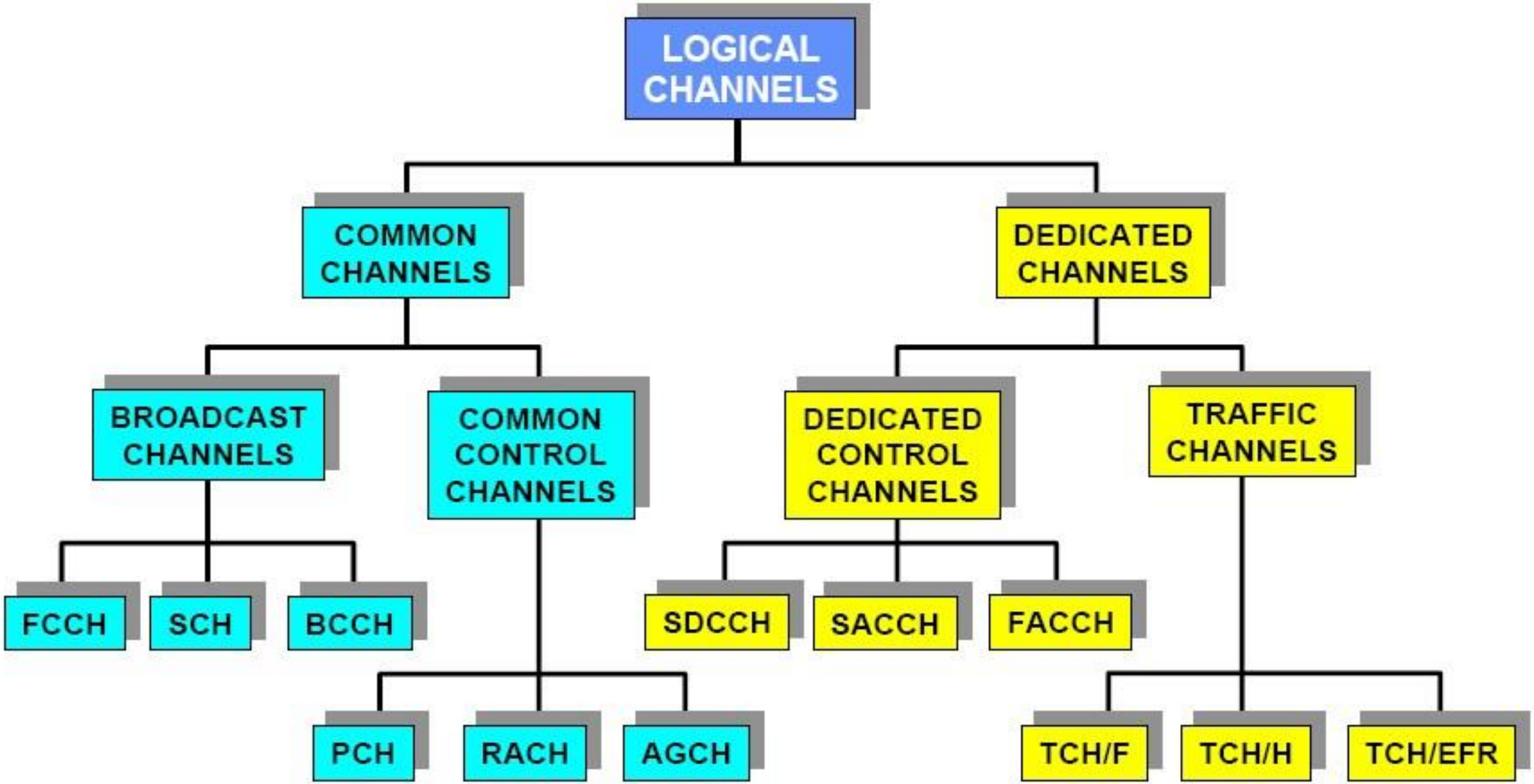
Traffic Channels

- Full Rate (TCH/F)
- Half Rate (TCH/H)
- Eighth Rate (TCH/8)



What is a logical channel?

GSM uses a variety of channels in which the data is carried. In GSM, these channels are separated into physical channels and logical channels. The Physical channels are determined by the timeslot, whereas the logical channels are determined by the information carried within the physical channel.


LOGICAL CHANNELS
COMMON CHANNELS
DEDICATED CHANNELS
BROADCAST CHANNELS
COMMON CONTROL CHANNELS
DEDICATED CONTROL CHANNELS
TRAFFIC CHANNELS
FCCH
SCH
BCCH
SDCCH
SACCH
FACCH
PCH
RACH
AGCH
TCH/F
TCH/H
TCH/EFR

What are the common control channels?

It is most often used in the context of a trunked radio system, where the control channel sends data which coordinates users in talk groups. In GSM networks, Control Channels can be broadly divided into 3 categories; Broadcast Control Channel (BCCH), Common Control Channel (CCCH), and Dedicated Control Channels (DCCH).

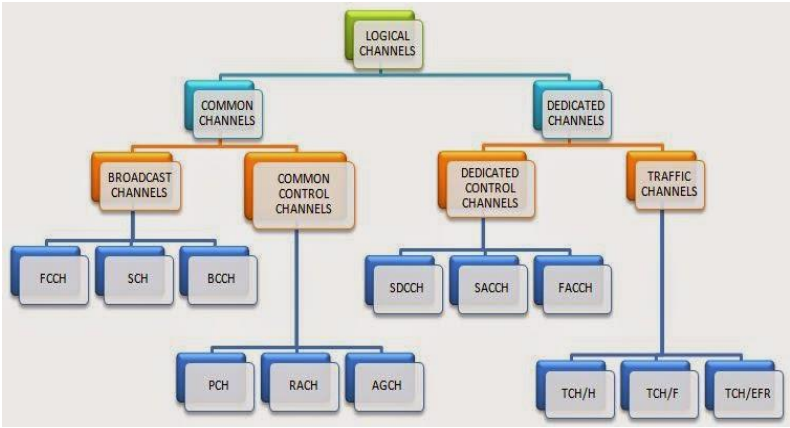
What is SDCCH in GSM?

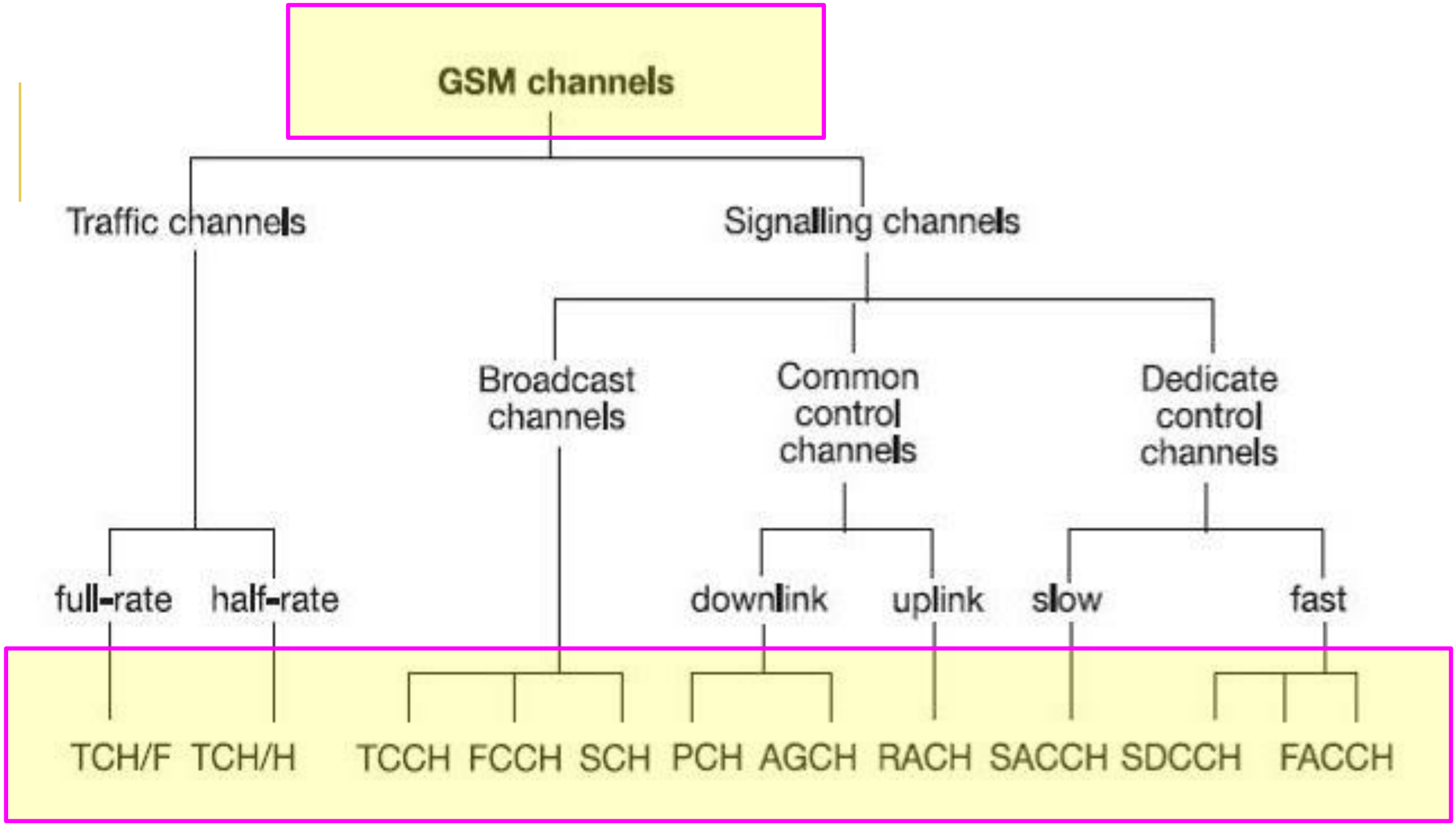
SDCCH. Stand-alone Dedicated Control Channel. This channel is used in the GSM system to provide a reliable connection for signalling and SMS (Short Message Service) messages. The SACCH (Slow Associated Control Channel) is used to support this channel.

What is the paging channel?

The Paging Channel (PCH) is a downlink transport channel. The PCH is always transmitted over the entire cell. The transmission of the PCH is associated with the transmission of physical-layer generated Paging Indicators, to support efficient sleep-mode procedures. Paging Channel selection.

- Logical Channel
  - Signaling Channel
    - Broadcast
      - Frequency Correction Channel (FCCH)
      - Synchronization Channel (SCH)
      - Broadcast control Channel (BCCH)
    - Common Control
      - Paging Channel (PCH)
      - Access grant Channel (AGCH)
      - Random access Channel (RACH)
    - Dedicated Control
      - Stand-alone dedicated control Channel
      - Slow associated control Channel (SACCH)
      - Fast associated control Channel (FACCH)
  - Traffic Channel
    - Half Rate
    - Full Rate




GSM channels
Traffic channels
Signalling channels
Broadcast channels
Common control channels
Dedicate control channels
full-rate
half-rate
downlink
uplink
slow
fast
TCH/F
TCH/H
TCCH
FCCH
SCH
PCH
AGCH
RACH
SACCH
SDCCH
FACCH

GSM: PHYSICAL AND LOGICAL CHANNELS

GSM uses a mix of Frequency Division Multiple Access (FDMA) and Time Division Multiple Access (TDMA). FDMA parts involves the division by frequency of the 25 MHz bandwidth in to 124 carrier frequencies (Also called ARFCN) spaced 200 KHz for GSM-900. For GSM-1800 frequency spectrum of 75 MHz bandwidth is divided in to 374 carrier frequencies spaced 200 KHz. TDMA further divides each carrier frequencies in to 8 time slots such that each carrier frequency is shared by 8 users. So in GSM, the basic radio resource is a time slot with duration of 577 µs. 8 Time slots of 577 µs constitutes a 4.615 ms TDMA Frame. GSM uses Gaussian Minimum Shift Keying (GMSK) modulation scheme to transmit information (data and signalling) over Air Interface.

GSM uses number of channels to carry data over Air Interface, these channels are broadly divided in to following two categories:

- ✓ Physical Channels
- ✓ Logical Channels

. PHYSICAL CHANNELS

A physical channel is determined by the carrier frequency or a number of carrier frequencies with defined hopping sequence and the Time Slot number.

8 Time Slots (1 Time Slot = 1 Physical Channel) of 577 µs constitutes a 4.615 ms TDMA Frame. In GSM standard data on a time slot transmitted in bursts, so time slot is often expressed in BP (Burst Period). 1 BP represents 1 TS. TDMA frame (4.615 ms of 8 TS) further structured in to multiframes. There are two types of multiframes in the system:

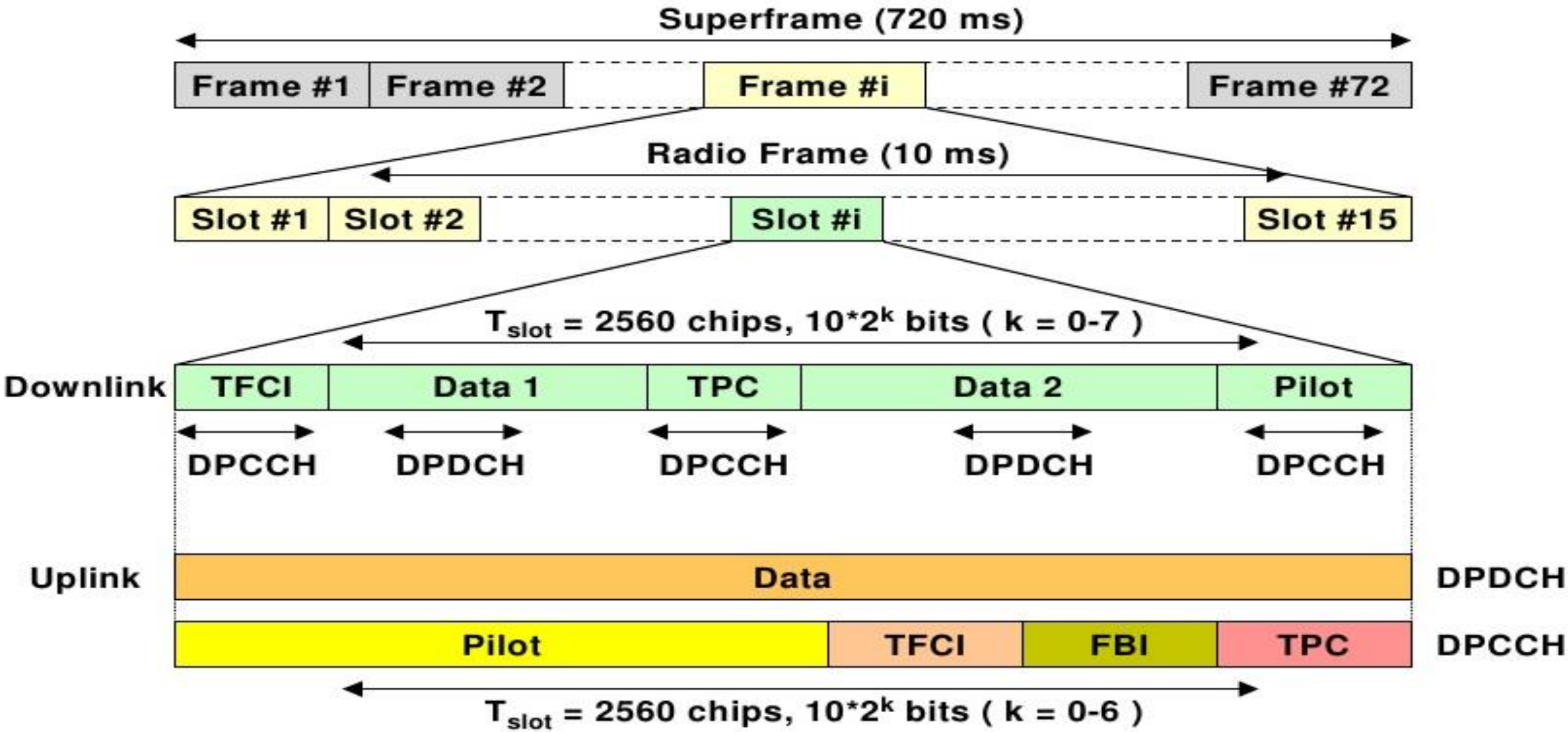
- ✓ 26 TDMA Multiframe: Consists 26 TDMA frames with duration of 120 ms and used to carry the Logical Channels TCH, SACCH, FACCH etc.
- ✓ 51 TDMA Multiframe: Consists 51 TDMA frames with duration of 234.5 ms and used to carry the Logical Channels FCCH, SCH, BCCH, CCCH, SDCCH, SACCH etc.

These multiframes further structured in to Superframe and Hyperframe.

SUPERFRAME: Superframe consists of 51*26 TDMA frames with duration of 6.12 sec. This is corresponding to the smallest cycle for which the organisation of all channels is repeated.

HYPERFRAME: Hyperframe consists 2048 superframes (2048*51*26 TDMA frames) with duration of 3 hrs, 28 min, 53 sec and 760 ms. It is in particular smallest cycle for frequency hopping, cyphering.

Dedicated Physical Channel Frame Structure



Dedicated Control Channels (DCCH)

Fast Associated Control Channel (FACCH)

- While Calls in progress and HO is required FACCH is used
- FACCH works in Stealing mode meaning that one 20ms segment of speech is exchanged for signaling information necessary for the HO



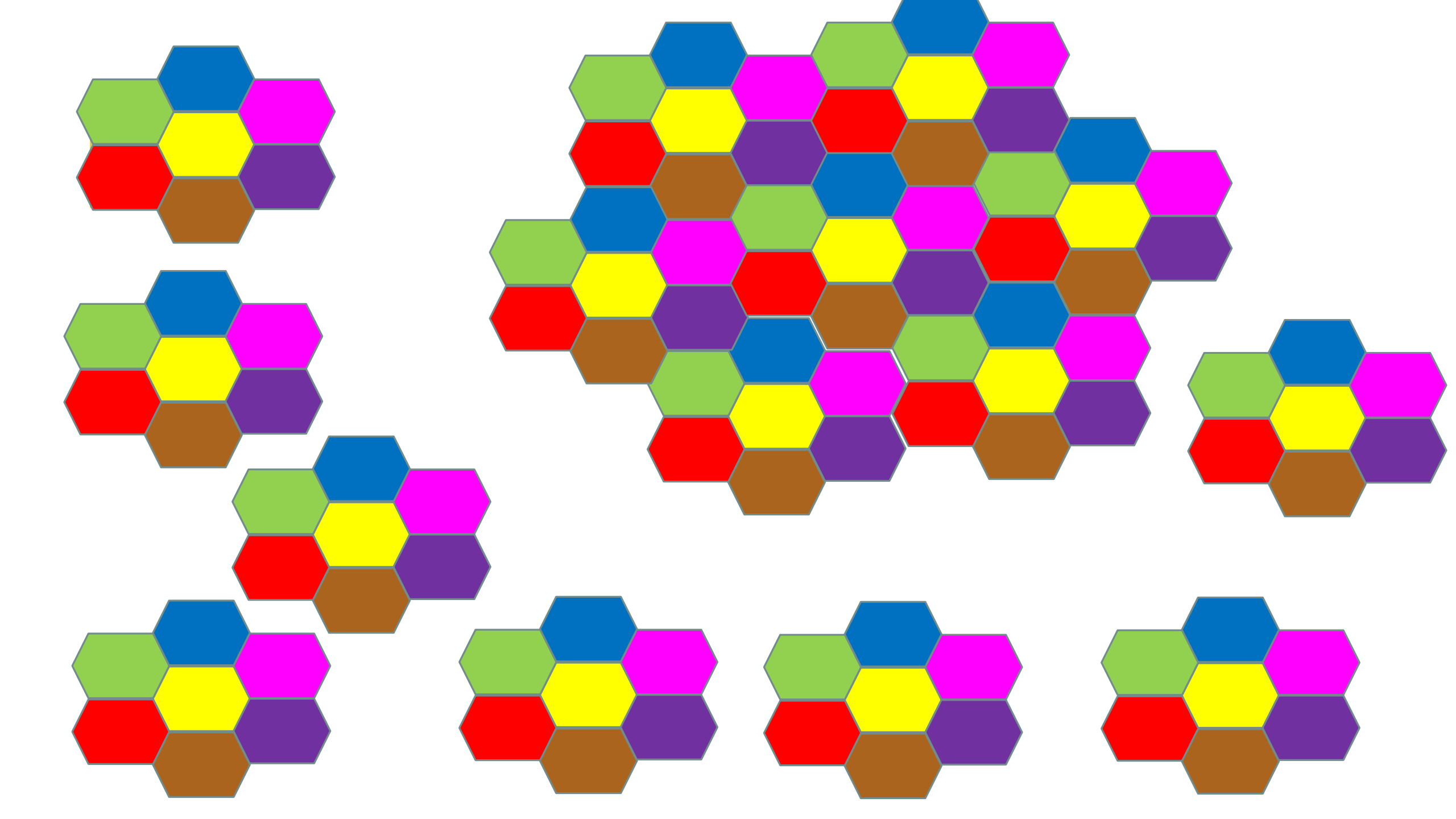
Cell Broadcast Channel (CBCH)

- CBCH is used in Downlink only
- It is used to carry Short Message Service Cell Broadcast (SMSCB) and uses the same physical channel as the SDCCH



Physical Vs. Logical Channel

- Physical channels are all the available time slots of a BTS
  - a BTS with 6 carriers has 48 physical channels
- Logical channels are piggybacked on the physical channels
  - logical channels are laid over the grid of physical channels
  - each logical channel performs a specific task

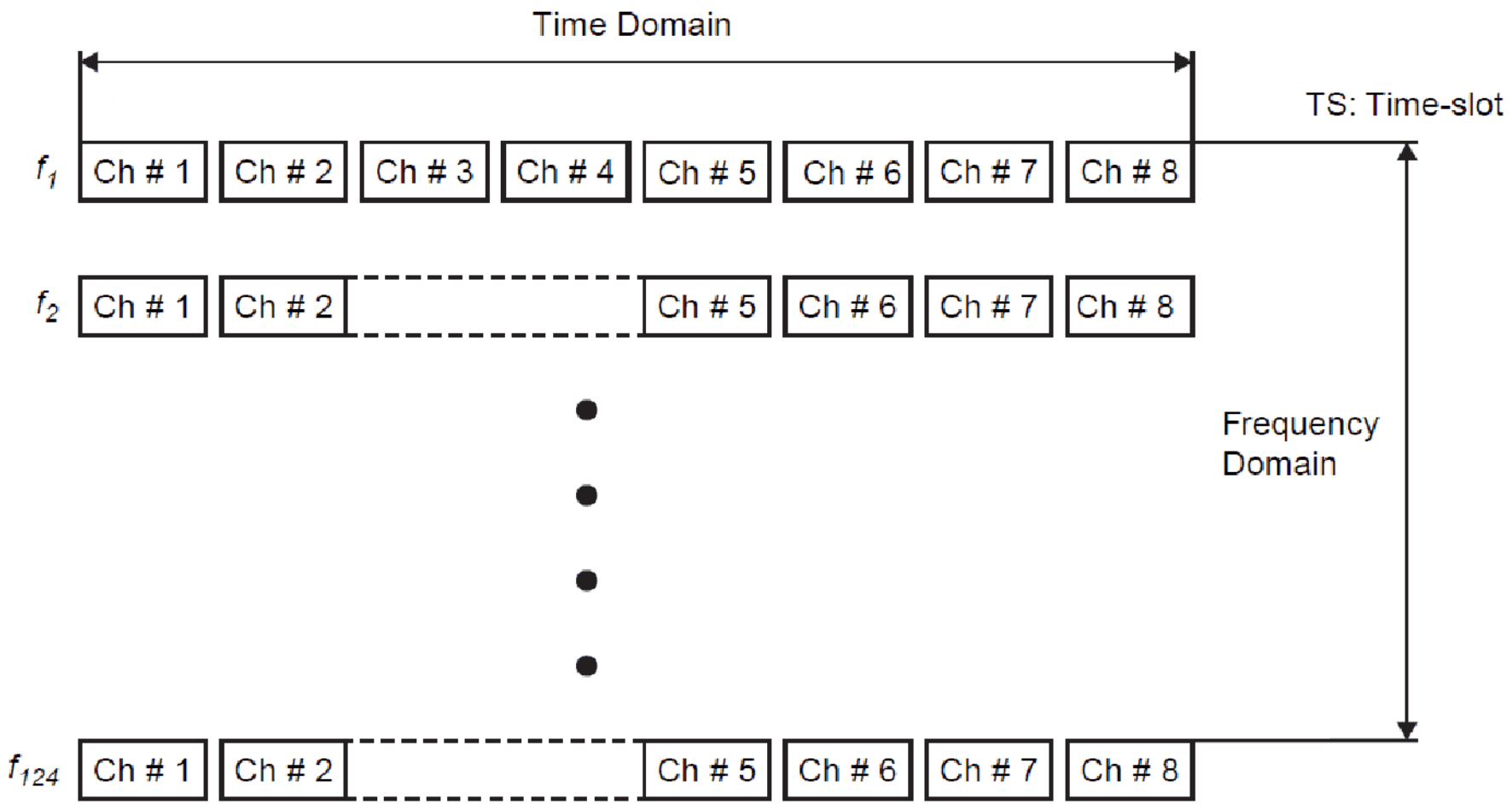


GSM

FRAME

STRUCTURE

GSM FDMA/TDMA Frame Structure

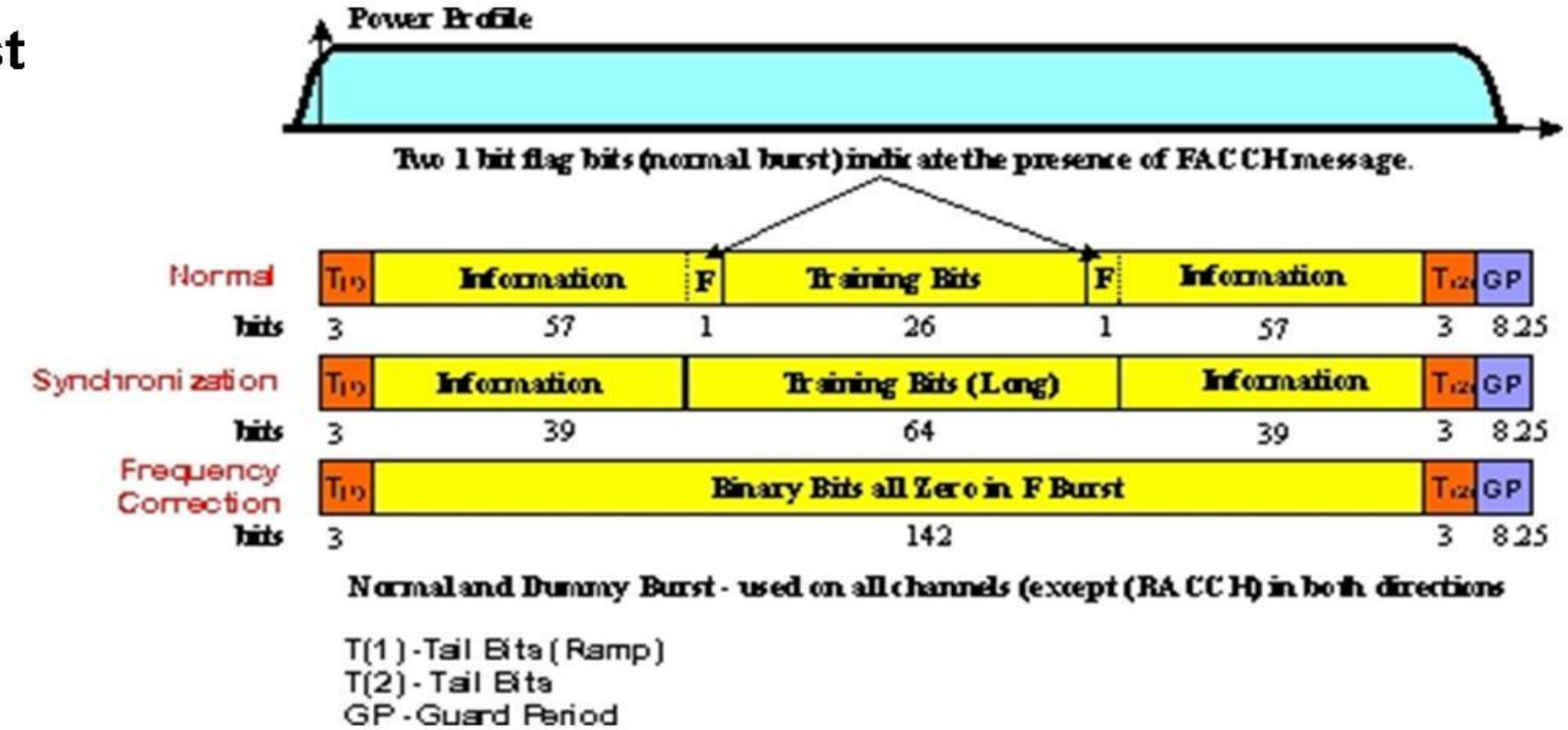


GSM Radio Channels

- Radio Channel Structure
- Time Slot Structure
- Frame Structure
- Dynamic Time Alignment

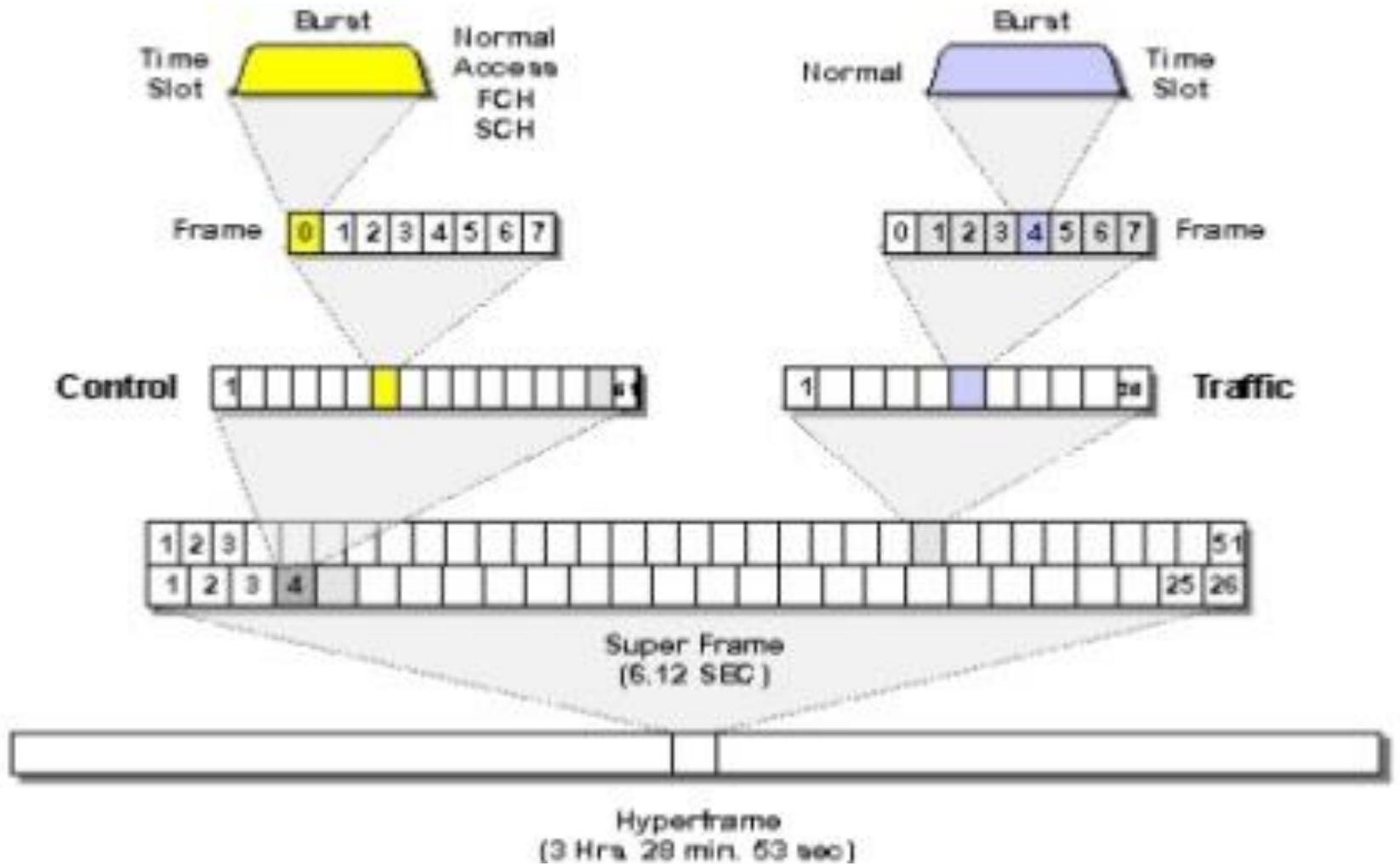
Time Slot Structure

- Normal Burst
- Random Access Burst (Shortened Burst)
- Frequency Correction Burst
- Synchronization Burst

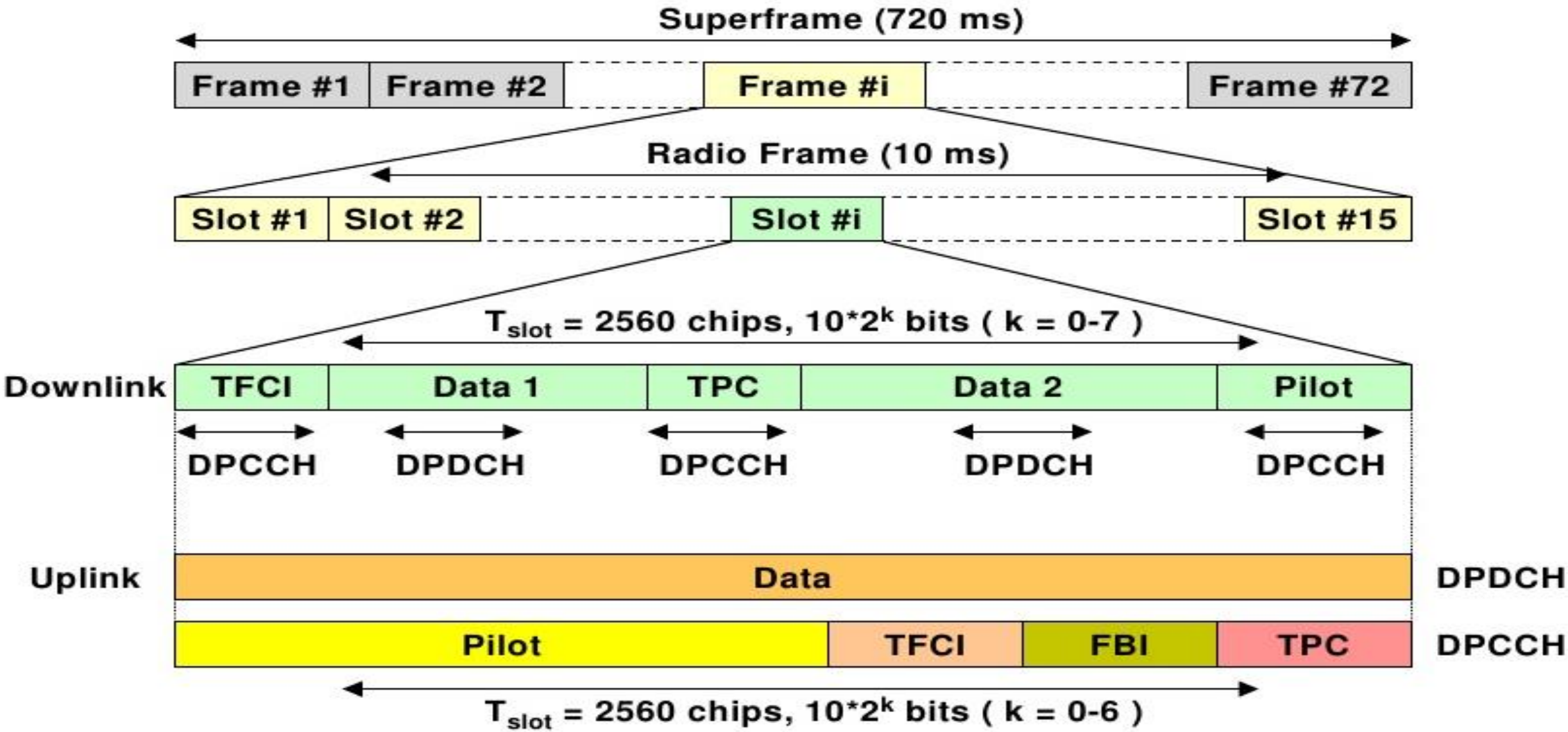


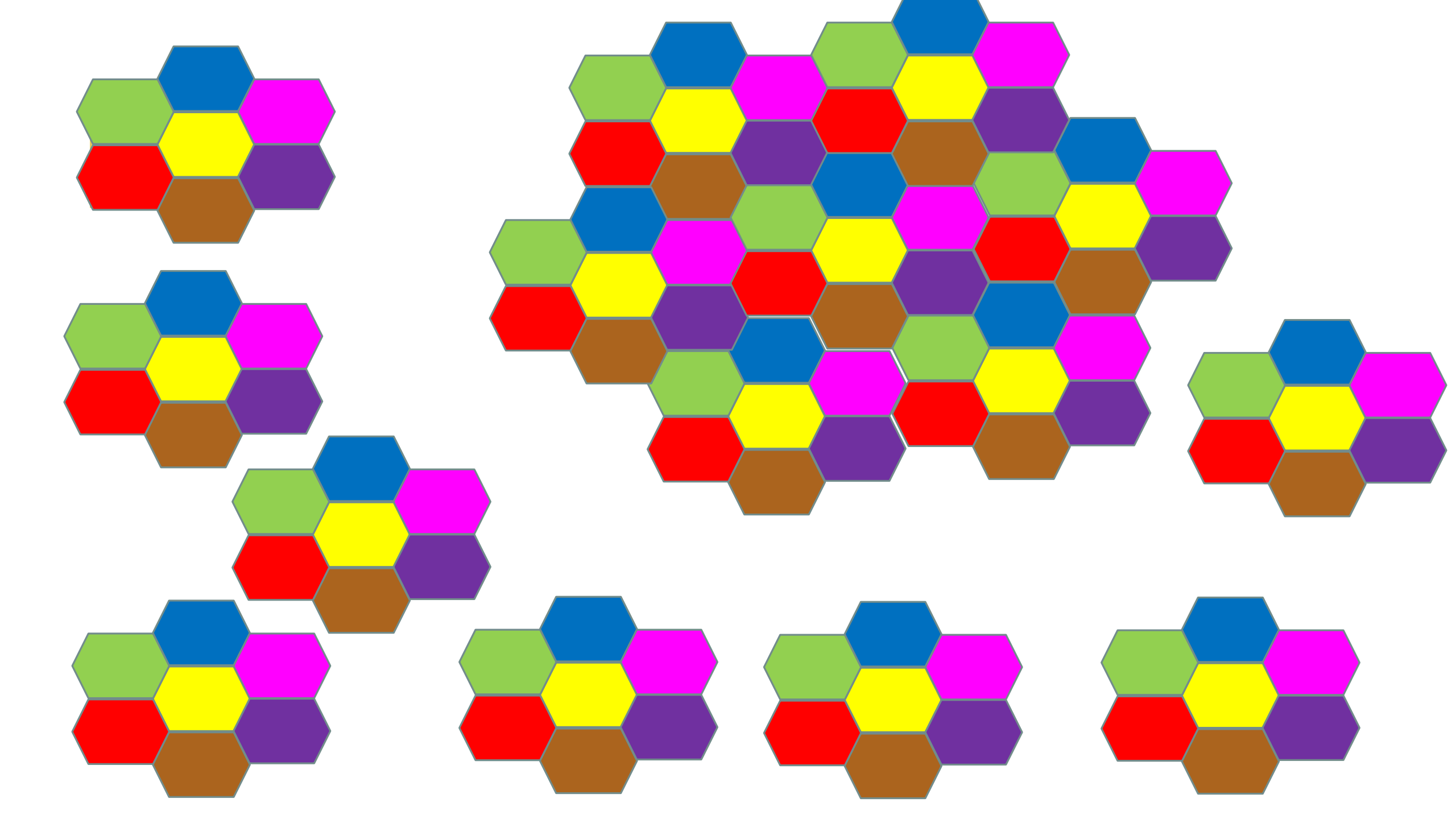
Frame Structure

- **GSM Frame**
  - 4.615 msec
- **Multiframes**
  - Traffic
  - Control
- **Superframes**



Dedicated Physical Channel Frame Structure





GSM

BURST

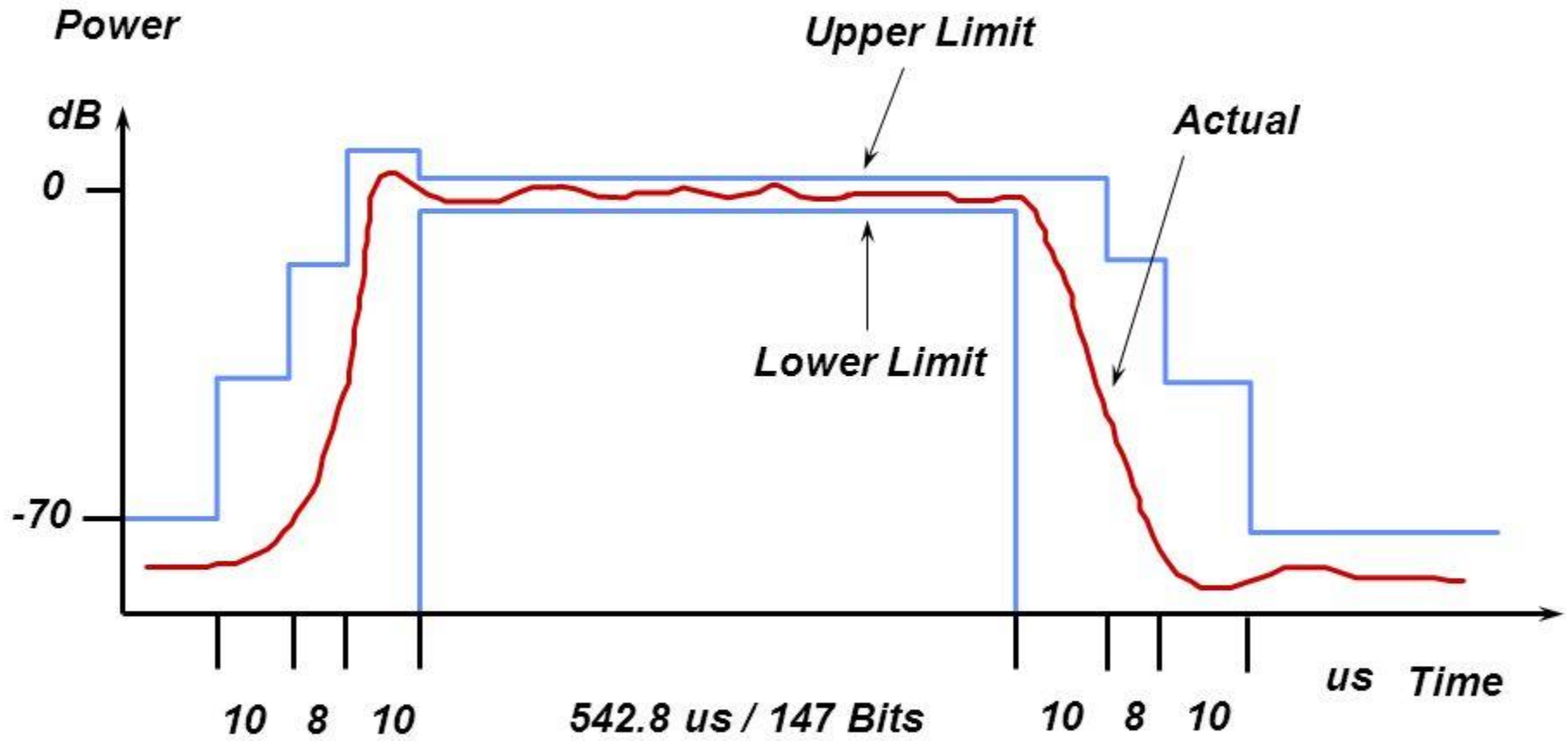
STRUCTURE

GSM Physical Channel Cntd....

BURST:

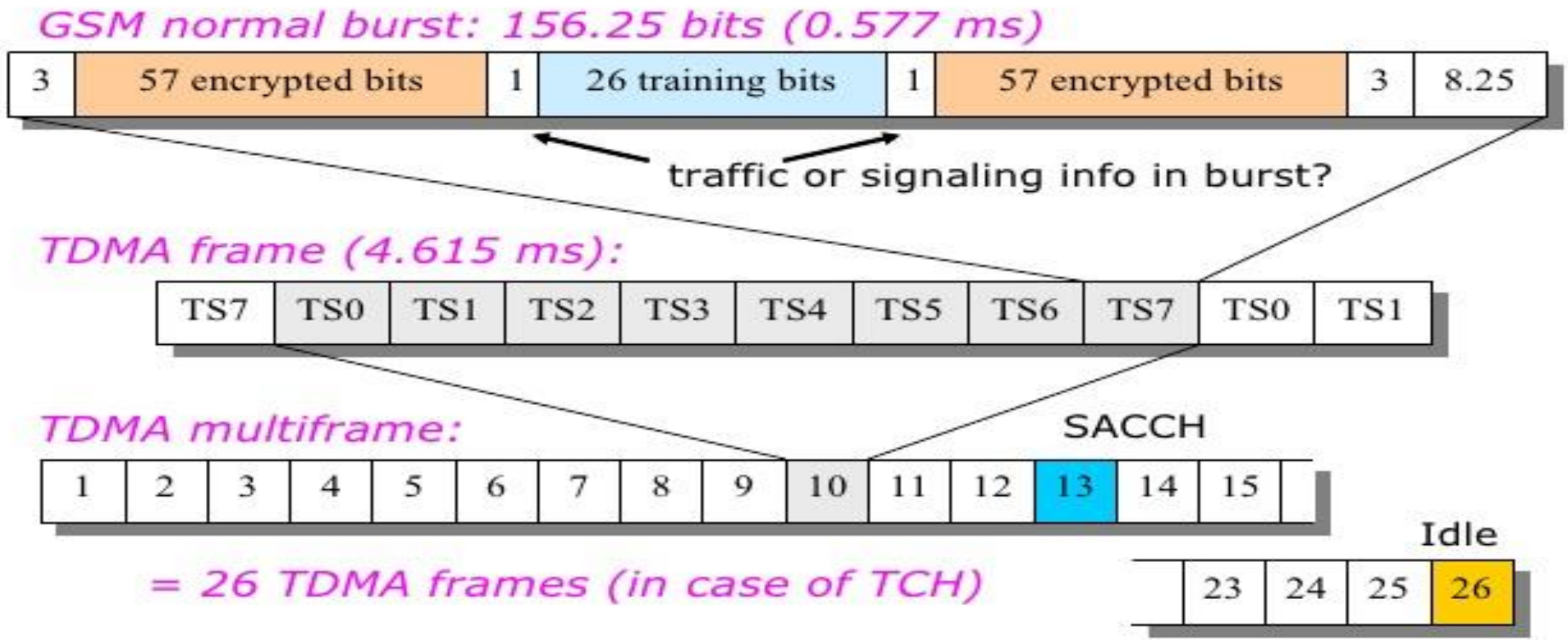
- The information carried in one *timeslot* is called a "*burst*".
- The timing of the burst transmissions to and from the mobiles is *critical*.
- Each time slot of a TDMA frame lasts for duration of *156.25* bit periods or
- *576.9 μsec* or *0.576 ms*
- so a frame takes *4.615 ms*
- GSM's data transmission rate is *270.83 kbps* per carrier frequency.
- Therefore one *bit* duration is *3.692 μsec*

Power Time Template



Normal Burst

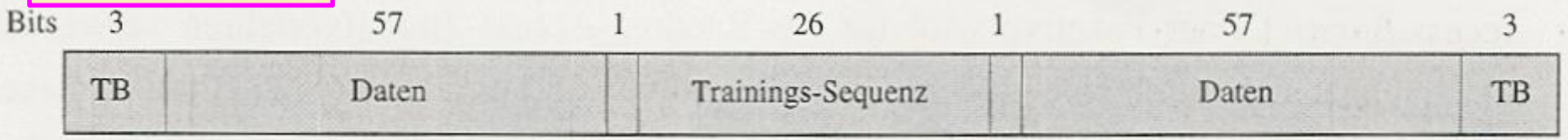
GSM burst structure



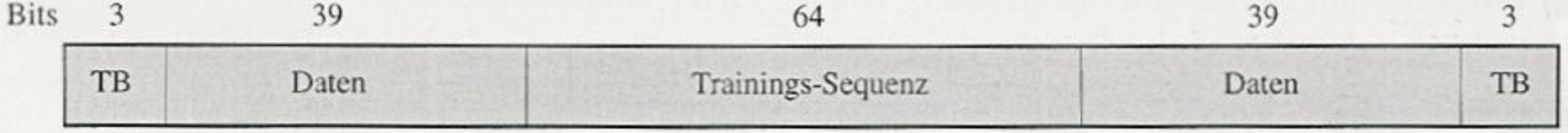
Burst structure

- As it has been stated before, the burst is the unit in time of a TDMA system. Four different types of bursts can be distinguished in GSM:
- The frequency-correction burst is used on the FCCH. It has the same length as the normal burst but a different structure.
- The synchronization burst is used on the SCH. It has the same length as the normal burst but a different structure.
- The random access burst is used on the RACH and is shorter than the normal burst.
- The normal burst is used to carry speech or data information. It lasts approximately 0.577 ms and has a length of 156.25 bits.

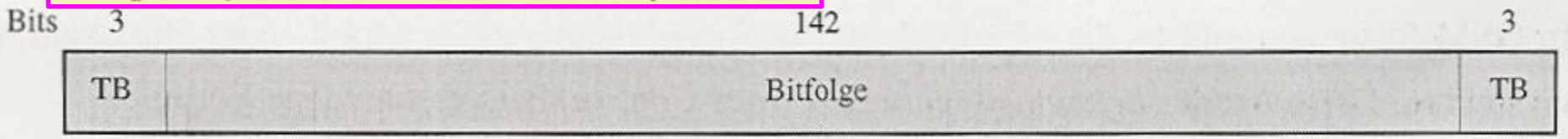
Normal Burst



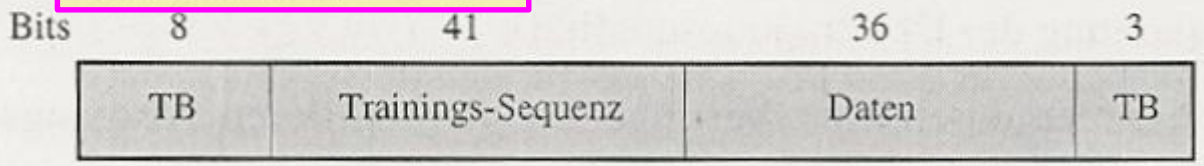
Synchronisation Burst



Frequency Correction bzw. Dummy Burst



Access Burst



TB = Trail Bits

Abb. 33: GSM-Frameformat

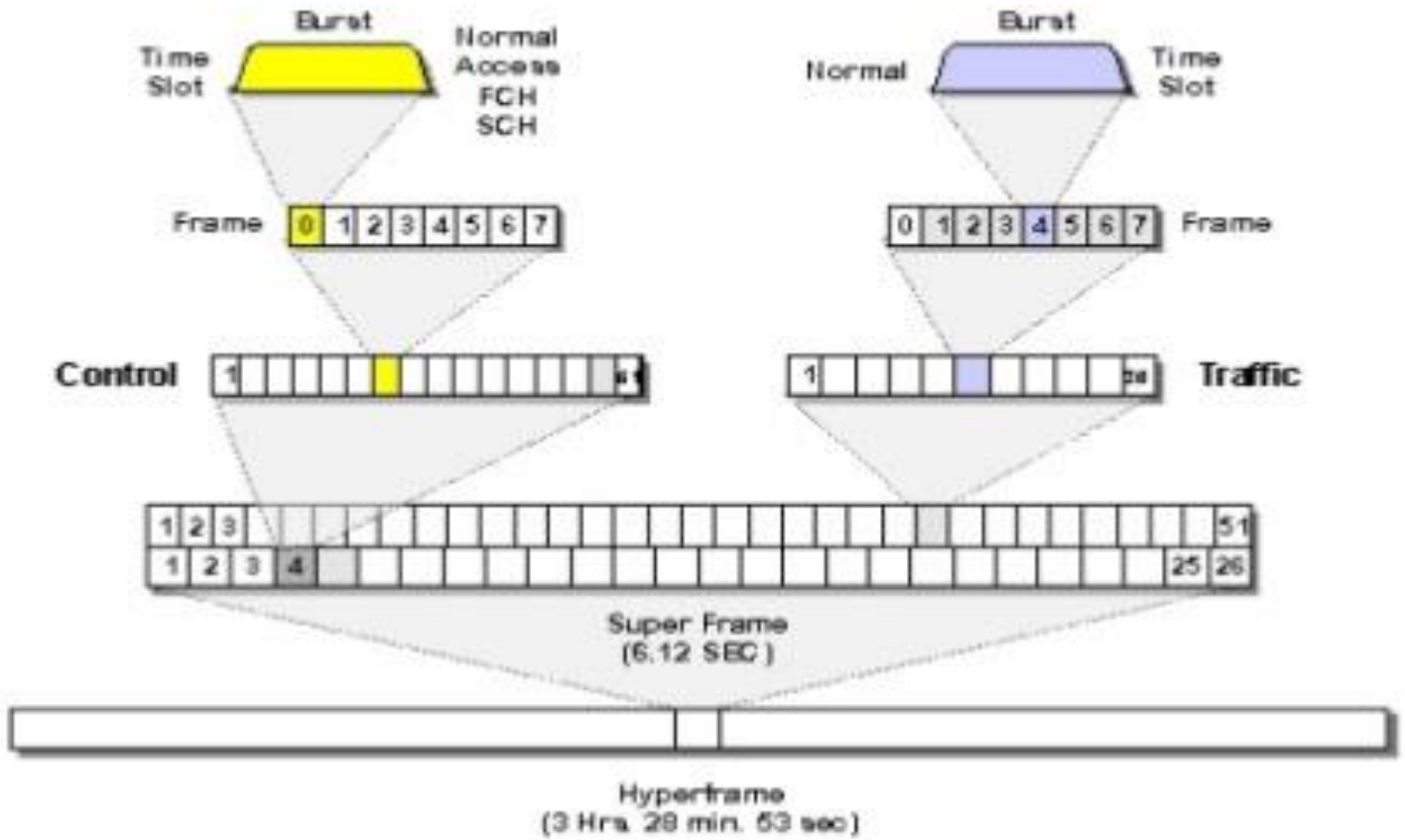
INTERLEAVING

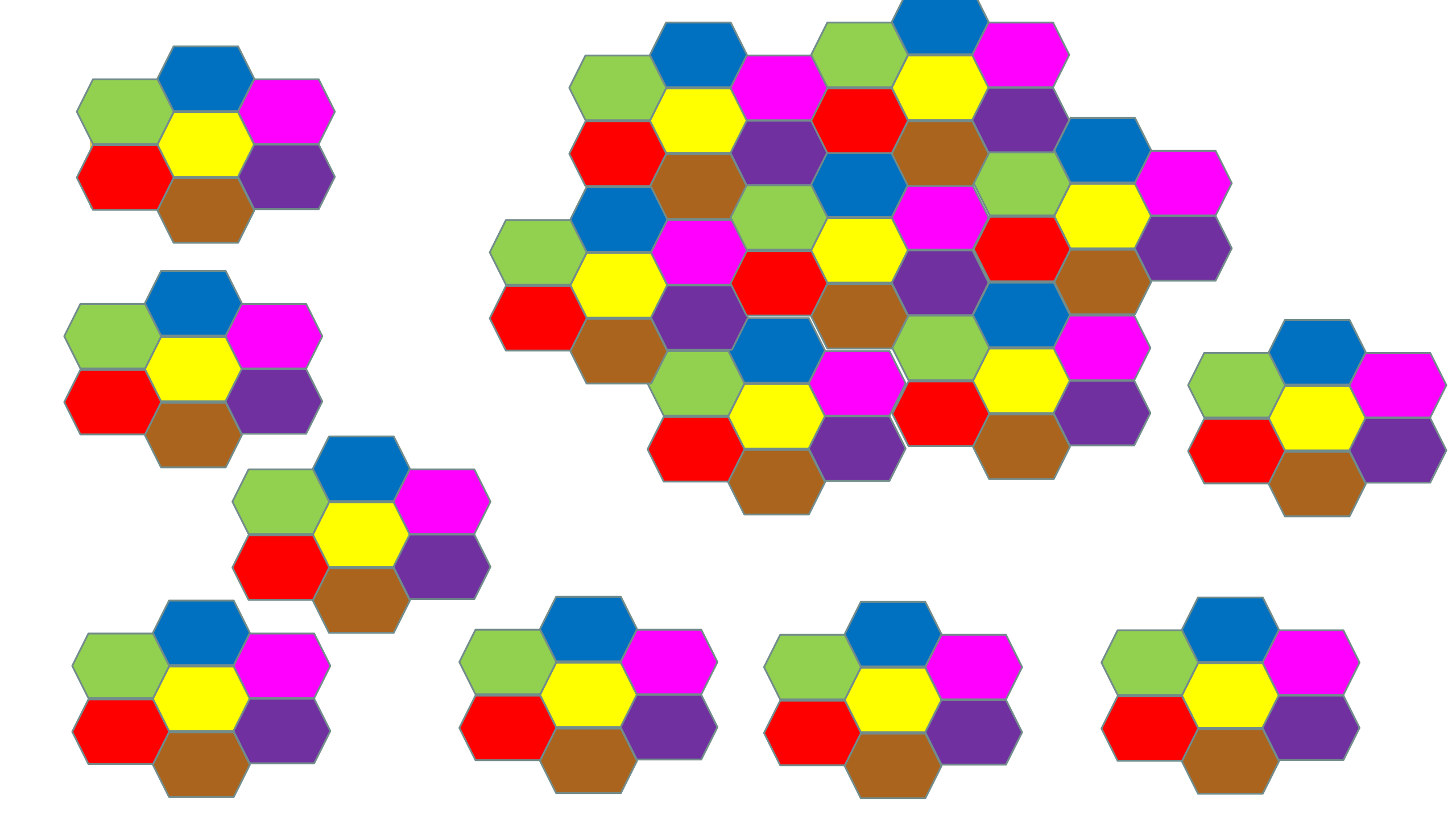
INTERLEAVING

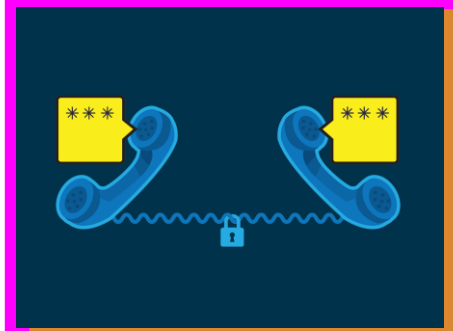
- An interleaving rearranges a group of bits in a particular way.
- It is used in combination with FEC codes(Forward Error Correction Codes) in order to improve the performance of the error correction mechanisms.
- The interleaving decreases the possibility of losing whole bursts during the transmission, by dispersing the errors.
- As the errors are less concentrated, it is then easier to correct them.

Frame Structure

- **GSM Frame**
  - 4.615 msec
- **Multiframes**
  - Traffic
  - Control
- **Superframes**



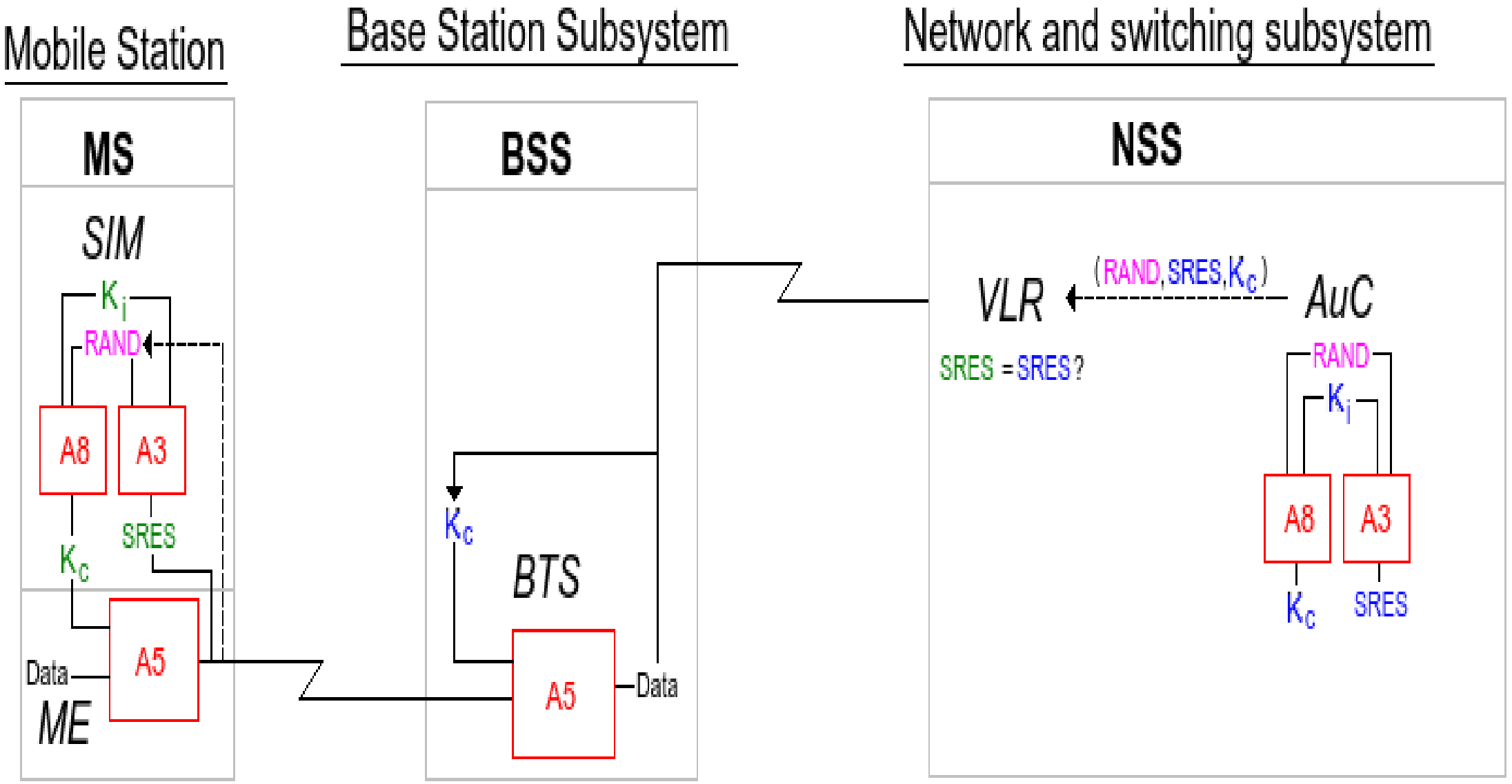




DATA ENCRYPTION IN GSM

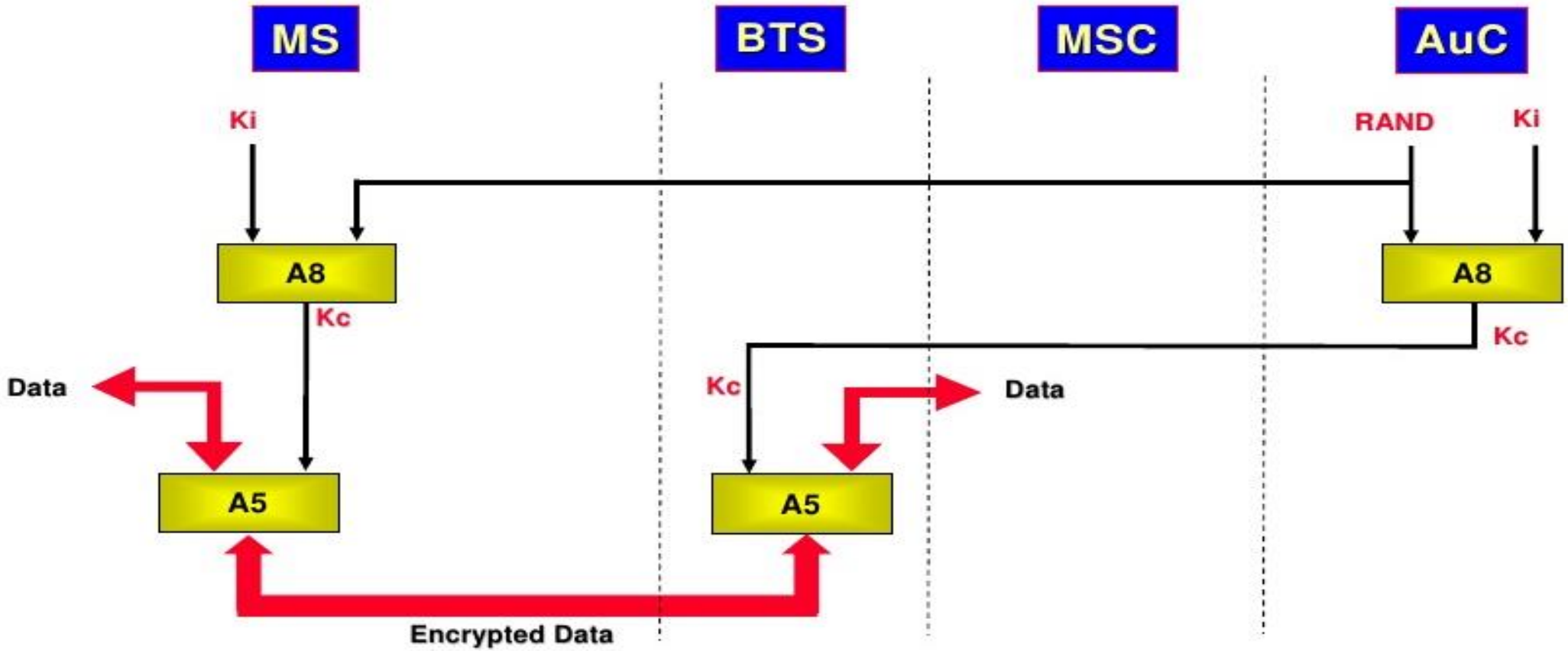
THUS GSM SECURITY FEATURES AS FOLLOWS :

- On air interface, GSM uses encryption and TMSI instead of IMSI.
- SIM is provided 4-8 digit PIN to validate the ownership of SIM
- 3 algorithms are specified :
  - A3 algorithm for authentication
  - A5 algorithm for encryption
  - A8 algorithm for key generation


Mobile Station
Base Station Subsystem
Network and switching subsystem
MS
SIM
Ki
RAND
A8
A3
SRES
Kc
A5
Data
ME
BSS
Kc
BTS
A5
Data
NSS
VLR
(RAND,SRES,Kc)
AuC
SRES =SRES?
RAND
Ki
A8
A3
Kc
SRES

- TO ENSURE PRIVACY, ALL MESSAGES CONTAINING USER RELATED INFORMATION ARE ENCRYPTED IN GSM
- THE CIPHER KEY ‘Kc’ IS GENERATED USING THE INDIVIDUAL KEY ‘Ki’ AND A RANDOM VALUE BY APPLYING THE ALGORITHM ‘A8’
- THE CIPHER KEY ‘Kc’ IS NOT TRANSMITTED OVER THE AIR INTERFACE
- MOBILE AND BTS THEN ENCRYPT AND DECRYPT DATA USING THE ALGORITHM ‘A5’ AND THE CIPHER KEY ‘Kc’
- ‘Kc’ IS 64 BIT KEY – WHICH IS NOT VERY STRONG BUT PROVIDES GOOD PROTECTION AGAINST SIMPLE EAVESDROPPING

General GSM Encryption Procedure

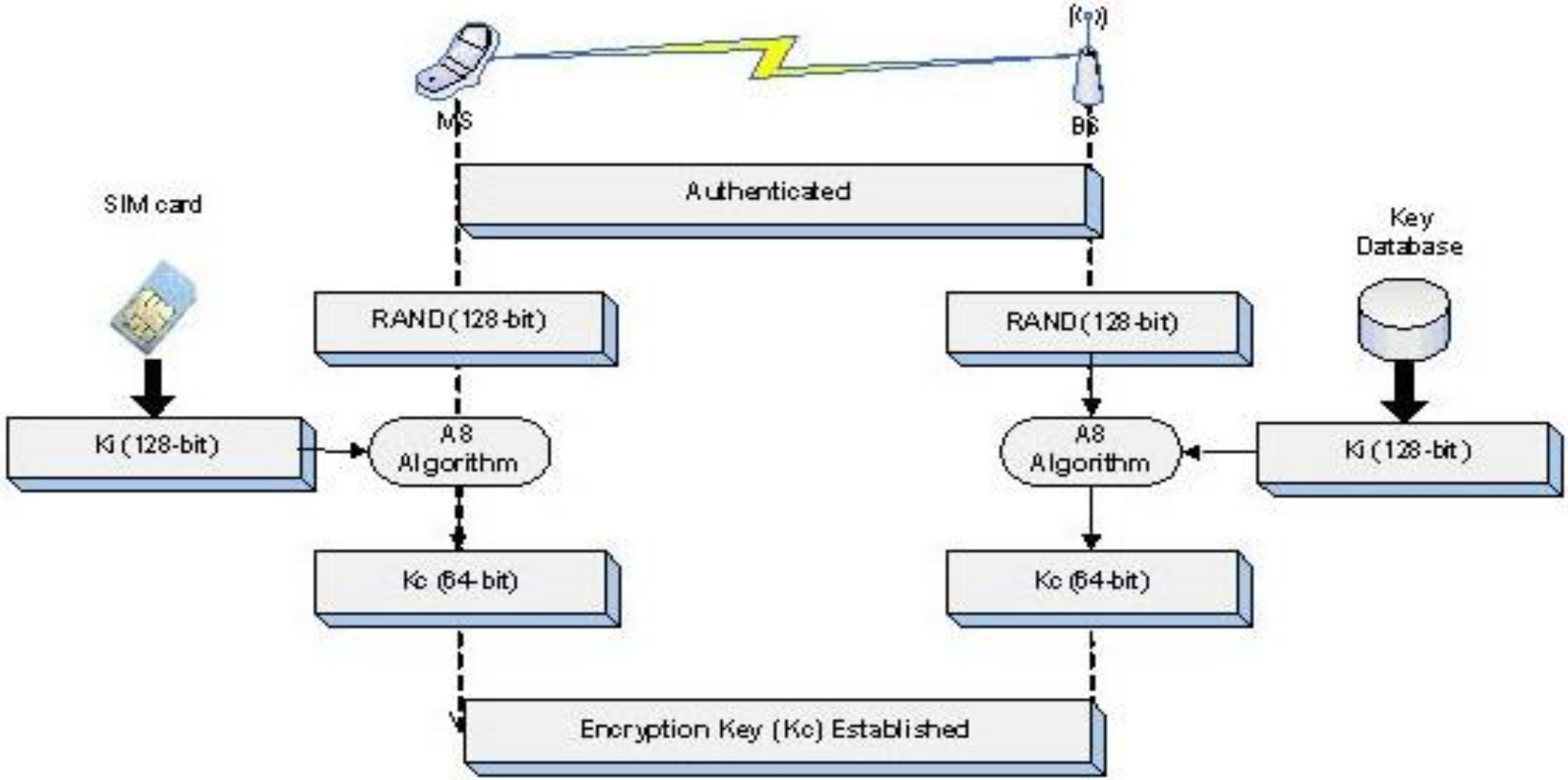


Confidentiality (Encryption)

- To ensure privacy .

  All user-related data is encrypted. After authentication, BTS (base transceiver station) and MS apply encryption to voice, data, and signaling by applying the cipher key Kc .

- Kc is generated using the individual key Ki and a random value by applying the algorithm A8.
- This confidentiality exists only between MS and BTS, but it does not exist end-to-end or within the whole GSM network.


MS
BS
Authenticated
SIM card
Key
Database
RAND (128-bit)
RAND (128-bit)
Ki (128-bit)
A8
Algorithm
A8
Algorithm
Ki (128-bit)
Kc (64-bit)
Kc (64-bit)
Encryption Key (Kc) Established

Security plays a very important part in wireless communication systems than in systems that use wired communication. This is mainly because of the ubiquitous nature of the wireless medium that makes it more susceptible to security attacks than wired communications.

In the wireless medium, any eavesdropper can get over to whatever is being sent over the network.

The presence of communication does not uniquely identify the originator. To make things worse, any tapping or eavesdropping cannot even be detected in a medium as ubiquitous as the wireless medium. **Thus security plays a vital role for the successful operation of a mobile communication system**.

In GSM, A5 stream cipher is used. Versions A5/1 and A5/2 were kept secret for a long period of time. A5/1 and A5/2 were reverse-engineered from a GSM handset. After which attacks were quickly found for these algorithms.

The primary problem is the small key length of the session key Kc. The actual length of Kc is 64 bits.

However, the last 10 bits of this key are specified to be 0 thus reducing the effective key size to 54 bits.

Even though this key size is big enough to protect against real-time attacks (decrypting packets being transmitted in real-time), the state of the hardware available today makes it possible to record the packets between the MS and the BTS and then decode them at a later time

A known-key stream attack on A5/1 requiring about two second of the key stream and recovers Kc in a few minutes on a personal computer after a somewhat large preprocessing stage.

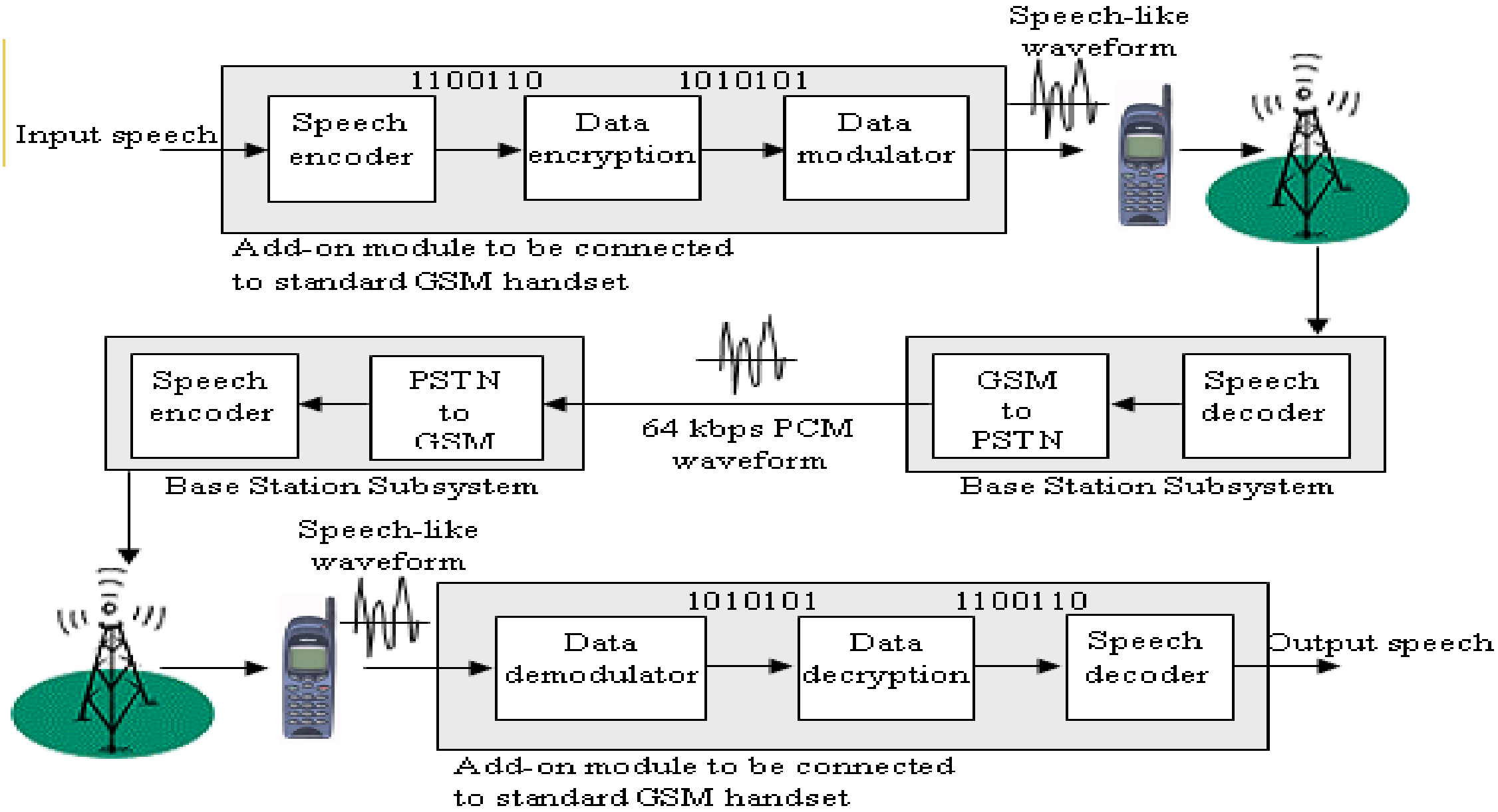
A cipher text-only attack on A5/1 that also recovers Kc using only four frames, but with a relative high complexity. A5/2 was also cracked and proved to be completely insecure. The attack required very few pseudo random hits and only 216 steps

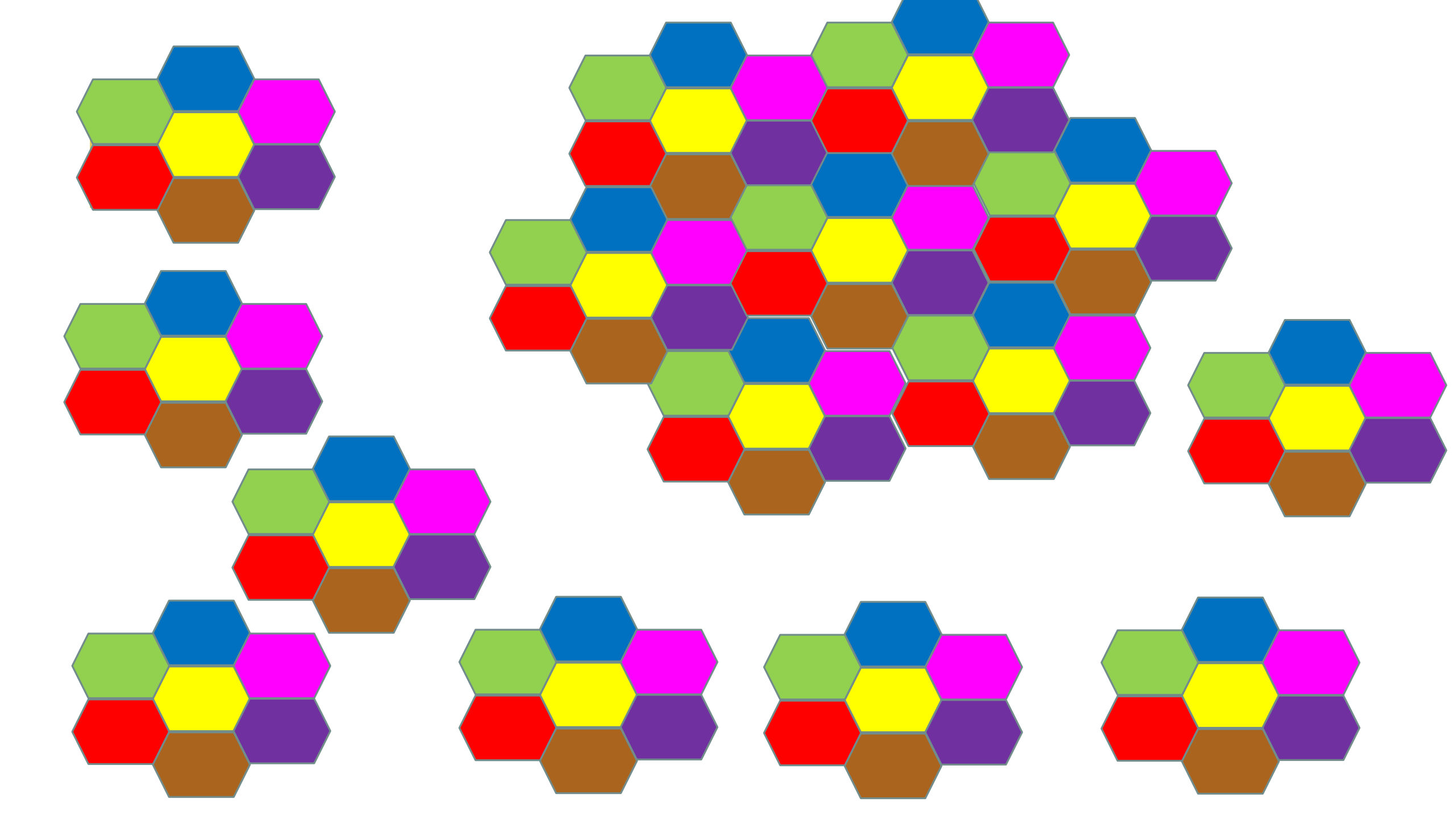
A new security algorithm, known as A5/3 provides users of GSM mobile phones with an even higher level of protection against eavesdropping than they have already .

A5/3 is based on the Kasumi algorithm, specified by 3GPP for use in 3G mobile systems.

The A5/3 encryption algorithm specifically supplies signaling protection, so that sensitive information such as telephone numbers is protected over the radio path, and user data protection, to protect voice calls and other user generated data passing over the radio path .

The algorithm is so far believed to be stronger than A5/1 and A5/2 but…….


Input speech
Speech encoder
1100110
Data encryption
1010101
Data modulator
Add-on module to be connected to standard GSM handset
Speech-like waveform
Speech encoder
PSTN to GSM
Base Station Subsystem
64 kbps PCM waveform
GSM to PSTN
Speech decoder
Base Station Subsystem
Speech-like waveform
1010101
Data demodulator
1100110
Data decryption
Speech decoder
Output speech
Add-on module to be connected to standard GSM handset



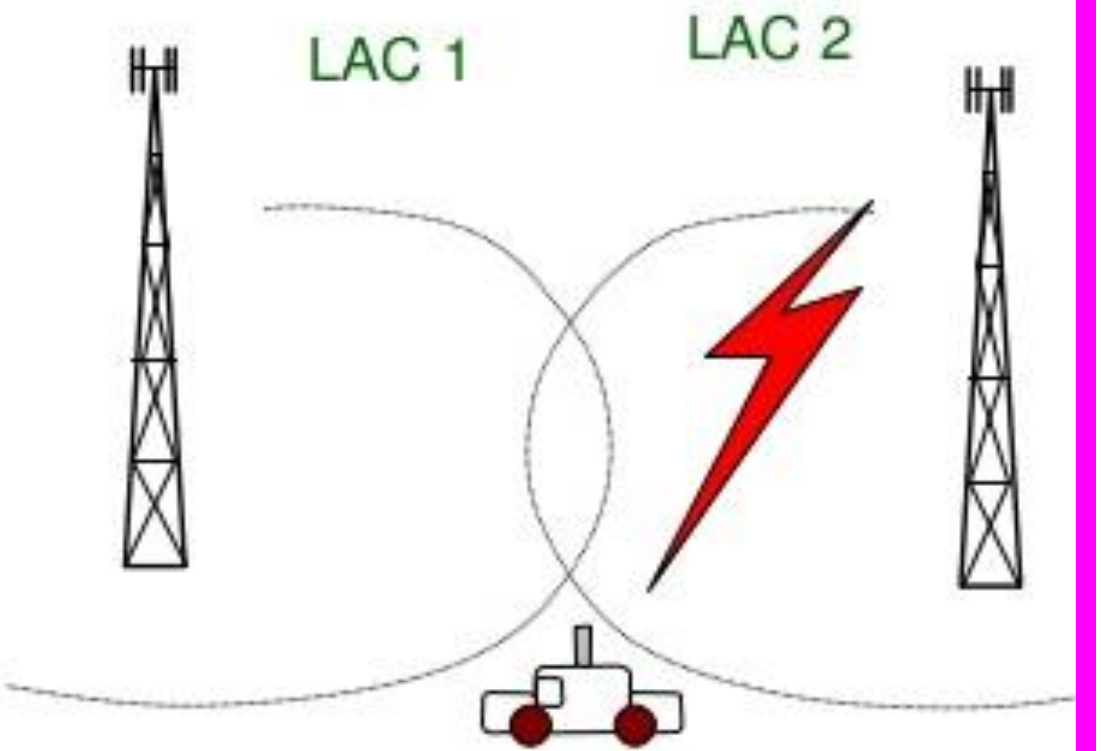
MOBILITY MANAGEMENT

- Mobility Management (MM) function handles the procedures that arise from the mobility of the subscriber and is in charge of all the aspects related to the mobility of the user, especially the roaming, the location management, and the security/authentication of the subscriber.

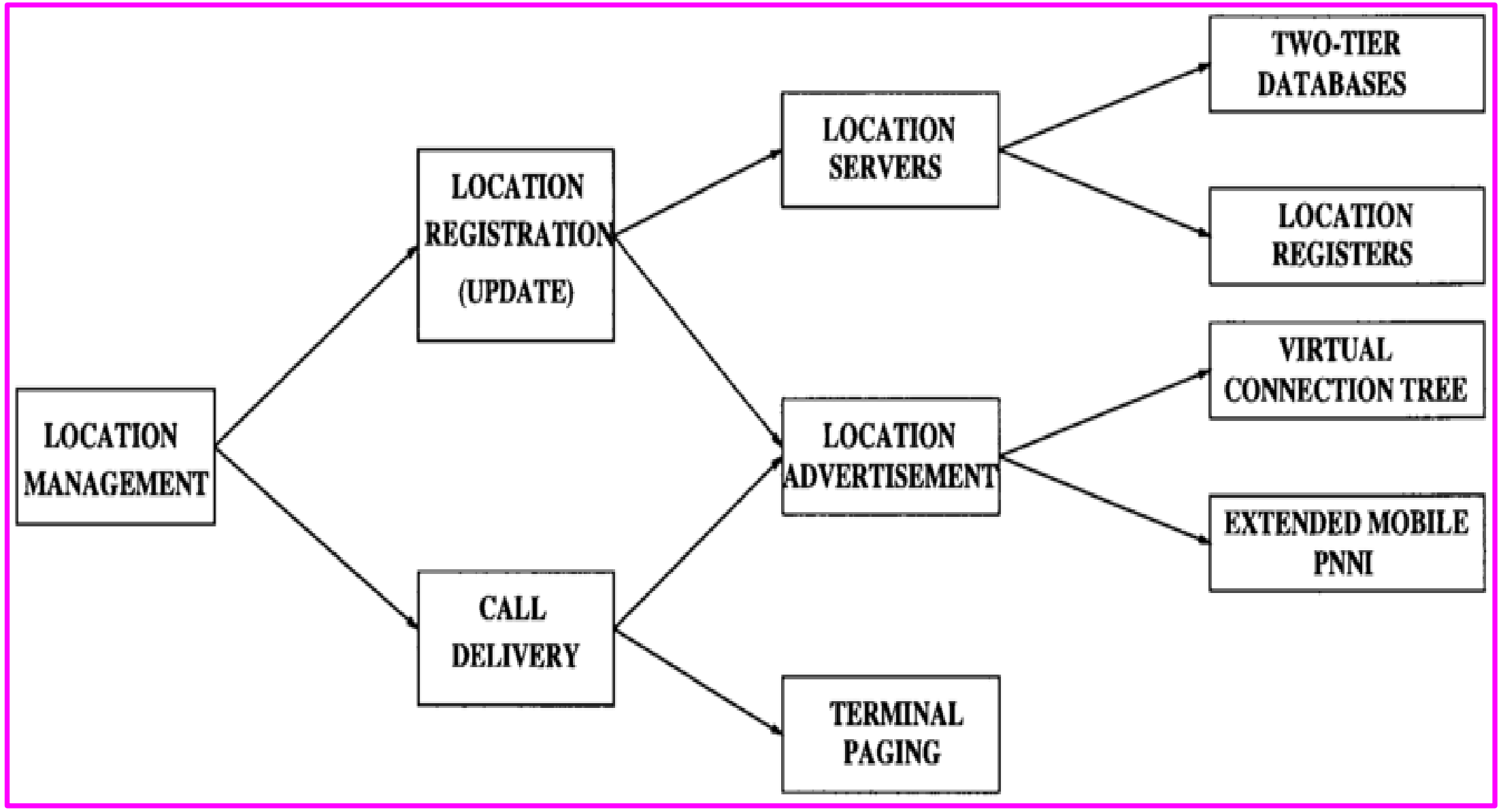
- First location update procedure is called the IMSI attach procedure where the MS indicates its IMSI to the network whereas when a mobile station is powered off, it performs an IMSI detach procedure in order to tell the network that it is no longer connected.

Mobility Management Messages

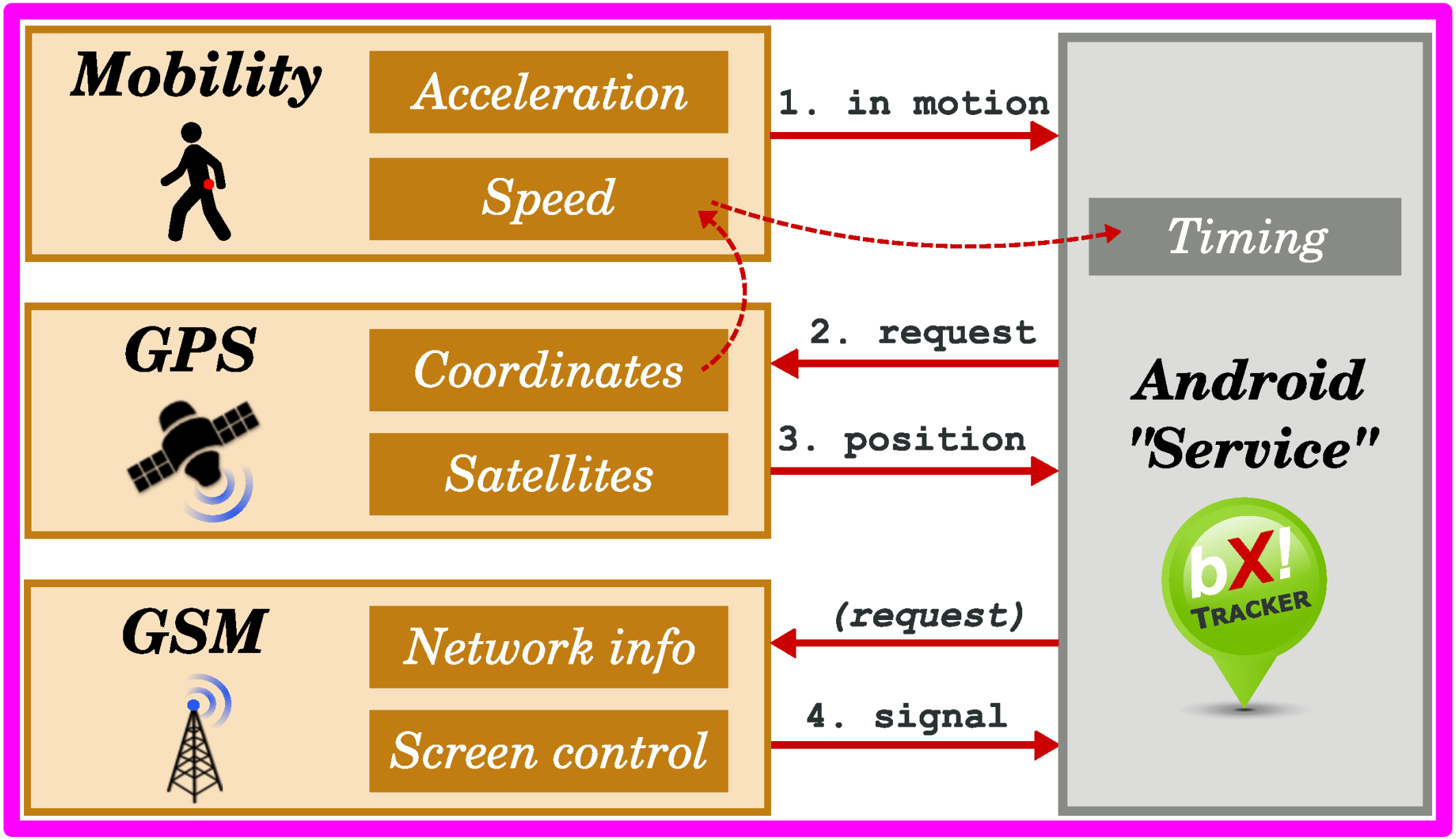
- **Registration Messages**
  - IMSI Detach
  - Location Updating Request
  - Location Updating Accept
  - Location Updating Reject
- **Security Messages**
  - Authentication Request
  - Authentication Response
  - Authentication Reject
  - Identity Request
  - Identity Response
  - TMSI Reallocation Command
  - TMSI Reallocation Complete



- **Mobility Types**
- Track location of users for incoming calls/SMS
  - Location registration/authentication/paging
  - Divide coverage area into non-overlapping groups of cells – assign each a unique id
  - Location Area ID periodically broadcast by each cell
  - As a mobile moves/turns phone on – it listens to location area id – if different from last one registered in – performs a location update/authentication procedure with VLR and possibly HLR
- Call in progress mobility
  - Handoff call from one BTS to another BTS
  - MAHO by mobile reporting measurements of signal strength

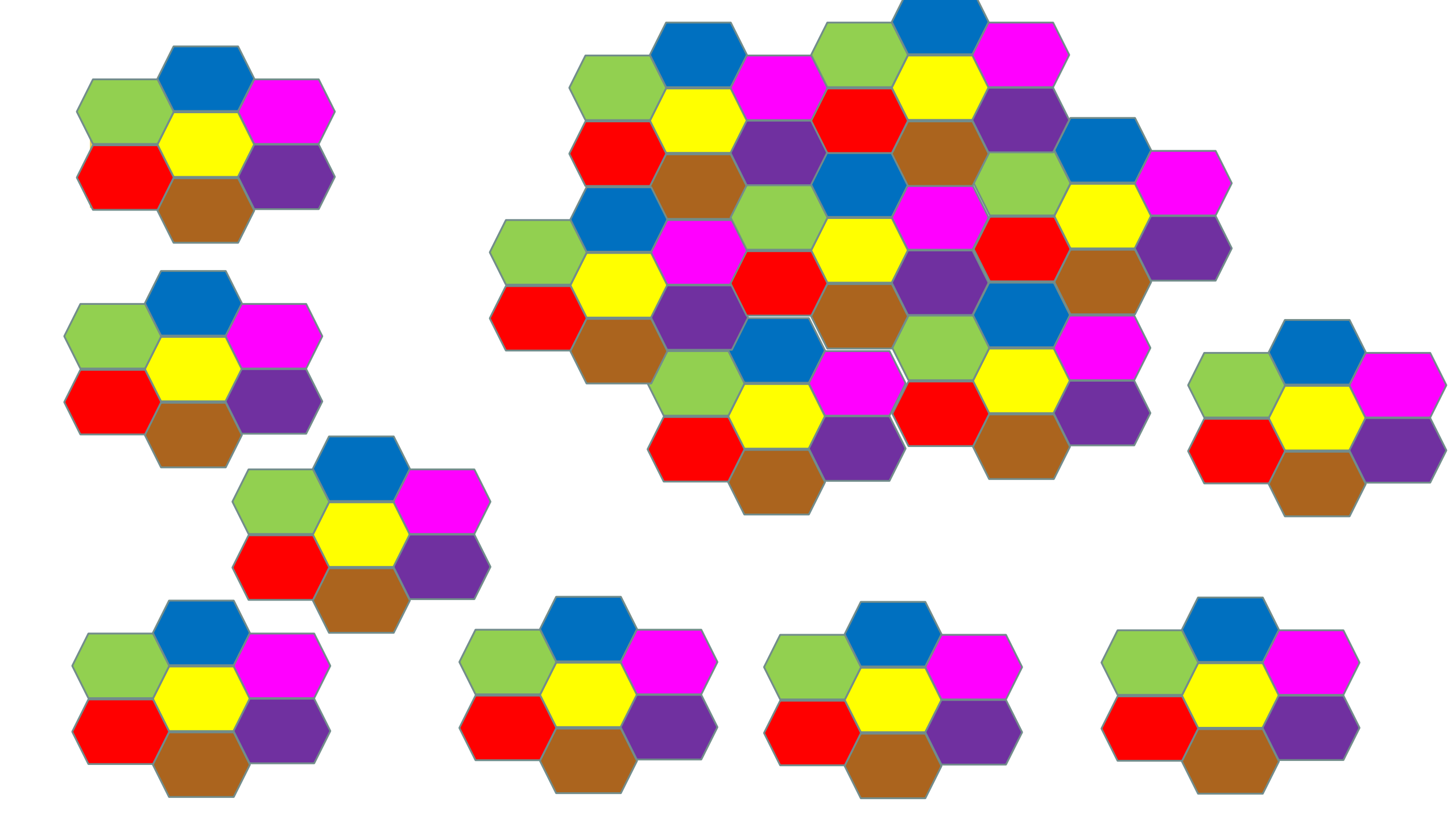

LOCATION MANAGEMENT
LOCATION REGISTRATION (UPDATE)
CALL DELIVERY
LOCATION SERVERS
LOCATION ADVERTISEMENT
TERMINAL PAGING
TWO-TIER DATABASES
LOCATION REGISTERS
VIRTUAL CONNECTION TREE
EXTENDED MOBILE PNNI

- Mobility management is essential for keeping track of user's current location
- Many schemes proposed for cellular networks
- Next-generation wireless/mobile network will be unified networks based on IP technology
- Design of IP-based mobility management schemes has become necessary


Mobility
Acceleration
Speed
GPS
Coordinates
Satellites
GSM
Network info
Screen control
1. in motion
2. request
3. position
(request)
4. signal
Timing
Android
"Service"
bX!
TRACKER

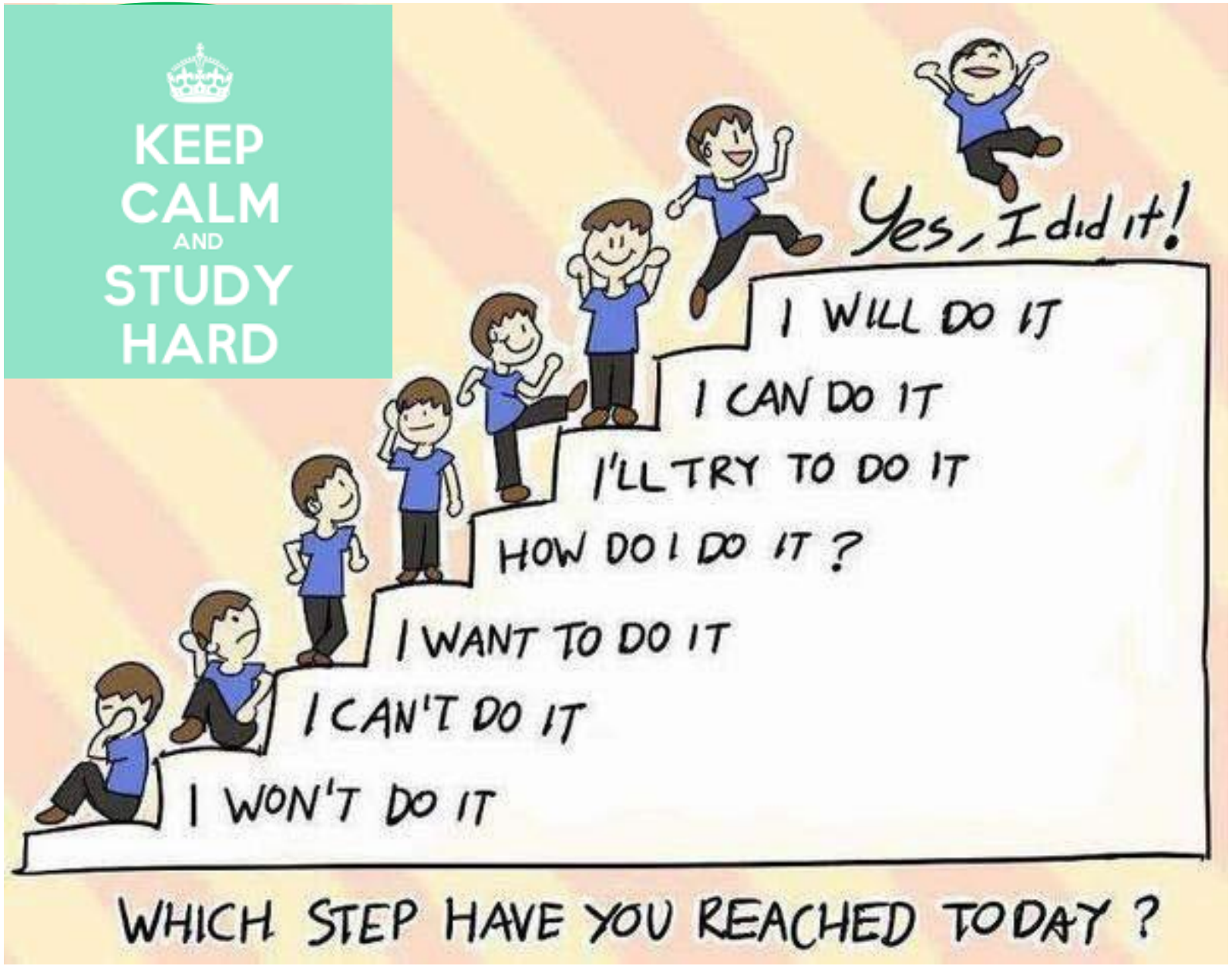
GSM Protocols

- CM - Connection Management
- MM - Mobility Management
- RR - Radio resource
- LAPDm - LAPD for mobile
- LAPD - Link Access Procedure for D channel
- BTSM - BTS Management Part
- BSSAP - BSS Application Part (BSC - MSC)
- DTAP - Direct Transfer Application Part (MS - MSC)
- MAP - Mobile Application Part
- MTP - Message Transfer part of SS7
- SCCP - Signalling Connection Control Part of SS7
- TCAP - Transaction Capabilities Application Part
- ISUP - ISDN User Part

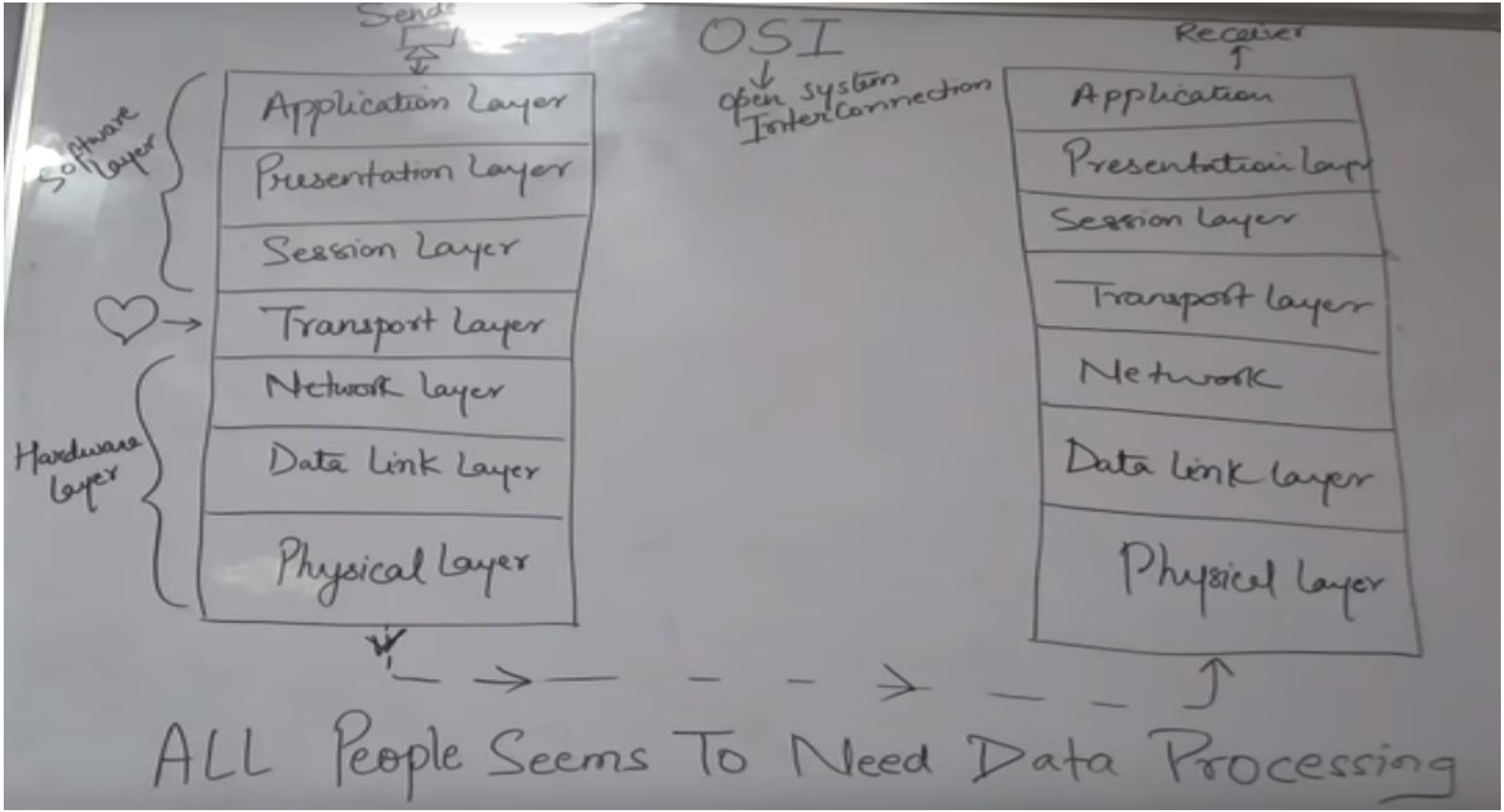


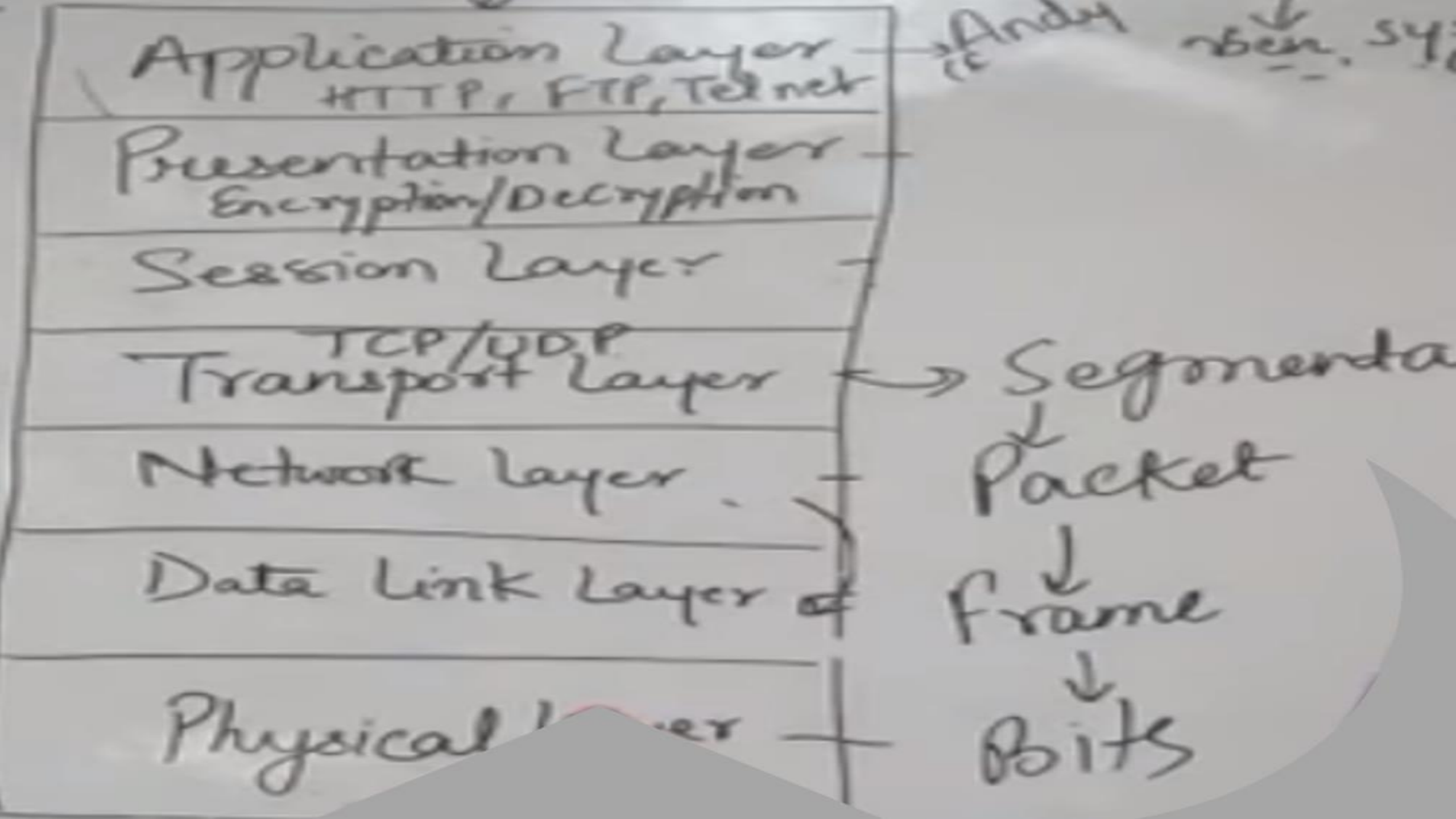


Further slides are for reference



Global Systems for Mobile : System Architecture, GSM frequency bands, GSM PLMN, GSM subsystems, GSM interface, Mapping of GSM Layer on to OSI Layers, GSM Logical Channel and Frame, Structure, GSM Burst, Data encryption in GSM, Mobility management.


OSI
Open system Interconnection
Sender
Receiver
Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer
Physical Layer
Application
Presentation layer
Session Layer
Transport layer
Network
Data link layer
Physical layer
Software layer
Hardware layer
ALL People Seems To Need Data Processing


Application Layer
HTTP, FTP, Telnet
Presentation Layer
Encryption/Decryption
Session Layer
TCP/UDP
Transport Layer
Network Layer
Data Link Layer
Physical Layer
Segments
Packet
Frame
Bits

Circuit switching is a type of switching in which a dedicated channel or circuit is established for the duration of communications,

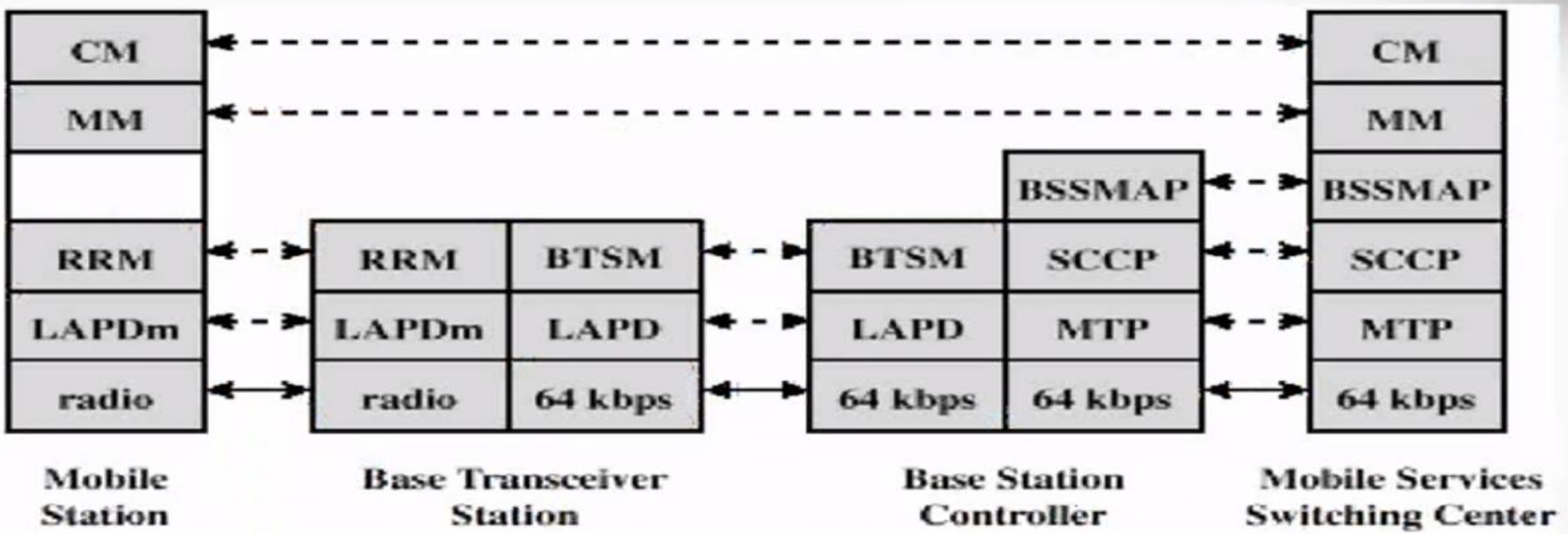
A method used by the old traditional telephone call -carried over the **public switched telephone network** (PSTN)

It is also referred to as the **Plain Old Telephone Service** (POTS).

GSM Air interface

- Layer 1 (GSM Rec. 04.04): Um interface
- Layer 2 (GSM Rec. 04.05/06): LAP-D_m protocol (similar to ISDN LAP-D):
  - connectionless transfer of point-to-point and point-to-multipoint signaling channels
  - Setup and tear-down of layer 2 connections of point-to-point signaling channels
  - connection-oriented transfer with in order delivery, error detection and error correction
- Layer 3 (GSM Rec. 04.07/08) with sublayers for control signaling channel functions (BCH, CCCH and DCCH):
  - Radio resource management (RR): to establish and release stable connection between mobile stations (MS) and an MSC for the duration of a call and to maintain connection despite user movements - functions of MSC:
    - cell selection
    - handover
    - allocation and tear-down of point-to-point channels
    - monitoring and forwarding of radio connections
    - enabling encryption
    - change transmission mode
  - Mobility management (MM) handles the control functions required for mobility:
    - authentication
    - assignment of TMSI,

GSM Protocol Architecture



BSSMAP = BSS mobile application part
BTSM = BTS management
CM = connection management
LAPD = link access protocol, D channel
MM = mobility management
MTP = message transfer part
RRM = radio resources management
SCCP = signal connection control part

GSM Signaling Overview

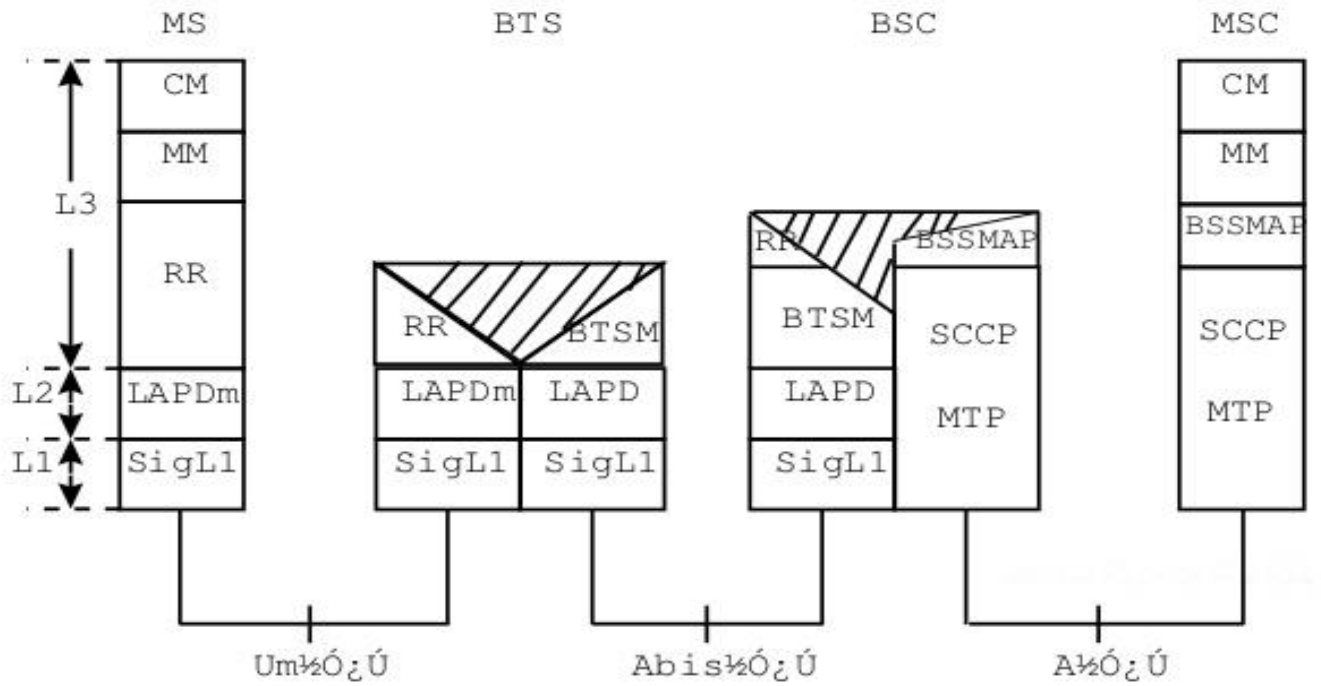
- GSM signaling defines the communications between the mobile and the network
- Signaling has to be carried through the network and across the air-interface to the mobile
- Different protocols are used across different interfaces
- All GSM signaling is based on the Open Systems Interconnect (OSI) model used for computer systems
- The MSC uses ITU's Signaling System No:7, (SS7 or CCS7) for all its signaling; to the BSC, HLR, VLR and to other networks
- The Air-interface and A-bis signaling is based on ISDN Link Access Protocol (LAP-D and LAP-Dm)
- Signaling channels are integrated into the air-interface as the BCCH, PCH, AGCH, SDCCH, SACCH and FACCH
- Signaling across all interfaces is specified in the standards to allow different equipment manufacturers for each unit

Mobile Signaling

- When the mobile is first switched on it is under the control of its' own internal algorithms.
- It performs the following sequence:
  - Search for GSM frequencies
  - Find a FCCH and synchronize frequency
  - Find a SCH and synchronize time
  - Listen for a BCCH and download information about the system
  - Download idle neighbor list and make measurements
  - Select a "best server" and camp on it
  - If the user wants to make a call, the mobile has to signal to the network.
  - It does this by sending a "service request" to the network on the RACH.

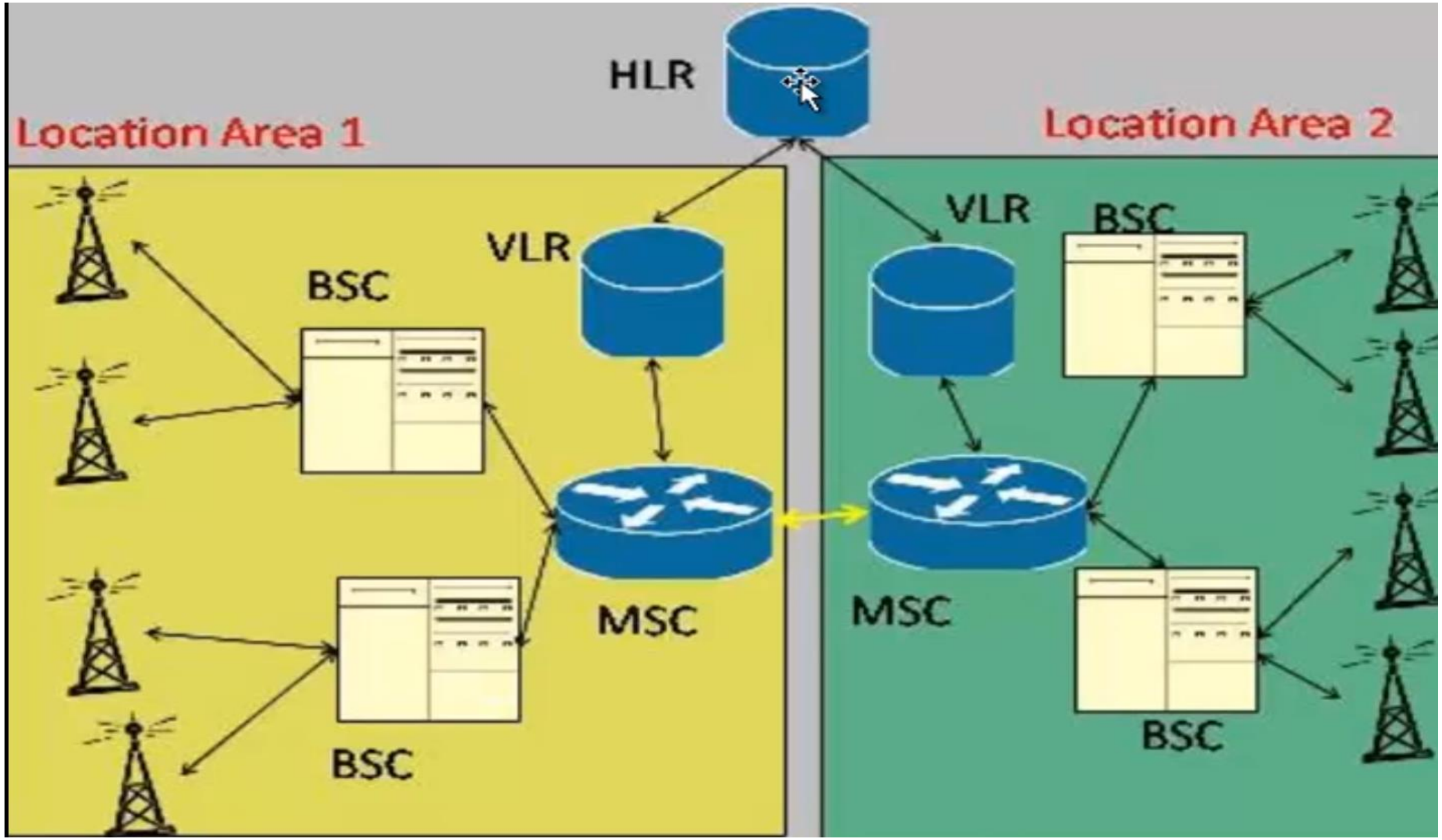


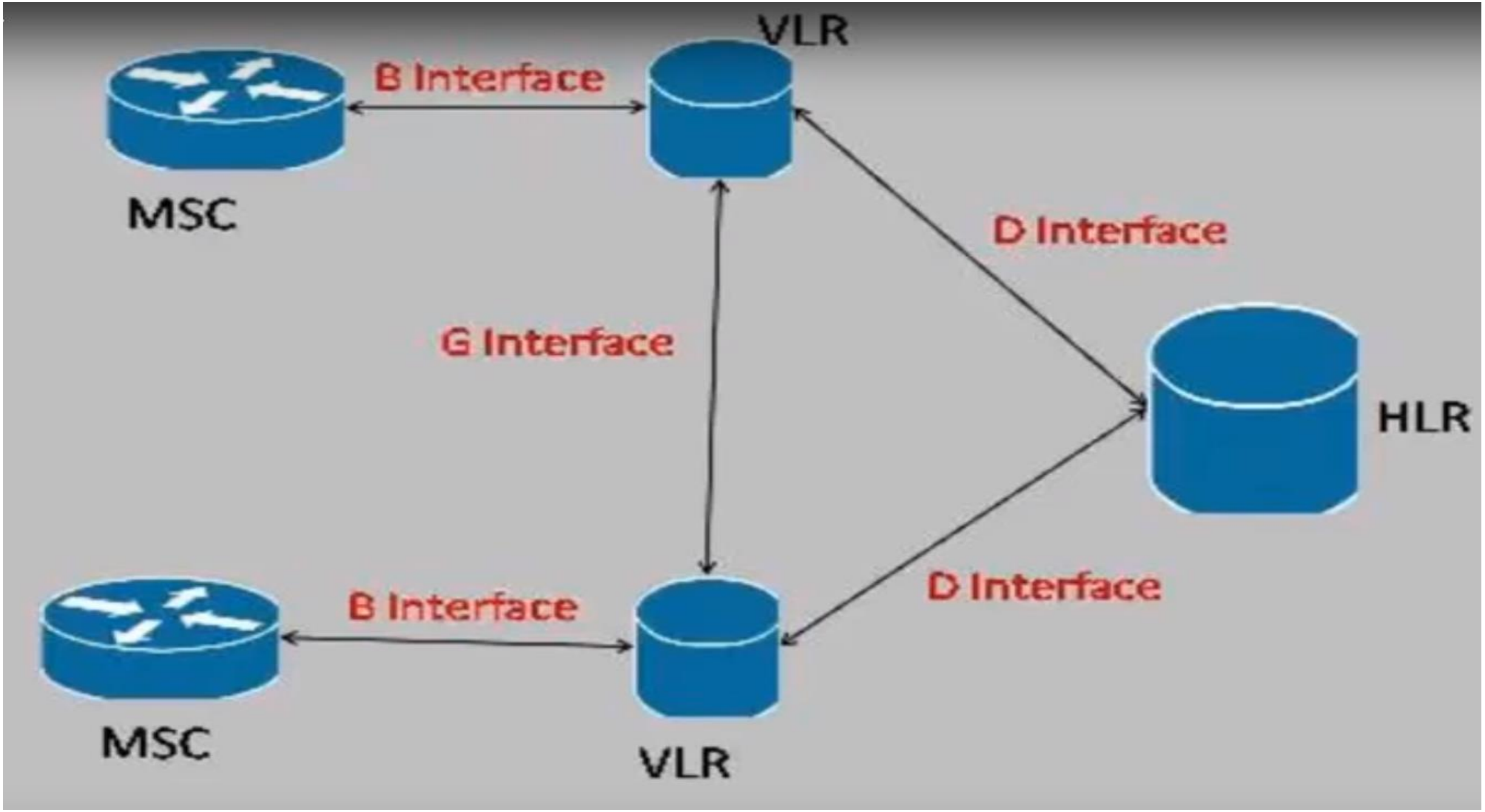
GSM Protocol



CONFIDENTIAL LEVEL

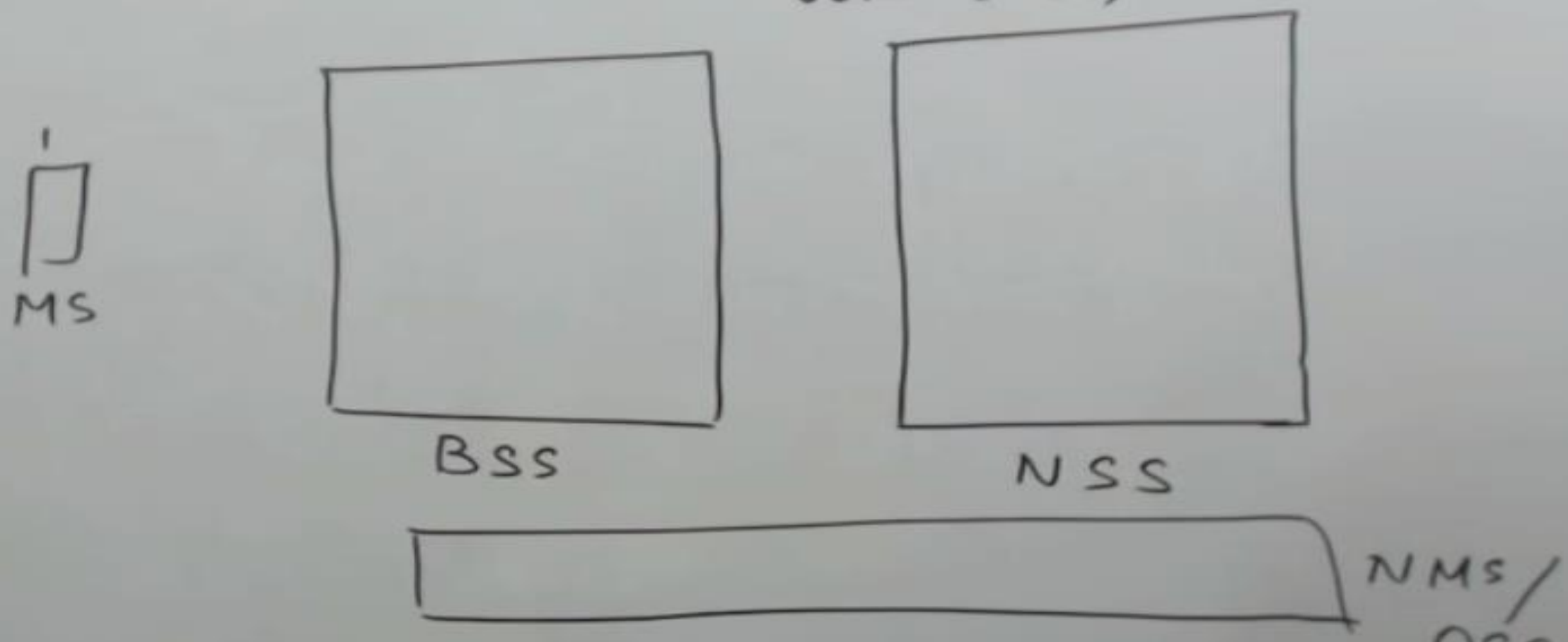
SECURITY LEVEL


HLR
Location Area 1
Location Area 2
VLR
BSC
VLR
BSC
MSC
MSC
BSC
BSC

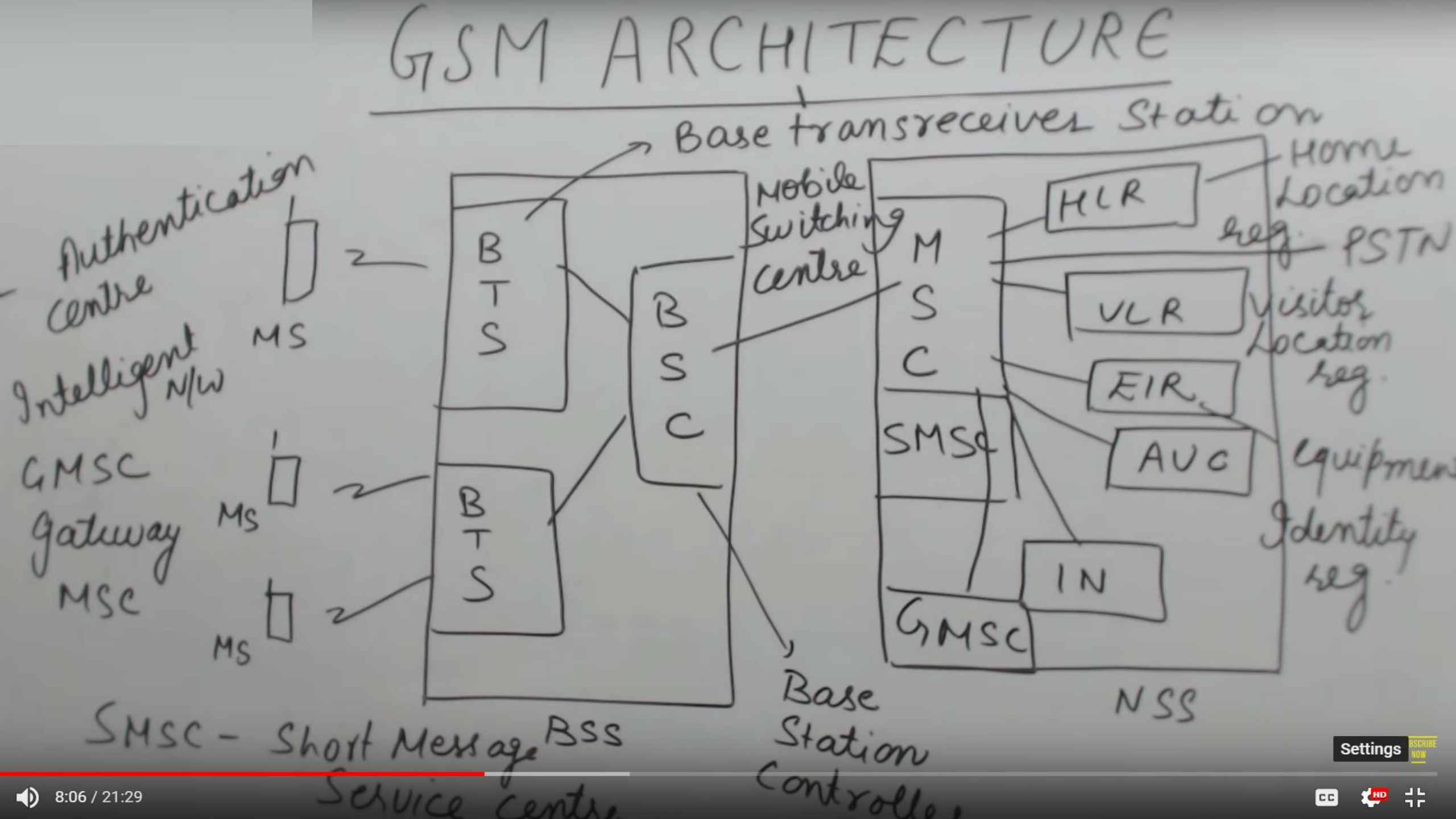

VLR
B Interface
MSC
D Interface
G Interface
HLR
B Interface
D Interface
MSC
VLR

GSM ARCHITECTURE

Voice (2G)

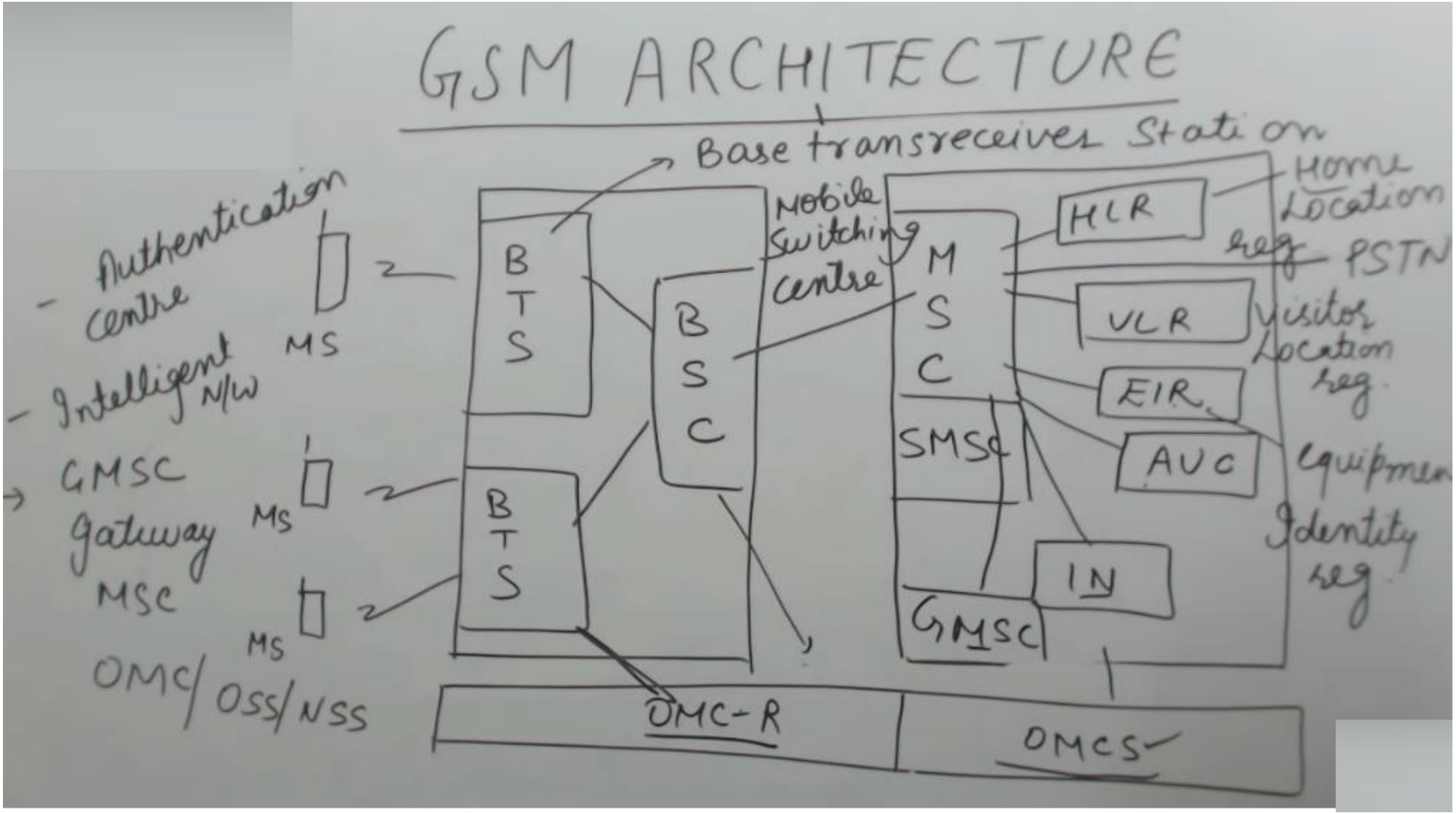


GSM ARCHITECTURE

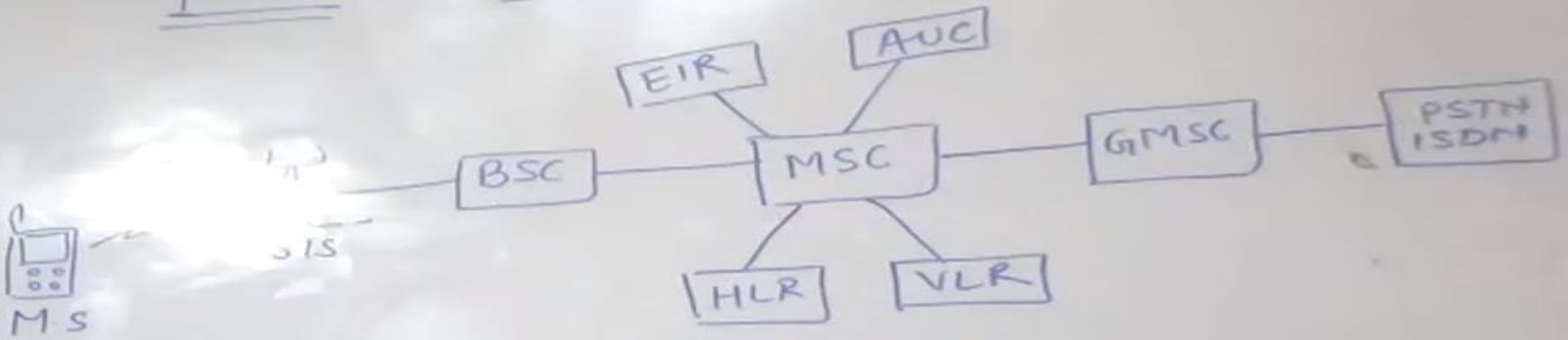


Settings
SUBSCRIBE NOW

GSM ARCHITECTURE



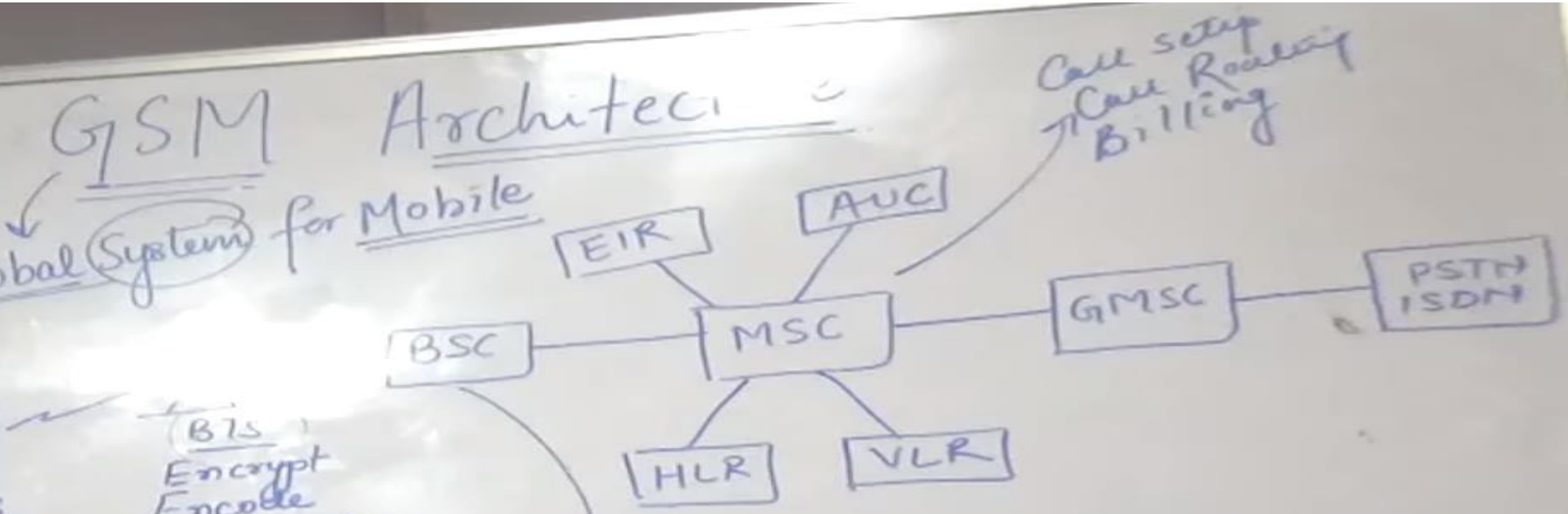
GSM Archi. ure



MS → Mobile Station
BTS → Base Transceiver station
BSC → Base Station Controller
MSC → Mobile Switching Centre
GMSC → Gateway MSC
HLR → Home Location Register
VLR → Visitor Location Register
AuC → Authentication Centre
EIR → Equipment Identity Register

(2G) GSM Architecture

Global System for Mobile



Equipment + SIM

Subscriber Identity Module

M.S

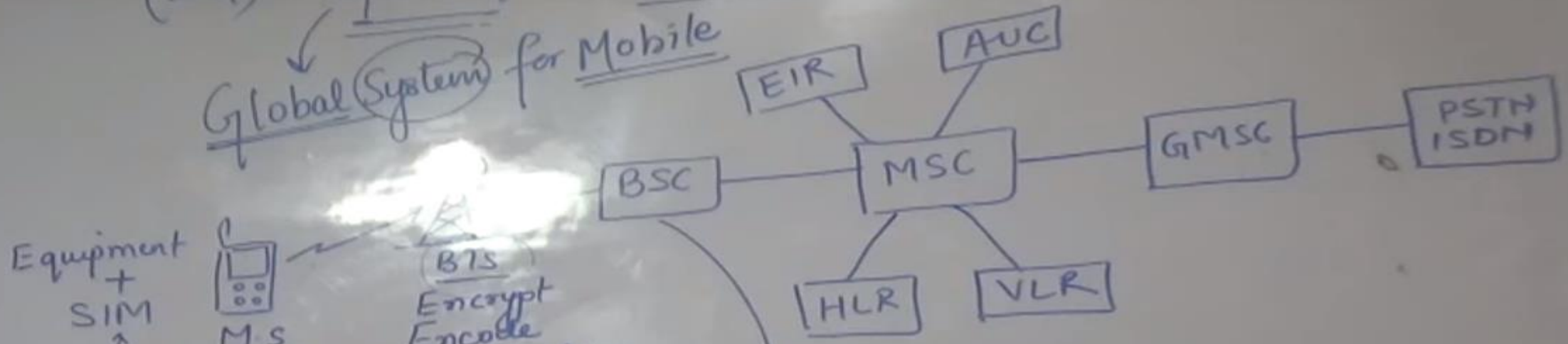
BTS
Encrypt
Encode
Modulation
Frequency hopping

Manages Radio Resources
Handles Call Set up
Handover
Radio Power control

M.S → Mobile Station
BTS → Base Transceiver station
BSC → Base Station Controller
MSC → Mobile Switching Centre
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(2G) GSM Architecture

Global System for Mobile

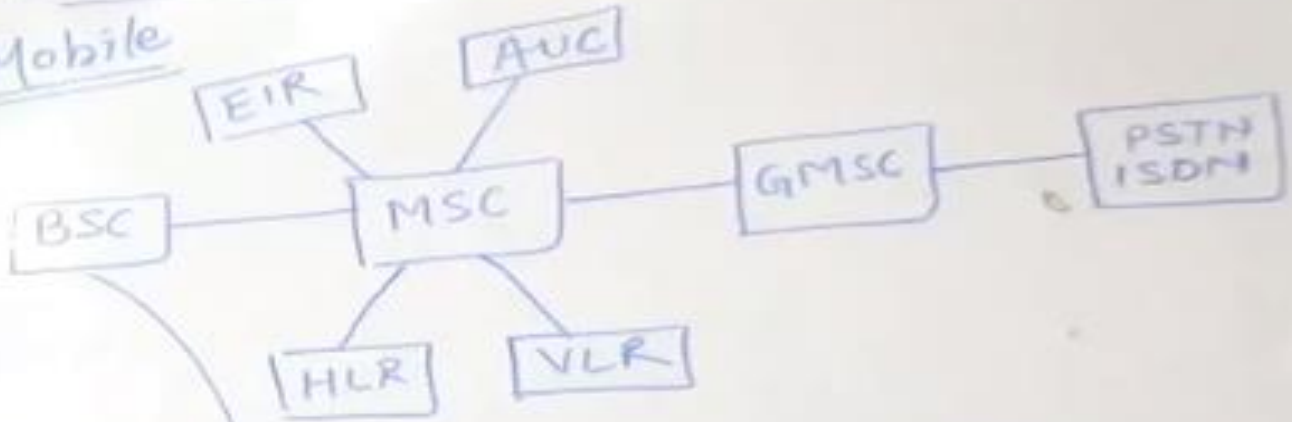


Manages Radio Resources
Handles Call Set up
Handover
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(2G) GSM Architecture

Global System for Mobile



Equipment + SIM

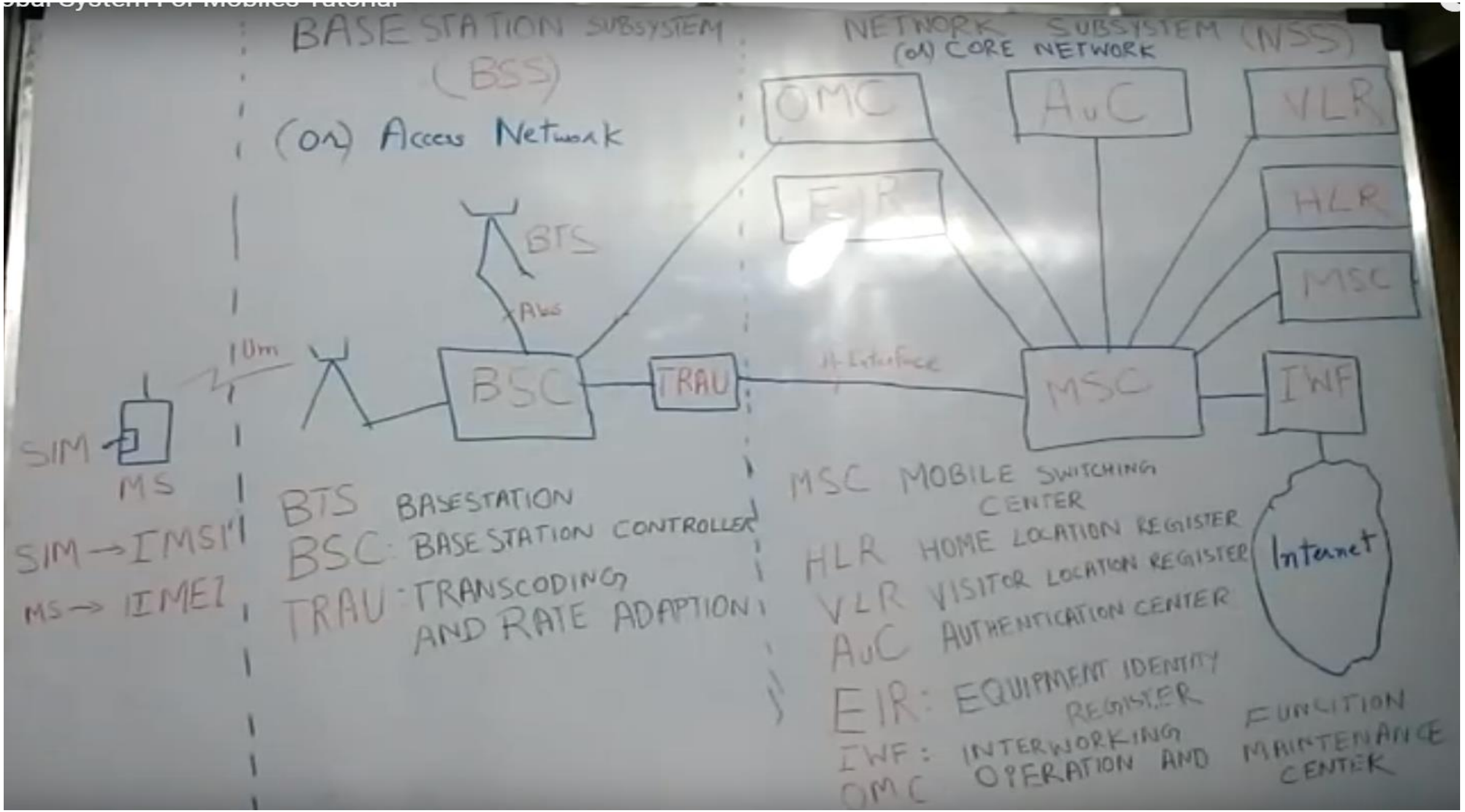
Subscriber Identity Module

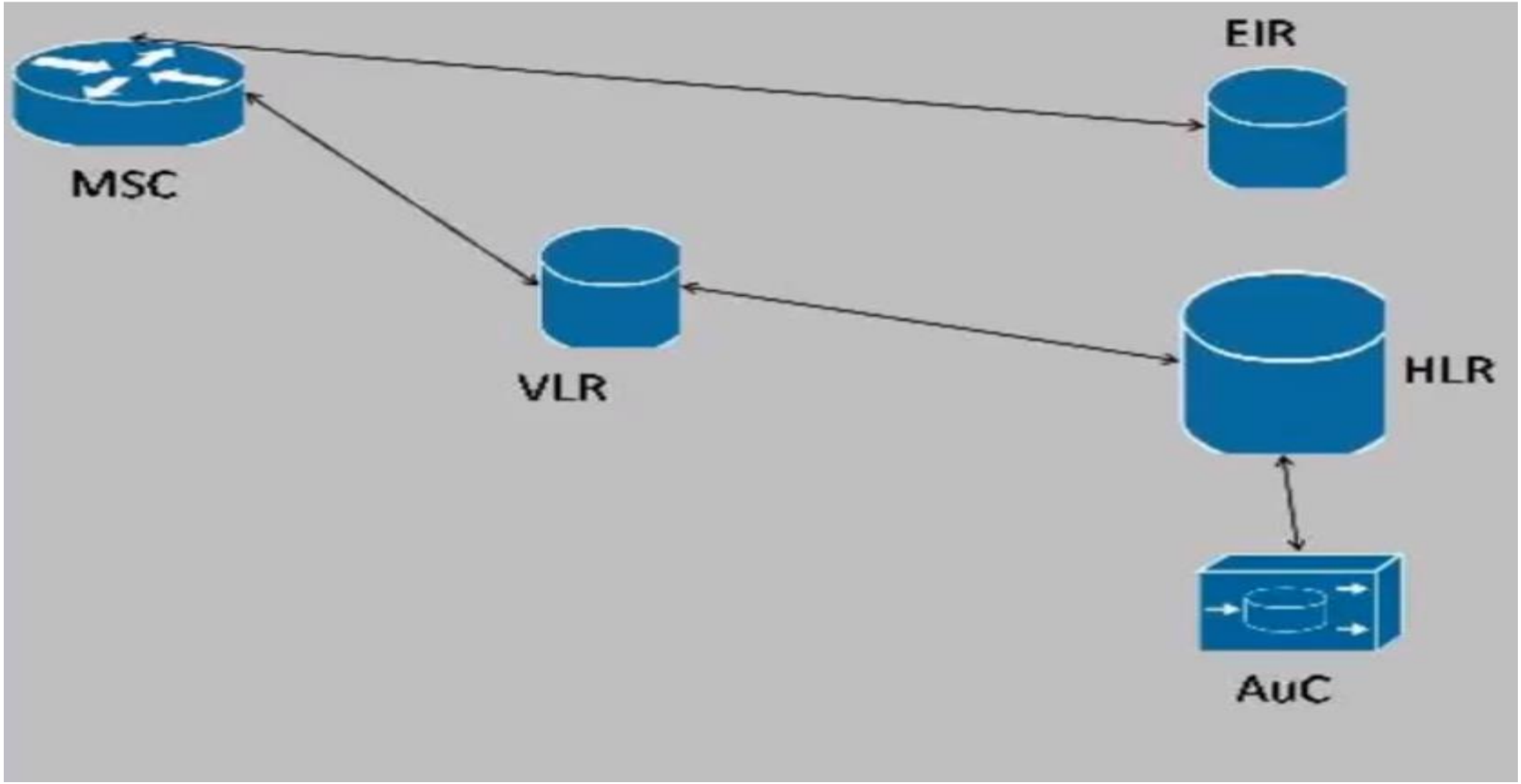
MS

BTS
Encrypt
Encode
Modulation
Frequency hopping

Manages Radio Resources
Handles Call Set up
Handover
Radio Power control

MS → Mobile Station
BTS → Base Transceiver station
BSC → Base Station Controller
MSC → Mobile Switching Centre
GMSC → Gateway MSC
HLR → Home Location Register
VLR → Visitor Location Register
AuC → Authentication Centre
EIR → Equipment Identity Register


BASE STATION SUBSYSTEM
(BSS)
(or) Access Network
NETWORK SUBSYSTEM (NSS)
(or) CORE NETWORK
OMC
AuC
VLR
EIR
HLR
MSC
BTS
Abis
10m
BSC
TRAU
A-Interface
MSC
IWF
SIM
MS
Internet
SIM → IMSI
MS → IMEI
BTS BASESTATION
BSC: BASESTATION CONTROLLER
TRAU: TRANSCODING AND RATE ADAPTION
MSC MOBILE SWITCHING CENTER
HLR HOME LOCATION REGISTER
VLR VISITOR LOCATION REGISTER
AuC AUTHENTICATION CENTER
EIR: EQUIPMENT IDENTITY REGISTER
IWF: INTERWORKING FUNCTION
OMC OPERATION AND MAINTENANCE CENTER


EIR
MSC
VLR
HLR
AuC

Serving General Packet Radio Service Support Node (SGSN)

- A serving general packet radio service support node is a switching node that coordinates the operation of packet radios that are operating within its service coverage range.
- **The SGSN operates in a similar process of a MSC and a VLR, except the SGSN performs packet switching instead of circuit switching.**
- The SGSN registers and maintains a list of active packet data radios in its network and coordinates the packet transfer between the mobile radios.

Voice Message System (VMS)

- The voice mail system is a telecommunications system that allows a subscriber to receive and play back messages from a remote location (such as a PBX telephone or mobile phone).
- The VMS consists primarily of memory storage (for messages), telephone interfaces (to connect to the communication system), and message recording, playback, and control features (typically via DTMF tones).

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- The VMS consists primarily of memory storage (for messages), telephone interfaces (to connect the communication system), and message recording, playback, and control features (typically via DTMF tones).

Gateway GPRS Support Node (GGSN)

- A gateway GPRS support node is a packet switching system that is used to connect a GPRS packet data communication network to other packet networks such as the Internet.

Public Packet Data Network (PPDN)

- A packet data network is a network that is generally available for commercial users (the public).
- An example of a PPDN is the Internet.

NETWORK DATABASES

- Network databases are information storage and retrieval systems that are accessible by a network.
- There are many network databases in the mobile
- network. Some of the key network databases include
  1. A master subscriber database (home location register),
  2. Temporary active user subscriber database (visitor location register),
  3. An unauthorized or suspect user database (equipment identity register),
  4. Billing database, and
  5. Authorization and validation center (authentication).

Billing Center (BC)

- A separate database (called the billing center) keeps records on billing.
- The billing center receives individual call records from MSCs and other network equipment. The switching records (connection and data transfer records) are converted into call detail records (CDRs) that hold the time, type of service, connection points, and other details about the network usage that is associated with a specific user identification code.
- These billing records are then transferred via tape or data link to a separate computer typically by electronic data interchange (EDI) to a billing system or company that can settle bills between different service providers (a clearinghouse company).

Number Portability Database (NPDB)

- Number portability is the ability for a telephone number to be transferred between different service providers. This allows customers to change service providers without having to change telephone numbers.
- Number portability involves three key elements: local number portability, service portability and geographic portability.
- To enable number portability, the mobile system maintains a number portability database (NPDB). This database helps to route calls to their destination which may have an assigned telephone number that is different (number has been ported) than the destination phone number.

IP Backbone Network

- A backbone network is the core infrastructure of a network that connects several major network components together. A backbone system is usually a high-speed communications network such as ATM or FDDI.
- The mobile system uses a backbone network that can provide end-to-end IP transmission capability.

BACKBONE NETWORK

- Backbone network is a communications network that connects the primary switches or nodes within the network.
- The backbone network is usually composed of high-speed switches and communication lines

ATM & FDDI

- Asynchronous Transfer Mode a network technology based on transferring data in cells or packets of a fixed size.
- Fiber Distributed Data Interface, a set of ANSI protocols for sending digital data over fiber optic cable.

Authentication, Authorization, and Accounting (AAA)

- Authentication, Authorization, and Accounting are the processes used in validating the claimed identity of an end user or a device, such as a host, server, switch, or router in a communication network.

AUTHENTICATION

- Authentication is a process of exchanging information between a communications device (typically a user device such as a mobile phone or computing device) and a communications network that allows the carrier or network operator to confirm the true identity of the user (or device).

Authorization

- Authorization is the act of granting access rights to a user, groups of users, system, or a process

ACCOUNTING

- Accounting is the method to establish who, or what, performed a certain action, such as tracking user connection and logging system users.

Interworking Function (IWF)

- Interworking functions are systems and/or processes that attach to a communications network that is used to process and adapt information between dissimilar types of network systems.
- IWFs in the mobile system may include data gateways that convert circuit switched data from the MSC to the Internet.

Message Center (MC)

- The message center is a node or network function within a communications network that accommodates messages sent and received via short messaging service (SMS).

Absolute power of a signal is measured in wattage.
The bel or decibel system can only describe relative power, but the dB scale doesn't define where you're starting from or what your 'zero' is.
So, we specify dBm.

dBm (sometimes dBmW) is an abbreviation for the power ratio in decibels (dB) of the measured power referenced to one milliwatt (mW). Indicating that our scale is relative to 1 milliWatt of power.
0 dBm = 1 mW.

A 3 dB increase represents roughly doubling the power, which means that 3 dBm equals roughly 2 mW. For a 3 dB decrease, the power is reduced by about one half, making −3 dBm equal to about 0.5 milliwatt or 500 microwatts.

0dbm = 1 milliwatt.
-3dBm = half a milliwatt
-6dBm = 1/4 of a milliwatt
-9dBm = 1/8 of a milliwatt.